What Are Intangibles Worth?

Production versus Market Power

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Abstract

How much of intangible capital's value comes from production efficiency versus market power? I estimate a dynamic firm model and quantify both channels for each U.S. public firm. Market power accounts for the majority of intangible value and nearly 30% of aggregate firm value. The two channels evolve differently: across firms, those with greater production value capture market share, whereas within firms, intangible value shifts toward market power as firms mature. Intangible-driven market power also helps explain the value premium decline and predicts intangible investment more strongly than production value does.

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1 Introduction

The rise of intangible capital has transformed both how firms produce and how they compete. Investments in R&D, software, and organizational know-how raise productivity, but they also strengthen market power by differentiating products and protecting firms from competition (Romer, 1990; De Ridder, 2024). While a large literature studies how intangibles affect valuation and firm decisions (e.g., Peters and Taylor, 2017; Belo et al., 2022), much less is known about the relative importance of production versus market power channels. Distinguishing the two roles is economically important. Productivity gains raise efficiency and support growth, whereas market power sustains higher markups and may weaken competition.

This paper separates and quantifies these two sources of intangible value. I construct a dynamic model in which intangible investment raises both productivity and market power. Combining the estimated model with U.S. firm data, I decompose the intangible value for each firm-year into production and market power components. I find that market power is the primary source of intangible value, accounting for close to 30% of aggregate firm value. I then further study how these components vary across firms, evolve over the firm life cycle, and relate to firm valuation and investment.

Because observed firm value reflects multiple interacting economic forces, the data alone cannot separately identify the contribution of intangible capital. I therefore rely on a model structure that separates the production and market power channels. Three features distinguish the model from the standard neoclassical framework. First, intangible capital raises markups, increasing the share of rents in profits and firm value. Second, firms face idiosyncratic markup shocks from changes in the competitive environment and other non-intangible forces, such as shifts in antitrust policy or rivals' strategies. Together, these two sources of market power introduce imperfect competition and non-constant returns to scale in profits. Third, firms differ in how efficiently they use physical and intangible capital, which generates additional variation in firm value.

I estimate the model using the Simulated Method of Moments (SMM) and target key

empirical moments from U.S. public firms over 1976–2025. I then combine the model's value function with state variables extracted from the data to recover model-implied value for each firm-year. The model fits the data well and explains over 80% of the cross-sectional variation in valuation. Next, I apply the Shapley decomposition method (Shapley, 1951) to decompose the firm value into four components: the value of physical capital, intangibles in production, intangibles in market power, and market power unrelated to intangibles.

I find that the majority of intangible capital's value comes from market power. By 2025, this channel accounts for nearly 30% of aggregate model-implied firm value, compared with 19% from production. Together, the two channels account for roughly half of aggregate firm value, broadly consistent with estimates in the prior literature. Thus, a large share of intangible value reflects firms' ability to earn higher margins rather than enhance production.

I validate the economic interpretation of the market power components using two external sources of variation. First, following a firm's first highly cited patent, the value share of intangible-driven market power rises, while non-intangible market power remains largely unchanged. As a complementary test, exposure to the China trade shock affects non-intangible market power but not the intangible-driven component. Together, these results support the interpretation of intangible-driven market power and suggest that the non-intangible component captures variation in the external competitive environment.

The two roles of intangibles evolve differently over time, across the firm life cycle, and in the cross section. Over the past five decades, the share of intangible market power has remained remarkably stable around 29% of the total market value, suggesting that the market power channel has been a persistent feature of intangible capital. In contrast, the value share of production rises from 7.5% in 1976 to 18.7% in 2025, potentially reflecting technological progress over this period. Moreover, this increase in production value is driven primarily by compositional changes, as firms with greater production value capture larger market shares.

Over the firm life cycle, the value share of intangibles in production declines while the market power contribution rises as firms mature. One interpretation is that young firms

initially benefit from proprietary technologies that enhance production. As competitors imitate these technologies or they become obsolete, mature firms increasingly derive intangible value from market power. In the cross section, although production value is smaller in level, it explains more variation in market value across firms than the market power component.

Incorporating intangible-driven market power is important for understanding both asset prices and firm investment. I first show that it helps explain the decline in the value premium (Fama and French, 2021). Book-to-market has become a noisier signal of expected returns because book equity primarily captures productive capital, whereas market equity also includes the value of rents. Consistent with this mechanism, adjusting market value to remove the value of market power yields a monthly return spread of 0.42% ($t = 2.60$), with no significant downward trend. Moreover, a value signal that combines book-to-market with model-implied value shares generates a stronger and more stable premium of 0.51% per month ($t = 3.20$). These results suggest that the value premium has not disappeared. Instead, growing differences in market power have weakened traditional value signals.

The value decomposition also allows me to identify the forces underlying intangible investment. Neoclassical investment theory predicts that q summarizes a firm's investment opportunities (Hayashi, 1982), but measures based on total firm value combine economically distinct sources of value. I therefore construct a separate q for each component of the decomposition and examine how each predicts investment. Both intangible components are strong predictors, but the market power component plays a larger role: its q carries a larger coefficient and explains more within-firm variation. These results suggest that market power is a stronger driver of intangible investment than production.

Beyond quantifying the value of intangible capital, the framework can also shed light on broader questions about market power and firm behavior. While prior work documents a broad rise in market power (Grullon, Larkin and Michaely, 2019; De Loecker, Eeckhout and Unger, 2020), the decomposition reveals that this increase is concentrated in its non-intangible component. I find that the value share of non-intangible market power rises by

more than 15 percentage points over the sample period. This increase is driven entirely by compositional changes, as firms with greater market power capture larger revenue shares. Moreover, non-intangible market power q contributes little to explaining either physical or intangible investment. These patterns suggest that the rise in market power is driven primarily by weaker competitive conditions rather than the returns to intangible investment.

This paper contributes to several literatures that, despite deep theoretical connections, have developed largely in parallel. A central insight from the endogenous growth literature is that intangible capital and market power are inherently linked. Monopoly rents provide incentives for intangible investments, and intangibles can strengthen market power by differentiating products and deterring entry (Romer, 1990; Grossman and Helpman, 1991; Aghion and Howitt, 1992; De Ridder, 2024). Yet little is known about how much of intangible capital's value comes from market power rather than production, or how this distinction matters for firms' investment decisions. This paper bridges this gap with a structural model that separates the production and market power value of intangibles.

This paper contributes to the literature on the rise of intangible capital and its impact on valuations and firm decisions (e.g., Peters and Taylor, 2017; Kogan et al., 2017; Falato et al., 2022). A closely related paper in this literature is Crouzet and Eberly (2023). They decompose the wedge between average and marginal q into intangibles, exogenous rents, and a nonlinear interaction term, highlighting key aggregate and sectoral patterns. My paper complements their analysis along two dimensions. First, I model the endogenous link between intangible capital and markups. Second, I provide firm-level value decompositions and focus on how the composition varies in the *cross section*.

This paper also contributes to investment-based asset pricing that studies asset prices through the lens of firm investment decisions (e.g., Merz and Yashiv, 2007; Liu, Whited and Zhang, 2009). Building on Belo, Xue and Zhang (2013), Belo et al. (2022) decompose firm value into components of physical capital, labor, and intangible inputs using a neoclassical investment model. This paper departs from the neoclassical framework by modeling non-

constant returns to scale and explicitly allowing intangibles to affect production and market power differently. The resulting decomposition quantifies the contribution of market power separately from that of production inputs. Additionally, my framework applies SMM estimation and recovers firm-specific value decompositions, extending beyond the portfolio-level estimation in Belo et al. (2022).

This paper is also connected to the recent literature on declining competition and rising market power. Several papers document a broad increase in market power over recent decades. For example, De Loecker, Eeckhout and Unger (2020) find that aggregate markups for U.S. public firms rose from 21% to 61% above marginal cost over the past four decades. Grullon, Larkin and Michaely (2019) show that industries have become more concentrated. Market power also affects firm risk and expected returns (Bustamante and Donangelo, 2017; Corhay, Kung and Schmid, 2020; Dou, Ji and Wu, 2021) and helps explain the growing divergence between small and superstar firms (Corhay, Kung and Schmid, 2025). This paper instead examines the sources of market power and shows that the rise in aggregate market power is driven mainly by non-intangible forces, consistent with declining competition.

The rest of the paper is organized as follows. Section 2 presents motivating facts and theories on the relation between intangible capital and market power. Section 3 describes a dynamic firm model with intangible capital and endogenous market power. Section 4 outlines the estimation and the decomposition method. Section 5 presents key quantitative results on value decomposition. Section 6 presents two applications: the decline in the value premium and the investment-q relation. Finally, Section 7 concludes.

2 Motivation

The relation between intangible capital and market power is pronounced in the data and is also well-established in economic theory. Section 2.1 presents motivating evidence on the relation between intangibles and markups. Section 2.2 discusses the economic mechanism

through which intangibles generate market power.

2.1 Empirical Facts

2.1.1 Data

I use accounting data on publicly traded firms from the Compustat annual file and market equity from the Center for Research in Security Prices (CRSP) monthly file. I keep only firms with a share code (SHRCD) of 10 or 11 and exchange code (EXCHCD) of 1, 2, or 3. I also exclude firms with non-positive sales, book assets, or firm value and firms with less than $5 million in physical capital (Compustat item PPEGT, deflated in 1990 dollars). I exclude firms from regulated utilities (SIC codes 4900–4999), financial firms (6000–6999), and firms categorized as public service, international affairs, or nonoperating establishments (9000+).

I use the measure of intangible capital from Peters and Taylor (2017), which capitalizes R&D and 30% of SG&A expenses along with balance-sheet goodwill.[1] I exclude firms with missing or zero intangible stock. The final sample contains 129,013 firm-year observations from 1976 to 2025, which correspond to 12,289 distinct firms. The sample starts in 1976 because it is the first year when the intangible capital measure is available.

In this section, I measure market power using firm-level markups based on the production function approach, which recovers the markup as the ratio of a variable input's output elasticity to its revenue share.[2] My main specification assumes Cobb-Douglas production with variable inputs, physical capital, and intangible capital. Internet Appendices A.1 and B.2 provide definitions and results for five alternative markup measures, including two model-free proxies and three production-based estimates under alternative technologies.

[1] I use the Peters and Taylor (2017) measure in the main specification because it is widely used in the literature (e.g., Eisfeldt and Papanikolaou, 2013; Crouzet and Eberly, 2023), so the results are easier to compare with prior work. Internet Appendix B.1.4 shows that the main results of the paper are similar using the alternative measure of Ewens, Peters and Wang (2024).

[2] While widely used, production-based markup estimates face concerns related to identification without quantity data and imperfect input proxies (Bond et al., 2021; Demirer, 2022; Raval, 2023; Traina, 2018). De Ridder, Grassi and Morzenti (2026) show that, although possibly biased in levels, they accurately capture cross-sectional markup dispersion.

2.1.2 Correlation between Intangibles and Market Power

Figure 1(a) plots the annual cross-sectional correlation between intangible capital share and markup. The correlation is consistently positive and significant throughout the sample period, ranging from 0.2 to 0.4. Thus, the link between intangibles and market power has been a persistent phenomenon in the economy. Figure 1(b) further illustrates this relation through binned scatter plots after purging out industry and year fixed effects. The relation is positive and approximately linear, with markups rising from roughly one at the lowest intangible share bins to around 1.75 at the highest.[3]

2.1.3 Patent Event Studies

The correlation results provide suggestive evidence linking intangibles to markups, but they may also reflect high-markup firms investing more in intangible capital. In this section, I use patent events to establish a clearer timing of the markup effects.

Patent events provide an observable outcome of prior intangible investment and a natural setting to study its market power effects. To identify economically meaningful innovations, I classify a patent as important if its forward citations or economic value from Kogan et al. (2017) exceeds the 90th percentile within its Fama-French 48 industry-year.

Figure 2 traces markup changes around the filing and grant dates of each firm's first important patent.[4] Following filing, markups rise by roughly 0.03 for highly cited patents and 0.07 for high-value patents. Importantly, markups remain largely flat before filing. The larger response for high-value patents reflects the fact that the economic value measure embeds investors' expectations of future profits and market power. Patent filing occurs after the underlying innovation has been developed. The innovation can therefore begin to create product differentiation and other competitive advantages before formal patent protection is

[3]Results are largely similar when using alternative markup measures and in a regression framework with firm-level controls (Internet Appendix B.2).

[4]I focus on the first important patent to avoid contamination from earlier patents. A stacked design using all important patents yields similar results but exhibits pre-trends; alternative markup measures also produce similar results (Internet Appendix B.2).

granted. Markups also rise around grant dates, but the response is weaker and preceded by pre-trends, suggesting that successful innovations affect markups before formal approval.

2.2 Why Intangibles Generate Market Power

Conceptually, intangible capital can generate market power because it is non-rival in use and partially excludable. Non-rivalry allows an intangible asset to be deployed across many products at little additional cost, making its benefits highly scalable. For example, a new product design can increase margins across all units sold without requiring investment to increase proportionally. Partial excludability then allows the firm to capture the resulting rents.

This mechanism is well-established in economic theory. In endogenous growth models, innovation generates monopoly rents that compensate firms for the fixed costs of creating knowledge (Romer, 1990; Grossman and Helpman, 1991; Aghion and Howitt, 1992; Klette and Kortum, 2004), while the industrial organization literature emphasizes patents and strategic innovation as sources of market leadership (Nordhaus, 1969; Loury, 1979; Dasgupta and Stiglitz, 1980).[5] Other intangibles, including branding, customer relationships, product differentiation, and organizational capital, can similarly reduce price sensitivity or deter entry (Gourio and Rudanko, 2014; Dixit and Stiglitz, 1977; Doraszelski and Markovich, 2007; De Ridder, 2024). These mechanisms, together with the evidence in Section 2.1, motivate a model in which intangible capital affects both production and market power.

3 Model

Time is discrete, and the horizon is infinite. Firms use physical capital, intangible capital, and costlessly adjustable inputs to produce heterogeneous products. Firms face imperfect competition in the product market and can invest in intangibles to raise their markup. In

[5] In these models, intangible inputs are modeled as disembodied ideas rather than as firms' own assets. However, the same logic carries over to dynamic firm models in which innovations are capitalized into an intangible capital stock.

each period, firms jointly choose the price and flexible inputs to maximize operating profits, subject to the market demand. Taking the operating profits as given, firms then choose physical and intangible investments to maximize the market value of equity.

3.1 Technology

Firm i produces under a Cobb-Douglas technology: $Y_{it} = \mathbf{K}_{it}^{\eta} L_{it}^{1-\eta}$, in which L_{it} is a bundle of flexible inputs, e.g., labor and materials. Similar to Crouzet and Eberly (2023), I specify a capital aggregator, $\mathbf{K}_{it}$, and assume that it follows the CES functional form:

$$\mathbf{K}_{it} = \mathbf{K}(z_{it}^K, z_{it}^N, K_{it}, N_{it}) = \left[\alpha \exp(z_{it}^K) K_{it}^{\rho} + (1-\alpha)\exp(z_{it}^N) N_{it}^{\rho}\right]^{\frac{1}{\rho}}, \tag{1}$$

in which K_{it} is physical capital and N_{it} is intangible capital. The parameter ρ governs the curvature of the CES aggregator. The corresponding elasticity of substitution between physical and intangible capital is $\sigma = 1/(1-\rho)$. Two capital-specific productivity shocks z_{it}^K and z_{it}^N capture the stochastic productivity of physical and intangible capital. They are firm-specific and thus contribute to cross-sectional heterogeneity in the model. I assume production exhibits constant returns to scale in capital and the flexible inputs. Intangible capital, N, includes all types of intangibles, such as R&D capital, organizational capital, and other intangible assets. Section 3.5 discusses these modeling choices in more detail.

Firms have pricing power in the product market, and markups reflect the degree of their market power. Firm i charges a markup of μ_{it}, defined as price over marginal cost. I assume that μ_{it} follows the functional form below:

$$\mu_{it} = \mu(N_{it}, \zeta_{it}^{\mu}) = \exp(\zeta_{it}^{\mu})(1 + \bar{\mu} \cdot N_{it}^{\phi}). \tag{2}$$

The exogenous shock ζ_{it}^{μ} captures the competitive environment the firm faces, and parameters $\bar{\mu}$ and ϕ govern the magnitude of intangible capital's effect on market power.

The markup specification embeds two distinct sources of market power. First, firms

endogenously accumulate market power through investment in intangible capital, in line with the empirical evidence and economic mechanisms in Section 2. Second, market power may shift due to non-intangible forces, captured by the exogenous shock ζ_{it}^{μ}. These shocks can be either positive, reflecting favorable market conditions such as a high entry barrier, or negative, reflecting increased competition or rivals' innovation. I do not allow physical capital to directly affect market power. Section 3.5 discusses this assumption.

In each period, the firm chooses its optimal level of flexible inputs and price to maximize profits, taking demand and the price of the input, P_{Lit}, as given. Let $\Pi_{it} = \Pi(z_{it}^K, z_{it}^N, \zeta_{it}^{\mu}, K_{it}, N_{it})$ denote the operating profits for firm i at time t (Appendix A provides details for derivations):

$$\Pi_{it} = \max_{L_{it}, P_{it}} P_{it}Y_{it} - P_{Lit}L_{it} = \begin{cases} \underbrace{\left[1 - \frac{1-\eta}{\mu_{it}}\right]\left(\frac{1-\eta}{\mu_{it}}\right)^{\frac{1-\eta}{\mu_{it}-1+\eta}} \bar{Y}}_{\bar{\Pi}(\mu_{it})} \cdot \mathbf{K}_{it}^{\frac{\eta}{\mu_{it}-1+\eta}} & \text{if } \mu_{it} > 1 \\ \eta(1-\eta)^{\frac{1-\eta}{\eta}} \bar{Y} \cdot \mathbf{K}_{it} & \text{if } \mu_{it} \leq 1, \end{cases} \tag{3}$$

where $\bar{Y}$ is the size of the market and is a constant.[6] When the markup is less than or equal to 1, the firm is assumed to operate under perfect competition and take prices as given.

The profit function shows two effects of market power. On one hand, firms earn more profits with higher price-setting power, as $\bar{\Pi}(\mu_{it})$ is increasing in μ_{it}. On the other hand, market power decreases the incentive to invest through the second term $\mathbf{K}_{it}^{\frac{\eta}{\mu_{it}-1+\eta}}$ as higher μ_{it} decreases the marginal benefit of capital. This feature is in line with the recent trend of declining capital share in the economy (Barkai, 2020).

[6] This profit function applies as long as the firm's implied revenue does not exceed the total market size $\bar{Y}$. If a firm is productive enough to hit this ceiling, it captures the entire market and its profit is governed by the capped-revenue regime detailed in Appendix A.

3.2 Investment

Capital equals investment plus beginning-of-period capital net of depreciation:

$$K_{it+1} = (1 - \delta^K)K_{it} + I^K_{it}, \tag{4}$$

$$N_{it+1} = (1 - \delta^N)N_{it} + I^N_{it}, \quad I^N_{it} \geq 0. \tag{5}$$

The depreciation rate δ^N for intangible capital can be interpreted as the rate of creative destruction. New technologies render old ones obsolete, and brand loyalty wears off over time. I do not allow disinvestment in intangible capital. Intangible investments are difficult to resell in secondary markets and are therefore largely sunk and irreversible once incurred.[7]

Investment in both types of capital incurs quadratic costs:

$$\Phi^K(I^K_{it}, K_{it}) = I^K_{it} + \frac{b^K}{2}\left(\frac{I^K_{it}}{K_{it}}\right)^2 K_{it}, \tag{6}$$

$$\Phi^N(I^N_{it}, N_{it}) = I^N_{it} + \frac{b^N}{2}\left(\frac{I^N_{it}}{N_{it}}\right)^2 N_{it}. \tag{7}$$

The quadratic adjustment cost for physical investment is standard in the q-theory literature. The same intuition can be generalized to intangible capital. The convex adjustment cost captures diminishing returns to intangible investment as well as constant returns to scale in I^N and N jointly, which is consistent with the stylized fact that the innovation rate is independent of firm size (Klette and Kortum, 2004).

3.3 Stochastic Processes and the Stochastic Discounting Factor

I jointly model aggregate and idiosyncratic productivity shocks as a single state variable, which implicitly assumes that they have the same persistence. This choice mitigates the curse of dimensionality and ensures that the structural estimation remains computationally

[7]The irreversibility assumption is also consistent with the construction of the data: because intangible capital is measured from capitalized expenditures, there is no observed disinvestment. Internet Appendix B.1.3 shows that the main results are similar when disinvestment is allowed.

feasible. Two capital-specific productivity shocks follow:

$$z_{it+1}^K = x_{t+1}^K + \zeta_{it+1}^K = \rho^K(x_t^K + \zeta_{it}^K) + \varepsilon_{x,t+1}^K + \varepsilon_{\zeta,it+1}^K, \tag{8}$$

$$z_{it+1}^N = x_{t+1}^N + \zeta_{it+1}^N = \rho^N(x_t^N + \zeta_{it}^N) + \varepsilon_{x,t+1}^N + \varepsilon_{\zeta,it+1}^N, \tag{9}$$

where, for $j \in \{K, N\}$, x_{t+1}^j is aggregate productivity, ζ_{it+1}^j is firm-specific productivity, and $\varepsilon_{x,t+1}^j$ and $\varepsilon_{\zeta,it+1}^j$ are i.i.d. aggregate and firm-specific shocks, respectively.

Following Zhang (2005), I specify the stochastic discount factor exogenously as:

$$M_{t+1} = \exp\left(-r_f - \gamma^K \varepsilon_{x,t+1}^K - \gamma^N \varepsilon_{x,t+1}^N - \frac{(\gamma^K)^2(\sigma_x^K)^2}{2} - \frac{(\gamma^N)^2(\sigma_x^N)^2}{2}\right), \tag{10}$$

where r_f, γ^K, γ^N correspond to the interest rate and the prices of risk for physical and intangible productivity shocks. σ_x^K and σ_x^N are the standard deviations of the aggregate productivity shocks for physical and intangible capital, respectively.

The exogenous shock to markup is given by

$$\zeta_{it+1}^\mu = \rho^\mu \zeta_{it}^\mu + \varepsilon_{\mu,it+1}, \tag{11}$$

in which $\varepsilon_{\mu,it+1}$ is independently and identically distributed and uncorrelated across firms.

3.4 Firms' Maximization Problem

The payout of firm i equals

$$D_{it} \equiv \Pi(z_{it}^K, z_{it}^N, \zeta_{it}^\mu, K_{it}, N_{it}) - \Phi^K(I_{it}^K, K_{it}) - \Phi^N(I_{it}^N, N_{it}) - f, \tag{12}$$

where f is the fixed operating cost. Firm i maximizes its market value of equity:

$$V_{it}^{\text{cum}} = \max_{\{I_{it+s}^K, I_{it+s}^N\}_{s=0}^\infty} \mathbb{E}_t\left[\sum_{s=0}^\infty M_{t,t+s} D_{i,t+s}\right]. \tag{13}$$

Firms are characterized by the vector of state variables $(z_{it}^K, z_{it}^N, \zeta_{it}^\mu, K_{it}, N_{it})$. Firms' maximization problem can then be written recursively as

$$V^{\text{cum}}(z_{it}^K, z_{it}^N, \zeta_{it}^\mu, K_{it}, N_{it}) = \max_{K_{it+1}, N_{it+1}} D_{it} + \mathbb{E}_t \left[M_{t+1} V^{\text{cum}}(z_{it+1}^K, z_{it+1}^N, \zeta_{it+1}^\mu, K_{it+1}, N_{it+1}) \right], \tag{14}$$

subject to capital accumulation equations (4) and (5). Lastly, I rewrite the value function as the ex-dividend value to match the data valuation:

$$V(z_{it}^K, z_{it}^N, \zeta_{it}^\mu, K_{it}, N_{it}) = V^{\text{cum}}(z_{it}^K, z_{it}^N, \zeta_{it}^\mu, K_{it}, N_{it}) - D_{it}. \tag{15}$$

3.5 Discussion of Modeling Assumptions

Constant returns-to-scale. The assumption of constant returns to scale in production is motivated by both identification and the prior literature. First, returns to scale and markups generally cannot be separately identified (Flynn, Gandhi and Traina, 2019). Second, CRS is a standard assumption in a large body of empirical and theoretical work (e.g., Atkeson and Burstein, 2008; Edmond, Midrigan and Xu, 2023). Flynn, Gandhi and Traina (2019) further document that estimated returns to scale in U.S. firm-level data are centered close to one, supporting CRS as a reasonable empirical approximation.

One concern is that the scalability of intangible capital may give rise to increasing returns. Although I impose constant returns to scale in production, the model allows profits to scale more than proportionally because intangible capital also raises market power. As firms scale their intangible stock, its productive benefits expand with output while the associated markup applies across units sold. Thus, the model captures the scalability of intangible capital without imposing increasing returns in the production technology.

Intangible capital as a single state variable. I model intangible capital as a single firm-level input because the paper decomposes its value by economic channel rather than

by asset type. This choice also reflects measurement constraints: accounting data do not fully capture intangible assets, and measures of individual components often lack broad coverage. Moreover, different intangibles can operate through both channels. For example, R&D and organizational know-how may simultaneously raise productivity and strengthen market power. Aggregating these components therefore keeps the model tractable while preserving the distinction between production and market power. The model captures the average effect of intangible capital along each channel rather than separately identifying the effects of different intangible asset types.

Markup and physical capital. I do not allow physical capital to directly affect markups. This modeling choice is motivated by the empirical relationship between intangibles and markups, as well as by the economic properties of intangibles discussed in Section 2. Other firm characteristics, including physical capital, may also contribute to market power—for example, through equipment scarcity or entry barriers. While these mechanisms are not modeled explicitly, their contribution to observed market power is absorbed empirically by the residual markup component associated with ζ^{μ}_{it}.[8]

Entry and exit. The model is designed to value individual firms and therefore abstracts from an explicit entry decision and an endogenous firm population. This modeling choice is less consequential for the value decomposition in the next section because I evaluate the value function at each observed firm's realized state. Entry and exit can affect aggregate value through changes in the composition of firms, but these changes are reflected in the observed cross section to which the decomposition is applied. In addition, competitive pressure from entry can affect incumbent firms through the exogenous market power state, ζ^{μ}, which summarizes changes in the competitive environment. Its estimated dynamics therefore

[8]In the model, ζ^{μ}_{it} represents exogenous changes that are outside the firm's control. Thus, endogenous sources of market power other than intangibles, such as those arising from physical capital, are not literally part of the model definition of ζ^{μ}_{it}. Empirically, however, it is inferred as markup variation not explained by intangible capital, so omitted non-intangible sources of market power are also absorbed into this component.

capture these effects on incumbent values in reduced form.

4 Estimation and Decomposition

In this section, I first outline the strategy for identifying model parameters in Section 4.1 and show the estimation results in Section 4.2. Then I discuss the decomposition framework and the measurement of state variables in Section 4.3. Finally, Section 4.5 evaluates the model fit and Section 4.6 validates the decomposition.

4.1 Identification

There are 22 parameters in the model. I calibrate four parameters as reported in Panel A of Table 1 and estimate the remaining 18 parameters. Specifically, I set the constant annual risk-free rate r_f to be 4%, matching the data value during the sample period 1976–2025. The size of the product market, $\bar{Y}$, is set to be 20, which normalizes the long-term average level of total capital to be around four. The annual depreciation rate of physical capital, δ^K, is set to be 0.12, following Gomes (2001) and Cooper and Haltiwanger (2006). The depreciation rate for intangible capital, δ^N, is calibrated to 0.2, close to both the average rate in Peters and Taylor (2017) and the rate calibrated in Falato et al. (2022).

I estimate the remaining 18 parameters using the Simulated Method of Moments (SMM). Identification relies on selecting moments that are informative about underlying structural parameters. To discipline the estimation, I group the parameters into four categories according to the margins they primarily govern. The first group concerns the production technology. I use the mean of the profits-to-total-capital ratio to identify the share of capital in production, η. I include the average ratio of intangible capital to total capital to pin down the share of physical capital in production, α.[9] To identify the CES substitution

[9]In the main specification, I use the intangible measure from Peters and Taylor (2017). Internet Appendix B.1.4 reports the estimated parameters and main results using the alternative measure from Ewens, Peters and Wang (2024). The results are similar.

parameter, ρ, I rely on the standard deviation of the intangible-to-total-capital ratio. When the two capital types are complementary, intangible intensity should remain close to its optimal ratio and exhibit smaller variation in the data. Lastly, I include the average total value-to-total-capital ratio to identify the fixed operating cost f.

The second group includes parameters that govern investment in physical and intangible capital. I use the standard deviations of the physical and intangible investment rates to pin down the adjustment cost parameters, $\{b^K, b^N\}$, respectively.[10] For the parameters that govern two productivity processes, I first match the AR(1) coefficient and the standard deviation of profits scaled by physical capital to identify the persistence and volatility of the idiosyncratic physical productivity shock, ρ^K and σ_ζ^K. For the intangible-productivity shock, I use the AR(1) coefficient and the standard deviation of profits scaled by intangible capital to identify the persistence and volatility, $\{\rho^N, \sigma_\zeta^N\}$, respectively. These two profitability moments load differently on the two shocks and help disentangle their respective contributions. Since I jointly model the persistence of aggregate and idiosyncratic productivity shocks, I additionally target the AR(1) coefficient of both firm-level and aggregate profits scaled by total capital to discipline the persistence of shocks.

The third group of parameters captures the interaction between intangible capital and markup. I use moments of the ratio between sales and cost of goods sold (Compustat item SALE/COGS) to discipline markup-related parameters.[11] I identify the average effect of intangibles, $\bar{\mu}$, and the curvature of the markup function, ϕ, using the average SALE/COGS and the cross-sectional correlation between intangible capital and SALE/COGS. I then rely on the AR(1) coefficient of SALE/COGS to identify the persistence of exogenous markup shocks, ρ^μ, and the standard deviation of SALE/COGS to pin down their volatility, σ^μ.

[10] In the main specification, I use the Compustat item CAPX to measure physical investment. This definition is consistent with the investment-q analysis in Section 6. Internet Appendix B.1.2 reports the estimated parameters and main results using changes in PPENT plus depreciation as investment. The results are similar.

[11] Under the assumption of Cobb-Douglas technology, the ratio of sales to variable costs maps directly to markups. However, COGS may not be an accurate proxy for variable costs. Internet Appendix B.1.5 follows Ayyagari, Demirgüç-Kunt and Maksimovic (2024) and includes all uncapitalized SG&A expenses in variable costs in addition to COGS. The main results are similar.

The last group governs the stochastic discounting factor and the aggregate productivity. I use the average market return to identify the risk price of physical productivity shocks, γ^K, and the average return spread between high- and low-profitability firms to identify the risk price of intangible productivity shocks, γ^N. I include two additional return moments to help discipline the risk prices: the average return spread sorted on markup and investment. I then use the volatility of aggregate physical and intangible investment to identify the standard deviation of the two aggregate productivity shocks, $\{\sigma_x^K, \sigma_x^N\}$. I also include the volatility of aggregate profitability to help identify the volatility of these aggregate shocks.

4.2 Estimation Results

Panel B of Table 1 reports the estimated parameters. Overall, the estimated parameters are economically plausible and broadly consistent with values used in the prior literature. Regarding production technology, physical capital accounts for around 83% of total capital in production, in line with recent studies such as Crouzet and Eberly (2023). The total capital share is around 0.26, close to that in Bloom (2009). The estimated CES curvature parameter is $\rho = 0.50$, suggesting that physical and intangible capital are more substitutable than under the Cobb-Douglas benchmark, for which $\rho = 0$. Although patents and machinery are not directly substitutable, firms can adopt different business models that rely more heavily on either intangible or physical capital. The estimated degree of substitutability captures this flexibility, consistent with the large dispersion in intangible intensity across firms and allowing the model to accommodate firms with little or no intangible capital.

In the second group, the adjustment cost for physical capital is estimated to be larger than that for intangible capital, as technological progress makes intangible investment less costly. Because I do not allow for disinvestment in intangibles, the overall adjustment friction is still higher for intangible capital.[12] In addition, intangible productivity shocks are larger

[12]Belo et al. (2026) find that intangible capital is substantially more costly to adjust than physical capital. Two methodological differences may explain this discrepancy. First, I identify adjustment costs using investment rates, whereas they use valuation ratios. Second, my baseline model assumes irreversibility

and more persistent than physical productivity shocks. The larger shocks are consistent with the scalable nature of intangibles and the uncertainty in innovations, which can generate more dispersed and skewed productivity outcomes than physical capital. The third group characterizes the behavior of markups and the interaction with intangible capital. The estimate of $\phi < 1$ implies diminishing marginal effects of intangible capital on markups. The exogenous markup shock is less persistent and less volatile than the productivity shocks, implying that changes in the external competitive environment are more transitory.

Table 2 presents the 23 moment conditions used in the estimation. Panel A reports moments based on firm-level characteristics and Panel B reports the aggregate moments. Simulated moments tend to be reasonably close to their empirical counterparts, with some gaps, as expected given the large number of moments I consider.[13]

4.3 Shapley Decomposition

Interactions among the state variables make their individual contributions to firm value inherently order-dependent. I therefore employ the Shapley decomposition (Shapley, 1951) to decompose firm value. The method averages each state variable's marginal contribution across all possible orderings, thereby fairly allocating firm value while accounting for interactions among the states.[14]

To fix ideas, consider a simplified example with only two state variables, physical capital K and intangible capital N. The Shapley decomposition of firm value is

$$V(K_{it}, N_{it}) = V(K_{\min}, N_{\min}) + V_{it}^{K} + V_{it}^{N}, \tag{16}$$

in intangible investment. Appendix B.1.2 considers an alternative definition of investment rates, while Appendix B.1.3 allows disinvestment in intangible capital. Both specifications yield higher estimates of intangible adjustment costs, while the main results remain similar.

[13]Return moments are measured with large standard errors and therefore receive smaller SMM weights under the inverse-variance weighting scheme, leading to larger gaps between data and simulated moments. Internet Appendix B.1.1 uses a two-stage procedure in which return moments identify risk prices separately. The estimated parameters and main results remain similar.

[14]Shapley decompositions have been widely used in economics and finance, including studies of income inequality (Shorrocks, 1982), worker- and firm-level wage dispersion (Abowd, Kramarz and Margolis, 1999), and heterogeneity in investor expectations (Couts, Gonçalves and Loudis, 2023).

where $K_{\min}, N_{\min}$ correspond to the minimum grid values for the two types of capital. There are two orderings in which the two variables can be added: K enters first, followed by N; or N enters first, followed by K. The Shapley values average the contribution of variables over these two orderings:

$$\begin{aligned} V_{it}^K &= \frac{1}{2}\Big[\underbrace{V(K_{it}, N_{\min}) - V(K_{\min}, N_{\min})}_{K \text{ enters first}} + \underbrace{V(K_{it}, N_{it}) - V(K_{\min}, N_{it})}_{K \text{ enters second}}\Big], \\ V_{it}^N &= \frac{1}{2}\Big[\underbrace{V(K_{\min}, N_{it}) - V(K_{\min}, N_{\min})}_{N \text{ enters first}} + \underbrace{V(K_{it}, N_{it}) - V(K_{it}, N_{\min})}_{N \text{ enters second}}\Big]. \end{aligned} \tag{17}$$

By construction, $V_{it}^K + V_{it}^N = V(K_{it}, N_{it}) - V(K_{\min}, N_{\min})$, so the decomposition is exact. Crucially, the Shapley approach accounts for complementarities between K and N as any interaction effect is split between the two variables, rather than being attributed entirely to whichever variable is added last.

Formally, I apply this same logic to the full value function, and decompose the contribution of intangible capital into production and markup channels:

$$\begin{aligned} V_{it} &= V(z_{it}^K, z_{it}^N, \zeta_{it}^\mu, K_{it}, N_{it}) \\ &= \underbrace{\bar{V} + V_{it}^{z^K} + V_{it}^K}_{\equiv V_{it}^{\text{prod-phy}}} + \underbrace{V_{it}^{z^N} + V_{it}^{N,\text{prod}}}_{\equiv V_{it}^{\text{prod-int}}} + \underbrace{V_{it}^{N,\text{mkp}}}_{\equiv V_{it}^{\text{mkp-int}}} + \underbrace{V_{it}^{\zeta^\mu}}_{\equiv V_{it}^{\text{mkp-non}}}, \end{aligned} \tag{18}$$

where $\bar{V} \equiv V(z^K = 0, z^N = 0, \zeta^\mu = 0, K = K_{\min}, N = N_{\min})$ is the value function evaluated at constant benchmark values for the five state variables. $\bar{V}$ represents the baseline value of a firm with close-to-zero levels of capital and subject to no exogenous shock. Each Shapley value term V_{it}^{comp} measures the marginal contribution of each variable to firm value, averaged over all possible orderings.

$V_{it}^{N,\text{prod}}$ and $V_{it}^{N,\text{mkp}}$ capture the contribution of intangible capital through production and markup, respectively. They cannot be separated directly in the current model because intangibles enter as a single state variable. I therefore solve an auxiliary model with separate intangible variables for production and markup. I then apply a six-variable Shapley decom-

position that treats N^{prod} and N^{mkp} as distinct. I use the original value function when the two variables are equal and the auxiliary value function when only one differs. The details are provided in Internet Appendix A.3.

I group the resulting six Shapley terms in Equation (18) into four economic components. The value of capital in production, $V_{it}^{\text{prod-phy}}$ and $V_{it}^{\text{prod-int}}$, contains both the capital-specific productivity and the capital stock. The value of physical capital, $V_{it}^{\text{prod-phy}}$, contains the constant term $\bar{V}$ as it represents the benchmark value of a firm without intangible capital and market power.[15] Two market power components, $V_{it}^{\text{mkp-int}}$ and $V_{it}^{\text{mkp-non}}$, capture the contribution of intangible capital and non-intangible shocks to market power, respectively.

Because the non-intangible markup shock can be either positive or negative, its contribution to firm value can also be negative relative to the zero-shock benchmark. A negative contribution indicates that the firm operates in a more competitive environment, which lowers its markup and firm value relative to the benchmark.

4.4 Measure of the State Variables

To carry out the decomposition, I need to map all five state variables $(z_{it}^K, z_{it}^N, \zeta_{it}^\mu, K_{it}, N_{it})$ for each firm-year onto the model's state space. Two capital inputs are directly observable in the data. I use Compustat item PPEGT for physical capital and the measure of intangible capital from Peters and Taylor (2017). I normalize total capital to match the model distribution:

$$\begin{aligned}
\widehat{K+N}_{it} &= \overline{K+N} + \sigma^{K+N} \cdot z(K_{it}+N_{it}), \\
\hat{K}_{it} &= \widehat{K+N}_{it} \cdot \frac{K_{it}}{K_{it}+N_{it}}, \\
\hat{N}_{it} &= \widehat{K+N}_{it} \cdot \frac{N_{it}}{K_{it}+N_{it}},
\end{aligned} \tag{19}$$

where $z(K_{it}+N_{it})$ is the cross-sectional z-score of firm i's total capital in the data, and $\overline{K+N}$ and σ^{K+N} are the mean and standard deviation of total capital in the model simulation. This

[15] Empirically, the constant term is small, accounting for around 3% of firm value on average.

transformation removes the empirical time trend in the level while preserving the observed capital composition.

In the model, the markup is $\mu_{it} = (1-\eta)P_{it}Y_{it}/P_{Lit}L_{it}$ under the assumption of Cobb-Douglas technology. I use Compustat item SALE to proxy for total sales, $P_{it}Y_{it}$, and COGS for variable costs, $P_{Lit}L_{it}$, following De Loecker, Eeckhout and Unger (2020).[16] I then back out the value of the exogenous markup shock $\hat{\zeta}^{\mu}_{it}$ using the estimated parameters and the markup function in Equation (2).

Next, I estimate the two productivity terms z^K_{it} and z^N_{it} by minimizing the following objective function for each firm-year:

$$\begin{aligned}\min_{z^K,\, z^N} \quad & \left[\frac{V}{K+N}(z^K, z^N, \hat{\zeta}^{\mu}_{it}, \hat{K}_{it}, \hat{N}_{it}) - \frac{V_{it}}{K_{it+1}+N_{it+1}}\right]^2 / \sigma^2_{V/(K+N)} \\ & + \left[\frac{N}{K+N}(z^K, z^N, \hat{\zeta}^{\mu}_{it}, \hat{K}_{it}, \hat{N}_{it}) - \frac{N_{it+1}}{K_{it+1}+N_{it+1}}\right]^2 / \sigma^2_{N/(K+N)},\end{aligned} \tag{20}$$

where $\frac{V}{K+N}(\cdot)$ and $\frac{N}{K+N}(\cdot)$ are the model-implied ratios, and the errors are scaled by their variances $\sigma^2_{V/(K+N)}$ and $\sigma^2_{N/(K+N)}$ in the data.[17] The system has two unknowns and two target moments, so in most firm-years the minimization achieves an exact match, though residuals remain for firms with extreme realizations. So, by construction, the model largely captures cross-sectional variation in valuations. The goal of the decomposition is to understand the composition of firm value rather than to explain its level.

With the estimated state variables $\hat{\mathbf{s}}_{it} \equiv (\hat{z}^K_{it}, \hat{z}^N_{it}, \hat{\zeta}^{\mu}_{it}, \hat{K}_{it}, \hat{N}_{it})$ for each firm-year, I de-

[16] Internet Appendix B.1.5 shows the results when 70% of SG&A expenses are also included as variable costs. The main results remain similar.

[17] Intangible capital may be measured with noise because capitalized R&D and SG&A do not fully capture assets such as brand loyalty or supplier relationships. Such noise is absorbed by productivity and markup shocks. This leaves the production component largely unaffected as productivity shocks are grouped with intangibles in production, but non-intangible market power may partly capture unmeasured intangibles. Section 4.6 discusses the interpretation of this component and shows that it plausibly captures the competitive environment, instead of unmeasured intangibles.

compose the firm value as:

$$V_{it} = V_{it}^{\text{model}} + V_{it}^{\varepsilon} \tag{21}$$

$$= V_{it}^{\text{prod-phy}} + V_{it}^{\text{prod-int}} + V_{it}^{\text{mkp-int}} + V_{it}^{\text{mkp-non}} + V_{it}^{\varepsilon}, \tag{22}$$

where

$$\begin{aligned} V_{it}^{\text{model}} &\equiv (K_{it+1} + N_{it+1}) \cdot \frac{V}{K+N}(\hat{\mathbf{s}}_{it}), \\ V_{it}^{j} &\equiv V^{j}(\hat{\mathbf{s}}_{it}) \cdot \frac{K_{it+1} + N_{it+1}}{K(\hat{\mathbf{s}}_{it}) + N(\hat{\mathbf{s}}_{it})}, \\ & j \in \{\text{prod-phy}, \text{prod-int}, \text{mkp-int}, \text{mkp-non}\}, \end{aligned} \tag{23}$$

and V_{it}^{ε} is actual firm value minus model-implied firm value.

4.5 Model Fit

Figure 3 plots the time series for the average estimated state variables as well as the standard deviation across firms in each year. Both z^K and z^N show fluctuations over the business cycle. Intangible productivity z^N has been steadily increasing over time and exhibits larger dispersion in the cross section, which suggests that firms differ more in the efficiency with which they utilize intangibles. Panels (c) and (d) plot the markup μ and the intangible capital share $N/(K+N)$, both of which trend upward over the sample. Moreover, firms become increasingly different in their market power, as indicated by the increasing cross-sectional dispersion of markups over time.

Figure 4(a) plots the cross-sectional R^2 in each year that measures how closely the model-implied firm value V_{it}^{model} matches the data counterpart V_{it}:

$$\hat{R}_t^2 = 1 - \frac{\sum_i \left(V_{it} - V_{it}^{\text{model}}\right)^2}{\sum_i \left(V_{it} - \bar{V}_t\right)^2}, \tag{24}$$

where $\bar{V}_t$ is the cross-sectional mean of V_{it}. Unlike the standard regression R^2, this measure does not allow for a free intercept or slope and therefore penalizes both level and scale

deviations between the model and the data. On average, the R^2 is 86%, and it is stable over time and above 80% for most years, except for periods with special economic conditions such as the dot-com bubble and the 2008 financial crisis. I also compute R^2 for value scaled by capital, $V_{it}/(K_{it+1} + N_{it+1})$. The average R^2 for the value-to-capital ratio is 75%, though it shows a slight decline in recent years, as shown in Figure 4(b).

Direct evaluation of model fit at the firm level remains relatively uncommon in the literature. A closely related and important benchmark is Belo et al. (2022), who obtain a portfolio-level R^2 of 94% in matching model-implied and observed valuations in the cross section. Building on this line of work, I assess fit at the individual firm level and obtain an R^2 of around 86%. This comparison is inherently demanding, as firm-level analysis preserves idiosyncratic variation that is naturally attenuated through portfolio aggregation.

4.6 Validation

4.6.1 Example Firms

Figure 5 decomposes the average market value of representative firms into four model-implied components, computed at the firm-year level and averaged over time.

Technology and pharmaceutical firms derive substantial value from intangible capital, but through different channels. Amazon and Meta benefit from both production and market power, whereas Johnson & Johnson derives more of its intangible value through market power, consistent with the importance of patent protection and pricing power in pharmaceuticals. By contrast, traditional firms such as Costco, 3M, and ExxonMobil derive more value from tangible capital.

The decomposition also reveals differences in non-intangible market power. Meta, for example, has a sizable positive contribution, potentially reflecting network effects, switching costs, or other competitive advantages not captured by its current intangible stock. Amazon and ExxonMobil instead face stronger competitive pressure and exhibit negative contributions from this component, indicating that their markups are below the levels implied by

their intangible assets alone.

Finally, Amazon has substantial residual values. Part of its market valuation is not explained by physical capital, intangible capital, or market power. This residual may reflect investor sentiment or other forces outside the model scope.

4.6.2 Interpreting the Two Sources of Market Power

The two market power components capture distinct economic forces. They should respond to different types of shocks and vary with industry dynamics in different ways. In this section, I provide external validation for the decomposition by examining how the two market power components respond to innovation and competition shocks.

Patents and intangible market power. I first examine changes in the two market power value shares around the filing and issuance of highly cited patents, using the same event definition and citation threshold as in Section 2.1.3. I use citations rather than the economic value measure of Kogan et al. (2017) to avoid the mechanical effects as that measure is itself constructed from stock market reactions, while the outcomes here are also value-based.

Figure 6 shows that the market power share rises around both filing and issuance, consistent with the markup response in Section 2.1.3. More importantly, the increase is concentrated in intangible-driven market power, while the non-intangible market power share remains broadly unchanged. This differential response supports the decomposition's attribution of patent-related market power to intangible assets.

Tariff and non-intangible market power. Next, I exploit the granting of Permanent Normal Trade Relations (PNTR) status to China as a plausibly exogenous increase in product market competition. Before PNTR, China's access to low Normal Trade Relations tariffs required annual renewal, creating uncertainty because nonrenewal would trigger much higher non-NTR tariffs. PNTR permanently removed this uncertainty in 2001 and accelerated Chinese import competition. Following Pierce and Schott (2016), I measure industry exposure

using the NTR gap, defined as the difference between non-NTR and NTR tariff rates. I then estimate a difference-in-differences specification that interacts the industry-level NTR gap with a post-PNTR indicator.

Table 3 reports the responses of market power value shares to the PNTR shock. Panel A presents static difference-in-differences estimates, while Panel B traces the dynamic effects. Consistent with Corsini (2025), greater import competition does not significantly reduce the level of market power. Instead, it slows the growth of the market power share, with the effect driven entirely by non-intangible market power. Intangible-driven market power remains unchanged, consistent with this component capturing innovation-based rents rather than changes in the competitive environment.

Taken together, the patent and PNTR results provide external validation for the decomposition. Intangible-driven market power responds to innovation, whereas non-intangible market power is more sensitive to shifts in the external competitive environment.[18]

5 An Anatomy of Firm Value

This section presents the main quantitative results on firm value decomposition. Section 5.1 shows how the value of intangibles varies in the time series. Section 5.2 documents the composition shifts over firms' life cycles. Section 5.3 focuses on the cross-sectional variation.

5.1 Time Variation

Figure 7(a) shows the aggregate decomposition, with value components summed across all firms. Figure 7(b) shows each component as a percentage of the total market valuation explained by the model. Panel A of Table 4 formalizes the results and statistically tests the difference between the early and late periods.

First of all, market power is the dominant source of intangible value, accounting for

[18]Internet Appendix B.3 provides additional validation relating firm characteristics to the value shares and product market characteristics to the market power components.

roughly two-thirds of the total on average over the sample. However, the two channels exhibit distinct time trends. The production value of intangibles grows steadily, increasing from 10.4% of aggregate market value before 2000 to 17.9% after 2000, whereas the contribution of intangible-driven market power remains relatively stable. As a result, the combined value of intangibles increases over time, reaching approximately half of aggregate market value by 2025. These patterns are consistent with technological advances increasing the productive value of intangibles, whereas their role in generating market power has long been an important source of firm value. Moreover, the market power component tends to be more valuable during economic downturns, consistent with higher margins providing some protection against adverse shocks.

At the same time, the value attributed to non-intangible market power also grows. During the first half of the sample period, firms operate in a relatively competitive environment, with negative value from non-intangible sources. After 2000, however, firms increasingly earn rents in excess of intangible-driven markups, as reflected by the increasing non-intangible market power component. This rise in non-intangible market power provides suggestive evidence of a decline in competition, consistent with findings in the IO and macroeconomic literature (De Loecker, Eeckhout and Unger, 2020).

Finally, the residual component is generally small and becomes more pronounced during market booms and busts. In the results that follow, I focus on the value components explained by the model and abstract from the residual term, which is not directly linked to the model's state variables and may reflect forces beyond the scope of this analysis.

Variation across time can be attributed to within-firm change or compositional change. For example, the value of market power can rise through either within-firm increases (all firms accumulate more market power) or compositional shifts (high-markup firms become larger). Following Paron (2025), I decompose the changes in the value components into these two channels. I use sales, $P_{it}Y_{it}$, to represent the firm revenue distribution. Define firm i's sales share as $w_{it} \equiv P_{it}Y_{it}/(P_tY_t)$ and its value-to-sales ratio as $v_{it} \equiv V_{it}/(P_{it}Y_{it})$. The

aggregate value-to-sales ratio can then be written as a sales-weighted average of firm-level value-to-sales ratios:

$$\frac{V_t}{P_t Y_t} = \sum_i w_{it} v_{it}. \tag{25}$$

The change in the value-to-sales ratio from $t-1$ to t can be decomposed into within-firm and compositional change:[19]

$$\Delta\left(\frac{V_t}{P_t Y_t}\right) = \underbrace{\sum_i \bar{w}_{it} \Delta v_{it}}_{\text{Within-firm change}} + \underbrace{\sum_i \Delta w_{it} \bar{v}_{it}}_{\text{Compositional change}} . \tag{26}$$

For either $a = w$ or $a = v$, the overbar denotes the midpoint of the variable across the two periods:

$$\bar{a}_{it} \equiv \frac{1}{2}\left(a_{it} + a_{i,t-1}\right). \tag{27}$$

Intuitively, the within-firm change focuses on changes in firm-level valuations while holding sales weight fixed. The compositional change measures the effect of changing sales shares while holding valuations fixed for each firm. The total value V in Equation (26) can be replaced with each value component to understand their drivers.

Figure 8 plots the cumulative change in the aggregate value-to-sales ratio and decomposes it into within-firm and compositional components. It also shows the same decomposition for the ratio of the value components to sales. Consistent with Paron (2025), Panel (a) shows that the rise in aggregate valuation is driven mainly by compositional change, indicating a reallocation of revenue toward higher-valuation firms. Panel (b) shows that compositional shifts dominate within-firm changes for the rise in production value of intangible capital, suggesting that firms that use intangible capital more effectively in production have captured a larger market share.

[19] I account for entry and exit dynamics following Paron (2025). See their Appendix B for details.

Panels (c) and (d) isolate intangible-driven and non-intangible market power, respectively. The rise in non-intangible market power comes entirely from compositional change, while its within-firm component declines. Thus, the average firm faces stronger competition, but revenue increasingly shifts toward firms facing weaker competitive environments. By contrast, the rise in intangible-driven market power is primarily a within-firm phenomenon, as firms accumulate intangibles that strengthen their own market power. Together, these patterns distinguish firm-level accumulation of intangible market power from the reallocation of market share toward firms earning higher pure rents.

5.2 Firm Life Cycle

The relative importance of productivity and market power may change as firms mature. Existing studies document systematic life cycle patterns along both margins. Young firms tend to have higher productivity (Foster, Haltiwanger and Syverson, 2008) and use low prices to attract customers (Gourio and Rudanko, 2014), while mature firms charge higher markups (Adam, Renkin and Züllig, 2025) and differ from young firms primarily on the demand side (Foster, Haltiwanger and Syverson, 2016). Because intangible capital contributes to both productivity and market power, its economic role may likewise evolve over the firm life cycle. In this section, I trace how the two components of intangible value change as firms mature.

I run the following regression and require firms included in the regression to be observed for at least 15 years to ensure a stable composition of firms in the sample:

$$\frac{V_{it}^{\text{comp}}}{V_{it}^{\text{model}}} = \sum_{j=1}^{26} \beta_j \cdot \mathbb{1}\{\text{age}_{it} = j\} + FE_t + FE_c + \varepsilon_{it}. \tag{28}$$

I define firm age as the number of years since the firm first appeared in CRSP, similar to Pástor and Pietro (2003) and Fama and French (2004). Following the literature on estimating life cycle effects (e.g., Fienberg and Mason, 1985), I include the year and cohort fixed effects in the regression. I group all observations at age 26 or older into a single age category. This

restriction breaks the exact linear dependence among firm age, calendar year, and cohort, allowing the age indicators to be identified jointly with year and cohort fixed effects. I also report a specification that replaces the age dummies with a natural cubic spline in Internet Appendix B.5. The results are qualitatively similar.

Figure 9 traces how the value components evolve over firm age. Each panel plots the average component share against firm age, with the shaded area indicating 95% confidence intervals. Panel (a) shows that the physical capital component remains relatively stable over the firm life cycle, with a slight decline over the first five years.

Panel (b) shows an overall decline in the production value share of intangible capital of around five percentage points. One interpretation is that newly public firms enter the sample with relatively fresh and productive intangible stocks, and these productivity advantages erode over time as competitors imitate or as the intangibles themselves become obsolete. Alternatively, young firms might use lower prices to attract customers and build their customer base. In contrast, Panel (c) documents that the value share of intangible-driven market power rises by roughly nine percentage points. This trajectory is consistent with firms progressively relying on their accumulated intangible capital to build market power, and the resulting rents become an increasingly important component of firm value.

Panel (d) shows that the non-intangible market power component declines over the firm life cycle. The decline indicates that older firms tend to operate in increasingly competitive environments. This may reflect the entry of new competitors over time, or the fact that firms tend to enter the market during favorable periods with less competition. This pattern is directly in contrast with the intangible-driven market power component, which rises over the life cycle, suggesting that firms progressively rely more on intangible assets to preserve their market power.

Taken together, the four panels provide a coherent picture of firm value dynamics over the life cycle. Young firms benefit from fresh productive intangibles and operate in less competitive environments. As firms age, a compositional shift occurs within the intangible

channel: the direct productivity value of intangibles decays, while firms increasingly convert their intangible stocks into durable market power.

5.3 Cross-sectional Heterogeneity

There is substantial heterogeneity in the composition of firm value across firms and across industries. I first perform a covariance decomposition to understand the contribution of value components to the cross-sectional variation. I scale both the actual and model-implied firm value by total capital, $K_{it} + N_{it}$, so that the decomposition is not dominated by large firms. Let $v_{it} \equiv V_{it}/(K_{it} + N_{it})$ denote the scaled value (and analogously for each component). Equation (18) then implies:

$$\begin{aligned} Cov(v_{it}, v_{it}^{\text{model}}) &= Cov(v_{it}, v_{it}^{\text{prod-phy}} + v_{it}^{\text{prod-int}} + v_{it}^{\text{mkp-int}} + v_{it}^{\text{mkp-non}}) \\ \Rightarrow 1 &= \frac{Cov(v_{it}, v_{it}^{\text{prod-phy}})}{Cov(v_{it}, v_{it}^{\text{model}})} + \frac{Cov(v_{it}, v_{it}^{\text{prod-int}})}{Cov(v_{it}, v_{it}^{\text{model}})} + \frac{Cov(v_{it}, v_{it}^{\text{mkp-int}})}{Cov(v_{it}, v_{it}^{\text{model}})} + \frac{Cov(v_{it}, v_{it}^{\text{mkp-non}})}{Cov(v_{it}, v_{it}^{\text{model}})}. \end{aligned} \tag{29}$$

This equation decomposes the cross-sectional variation of the scaled firm value, v_{it}, that can be explained by the model, v_{it}^{model}. Figure 10 shows the results of the covariance decomposition for each cross section and Panel B of Table 4 reports the estimates. Consistent with the level decomposition, intangible capital plays a larger role in driving firm-level variation while physical capital becomes less important. Moreover, though the level of intangible-driven market power value is higher than the value of intangibles in production, the contribution of the latter to cross-sectional variation is much larger. Firms differ more in their production intangibles and their efficiency in utilizing intangibles in production. On the other hand, the total contribution of market power has increased substantially across time, accounting for roughly 30% of the value variation in the past two decades.

There is also substantial heterogeneity in value composition across sectors. Figure 11(a) shows the relative importance of intangible capital in production and market power across Fama-French 48 sectors. Each point represents a sector, with its size proportional to the

sector's market capitalization. The dashed line shows the linear fit between production value and intangible-driven market power value. Sectors such as Pharmaceuticals and Computers lie above the fitted line, indicating relatively greater intangible value from market power. In contrast, sectors such as Machinery and Transportation lie below the fitted line, suggesting that their intangible value is more concentrated in production.

The level and composition of market power also vary substantially across sectors. Figure 11(b) compares intangible-driven and non-intangible market power. The fitted line is nearly flat, indicating little cross-sector relation between the two sources of market power. Sectors such as Pharmaceuticals and Business Services have substantial non-intangible market power, suggesting that competitive forces unrelated to intangible capital also support higher markups in these sectors. In contrast, sectors such as Retail and Machinery derive most of their market power value from intangibles, with relatively little contribution from non-intangible factors. This pattern is consistent with stronger external competitive pressures limiting the scope for non-intangible market power in these sectors.

6 Applications

Distinguishing the production and market power roles of intangibles helps explain asset prices and corporate decisions. In this section, I focus on two applications using the value decomposition in the previous section. Section 6.1 creates model-implied value signals and improves the performance of the value premium in recent decades. Section 6.2 constructs average q using each value component and explores the investment-q relation.

6.1 Explaining the Value Premium Decline

A value signal is designed to capture variation in expected returns by comparing a firm's fundamental value with its market value. Firms with lower market value relative to fundamentals are expected to earn higher subsequent returns, and vice versa. However, the value

premium based on the traditional book-to-market ratio has declined over time, approaching zero after 2000 (Fama and French, 2021). Several papers have argued that book equity no longer captures the fundamental value accurately, so the book-to-market ratio becomes a noisy return signal. For example, Eisfeldt, Kim and Papanikolaou (2022) and Park (2022) propose to include intangible capital in the book equity. Gonçalves and Leonard (2023) instead replace the book equity with fundamental value by discounting future cash flows at a constant rate. However, the book-to-market ratio adjusted for intangibles still shows a visible decline in recent years. Although the fundamental-to-market ratio of Gonçalves and Leonard (2023) restores the value premium, the authors do not explain the economic forces driving the wedge between book equity and fundamental equity.

I propose an alternative explanation. The book-to-market ratio has become less informative because market value increasingly reflects the value of market power, which is not captured by book equity. To fix ideas, consider a simple Gordon growth model with a constant growth rate and a firm-specific discount rate. Suppose the book equity reflects the value generated by production:

$$BE_i = \frac{CF_i^{\text{prod}}}{\bar{r} - g}, \tag{30}$$

while market equity contains both value from production and market power:

$$ME_i = \frac{CF_i^{\text{prod}} + CF_i^{\text{mkp}}}{r_i - g}. \tag{31}$$

Then the resulting book-to-market ratio is confounded by the share of cash flows attributable to production:

$$BM_i = \frac{BE_i}{ME_i} = \frac{r_i - g}{\bar{r} - g} \cdot \underbrace{\frac{CF_i^{\text{prod}}}{CF_i^{\text{prod}} + CF_i^{\text{mkp}}}}_{\text{Noise term}}. \tag{32}$$

6.1.1 Model-implied Proxies

To test this hypothesis, I construct two types of proxies. The first consists of model-implied valuation ratios that remove the value of market power from market value. I consider two variations: one makes a partial adjustment for market power and removes only the value of non-intangible markup from market value, while the other removes the total value of market power, including both intangible and non-intangible markups:

$$BM_i^{\text{model}} \text{ (ex. non-intangible markup)} = \frac{BE_i^{\text{total}}}{V_i^{\text{model}} - V_i^{\text{mkp-non}}}, \tag{33}$$

$$BM_i^{\text{model}} \text{ (ex. all markup)} = \frac{BE_i^{\text{total}}}{V_i^{\text{model}} - V_i^{\text{mkp-non}} - V_i^{\text{mkp-int}}}, \tag{34}$$

where BE^{total} contains both the book equity from Fama and French (1992) and intangible capital, following the adjustment in Eisfeldt, Kim and Papanikolaou (2022).

These model-implied measures directly test whether heterogeneity in market power explains the decline in the value premium. However, they may be noisy because they rely on a model that abstracts from leverage and other potentially relevant factors. Therefore, I also construct a fitted fundamental-to-market ratio using the following regression:

$$\begin{aligned} \log FM_{i,t} \; = \alpha_t + \beta_t \log\left(\frac{BE_{i,t}^{\text{total}}}{ME_{i,t}}\right) + \gamma_{1,t} \frac{V_{i,t}^{\text{prod-int}}}{V_{i,t}^{\text{model}}} \\ + \gamma_{2,t} \frac{V_{i,t}^{\text{mkp-int}}}{V_{i,t}^{\text{model}}} + \gamma_{3,t} \frac{V_{i,t}^{\text{mkp-non}}}{V_{i,t}^{\text{model}}} + \varepsilon_{i,t}, \end{aligned} \tag{35}$$

where FM_{it} is the fundamental-to-market ratio proposed by Gonçalves and Leonard (2023). The three additional regressors are the value shares of the corresponding components of the decomposition. I omit the value share of physical capital because it equals one minus the sum of the other three shares. I estimate the regression separately for each year t and use the fitted values, $\widehat{FM}_{i,t}$, to form portfolios for year $t+1$. Thus, portfolio sorts in the next section use only information available at the time of portfolio formation.

The fitted ratio flexibly isolates the information in the fundamental-to-market ratio that can be explained by the model-implied value components. If it restores the value premium similarly to the original FM ratio, this suggests that the model captures the return-relevant information in FM, especially that associated with market power.

The regression coefficients in equation (35) themselves can also shed light on the economic forces driving the wedge between FM and BM. Under the Gordon growth example in Equations (30) and (31), FE can be written as:

$$FE_i = \frac{CF_i^{\text{prod}} + CF_i^{\text{mkp}}}{\bar{r} - g}. \tag{36}$$

The variation in the wedge should therefore be driven primarily by the market power components. Table 5 reports estimates from a panel regression with year fixed effects.

The wedge between fundamental-to-market and book-to-market loads positively on both market power components and negatively on intangible production, consistent with book equity missing market power value while overstating production value. The market power loadings rise in the second half of the sample, especially for non-intangible market power. Accordingly, the R^2 of book-to-market alone falls sharply, whereas adding the model-implied value shares keeps explanatory power stable across subsamples.

6.1.2 The Value Premium Decline

I first look at the value premium in the full sample using different value signals. Panel A of Table 6 reports the return spreads from decile portfolio sorts from July 1977 to June 2026. The two book-to-market ratios constructed directly from the data generate monthly return spreads of 0.19% and 0.33%, neither of which is statistically significant. The model-implied ratio that removes only non-intangible market power also produces a small and insignificant spread of 0.18% ($t = 1.01$). In contrast, removing the value of both intangible and non-intangible market power yields a significant value premium of 0.42% per month

($t = 2.60$). The fitted fundamental-to-market ratio produces the largest spread, at 0.51% per month ($t = 3.20$), while the original fundamental-to-market ratio generates a spread of 0.45% ($t = 2.30$). Its stronger performance suggests that the model-implied value shares capture the information in the original fundamental-to-market ratio that is most relevant for predicting returns.

Next, I examine the time series of the return spread to assess whether it has declined over time. Figures 12a and 12b plot its ten-year and five-year trailing averages, respectively. The figures compare the intangible-adjusted book-to-market ratio with FM, the model-implied ratio that removes the value of all market power, and fitted FM. The data benchmark exhibits the clearest decline: its trailing average falls sharply after the mid-2000s and becomes negative in recent years. The value premium based on FM also weakens. By contrast, the model-implied ratio remains much more stable with a small decline after 2020. Most strikingly, fitted FM exhibits no persistent downward trend: both its ten-year and five-year trailing averages recover strongly after temporary declines in the early 2020s. Overall, the figure suggests that accounting for market power restores much of the time-series stability of the value premium.

To formally test whether the return spread has declined, I run a simple time trend regression on the monthly return series.[20] For each monthly return spread r_t I estimate

$$r_t \;=\; a + b \cdot \frac{t}{T} + \varepsilon_t, \qquad t = 1, \ldots, T, \tag{37}$$

where b is the total drift across the sample in percentage points per month. Panel B of Table 6 reports the results. Among these measures, fitted FM and the model-implied ratio removing all market power show no significant decline, with $b = -0.60$ ($t = -0.89$) and $b = -0.69$ ($t = -1.16$), respectively. This finding suggests that removing the value of market power

[20] I do not assess the decline in the value premium using subsample comparisons because the results are sensitive to the choice of the sample split date. Internet Appendix B.6 illustrates this sensitivity by reporting value premium estimates for three alternative split dates. Nevertheless, the fitted fundamental-to-market ratio delivers the largest value premium in the second subsample across all specifications.

attenuates the downward trend in the value premium, while the flexible adjustment embodied in fitted FM attenuates it further. Together with the decomposition results, this pattern is consistent with the growing importance of market power contributing to the decline of conventional value signals.

6.2 The Investment-q Relation

The neoclassical theory of investment predicts that Tobin's q, the ratio of capital's market value to its replacement cost, should fully capture investment opportunities (Hayashi, 1982). As firms increasingly rely on intangible assets, Peters and Taylor (2017) further incorporate intangible capital into the replacement cost and show that this total q measure better predicts investment. However, a large total q may reflect the market power rents, which raise the firm value without increasing the marginal return to capital (Gutiérrez and Philippon, 2017; Crouzet and Eberly, 2019). More importantly, firms may have very different investment incentives even when they share the same total q, because that market value may arise from fundamentally different economic sources. My decomposition framework provides market-power-adjusted and component-specific q measures and thus helps explain the different incentives for investments.

6.2.1 Summary statistics

Panel A of Table 7 reports summary statistics for the empirical and model-implied q measures. Standard q, defined as the market value of the firm relative to physical capital, has a mean of 5.17 and exhibits substantial dispersion and positive skewness. Incorporating intangible capital into the capital stock substantially lowers the level of total q, with the mean falling to 1.46, while maintaining a similar distribution.[21] The next three measures are

[21]Note that the definition of total q differs slightly from that in Peters and Taylor (2017), where they subtract current assets (Compustat item ACT) from firm value. Here I subtract cash (CHE) to be consistent with my definition of firm value throughout the paper. Internet Appendix B.7 reports the regressions in the next two subsections for the exact Peters-Taylor q. The results are similar to those obtained using this paper's total q measure.

model-implied counterparts to the data total q, where I purge out the value of residuals, non-intangible market power, or the total value of market power. They have similar distributions but different levels. The component q measures show that physical and intangible capital contribute comparably to firm value on average, with mean physical and intangible q of 1.62 and 1.40, respectively. Within intangible capital, the market power component (0.87) exceeds the production component (0.53). Overall, the level and distribution of model-implied q measures are reasonable and comparable to the data benchmarks.

Panel B of Table 7 reports the correlation matrix among these q measures. Model total q's covary with the data counterpart with a correlation of around 0.8. At the same time, the decomposed q measures are far from perfectly correlated. In particular, the production and market power components of intangible q have a correlation of only 0.25, and physical q is only weakly correlated with intangible market power q (0.08). These moderate correlations suggest that the decomposition isolates economically distinct sources of firm value rather than mechanically partitioning total q.

6.2.2 Horse race

Table 8 reports the horse race of model-implied total q's against the data benchmarks. Panel A regresses physical investment on q measures and Panel B uses intangible investment as the left-hand variable.[22] The first column reports the investment-q relation using standard Tobin's q. It explains some variation in both investment rates with very small coefficients. Under the neoclassical framework, the inverse q slope measures the adjustment costs. Therefore, the small coefficient implies an unreasonably large adjustment cost. Data total q in the second column predicts investment with a much larger R^2 and a more reasonable slope.

[22]Here I scale each investment rate by its own capital rather than by total capital. This is the main specification for two reasons. First, the model specifies separate adjustment costs for physical and intangible capital, so each first-order condition is expressed in terms of its corresponding investment rate. Second, scaling by total capital mechanically mixes investment intensity with capital composition. A firm with a larger intangible capital share will mechanically have lower physical investment relative to total capital, even if its physical investment intensity is unchanged. Because intangible value covaries with the intangible share, this denominator effect would bias its coefficient downward in the physical investment-q regression. Nevertheless, Internet Appendix B.7 reports the results for $(K+N)$-scaled investment rates.

If market power value and residual valuation components do not generate the same investment incentives as productive capital, removing them from total q should increase the estimated investment-q slope. Columns (3)–(5) of Table 8 examine this prediction. The adjusted model-implied total q measures deliver within-R^2 values comparable to those of the empirical benchmarks, but their coefficients are roughly twice as large. Under the standard quadratic adjustment-cost interpretation, these larger coefficients imply more moderate adjustment costs. The coefficients are smaller for intangible investment than for physical investment, consistent with greater adjustment frictions for intangible capital.[23]

6.2.3 Component q

Panel A of Table 9 reports investment-q regressions using the individual value components. Columns (1)–(3) examine physical investment. Physical q alone explains around 8% of the within-firm variation in physical investment. When all four components are included in column (2), physical q remains the dominant predictor, while the intangible production and market power components have substantially smaller coefficients. Column (3) adds total q to the component regression. The coefficient on non-intangible market power q becomes significantly negative, suggesting that total q overstates investment incentives when it assigns the same role to valuation arising from productive capital and from non-intangible market power. By contrast, physical q and intangible market power q remain positively related to physical investment, indicating that total q may underweight these sources of value.

Columns (4)–(6) report the corresponding results for intangible investment. Firms may invest in intangible capital either to improve production efficiency or to build market power. Consistent with this interpretation, both intangible-production q and intangible market power q strongly predict future intangible investment in columns (4) and (5). However, the market power component carries a larger coefficient. Moreover, in column (6), the coeffi-

[23] Table 1 reports a lower adjustment cost parameter estimate for intangible capital. However, I do not allow disinvestment in intangible capital, so the overall adjustment friction is higher for intangible capital, consistent with the smaller investment-q slope.

cient on intangible market power q remains strongly positive even after controlling for total q. Non-intangible market power q, in contrast, is economically small.

To quantify the explanatory power of the components, Panel B of Table 9 decomposes the within-firm R^2 of regressions in columns (2) and (5) in Panel A using Shapley values. Physical q explains over 70% of the variation in physical investment explained by the four-component regression. On the other hand, the explanatory power for intangible investment is concentrated in the two intangible components. Market power q accounts for 45% of the within-firm R^2, while production q contributes another 33%. Non-intangible market power q contributes little to the explanatory power of either regression.

Taken together, the results show that aggregate q obscures economically distinct investment incentives. Production components predict investment in the corresponding capital stock, while intangible-driven market power value provides an additional and even stronger incentive for intangible investment. Non-intangible market power value, by contrast, raises firm valuation without generating comparable investment incentives.

7 Conclusion

Firms actively acquire market power through intangible investments, such as marketing and product differentiation. This paper extends the neoclassical framework to explicitly incorporate the endogenous link between intangibles and market power. In the model, firms invest in intangible capital to both raise productivity and build market power. I estimate the model and provide a quantitative framework to decompose the intangible value into production and market power components. I show that, on average, two-thirds of intangibles' value comes from market power. The market power channel alone accounts for nearly 30% of aggregate market capitalization.

Over time, the value share of intangible-driven market power remains remarkably stable, whereas production value rises. This increase is driven largely by compositional change, as

firms with higher production value capture greater market share. Over the firm life cycle, young firms enter with productive intangible stocks and favorable competitive conditions. As firms mature, however, the direct productivity value of intangibles declines, while firms increasingly convert their intangible capital into durable market power.

Incorporating the intangible-driven market power is important for understanding asset prices and incentives for intangible investments. Specifically, the value of market power introduces noise to the book-to-market ratio, so the ratio no longer captures expected return signals. Removing the value of market power explains part of the value premium decline. Moreover, I construct a separate q measure for each economic component and show that firms' intangible investment reacts strongly to market power incentives. In contrast, non-intangible market power value raises firm valuation but does not predict investments.

This framework offers several avenues for future research. First, richer data could be used to examine how different types of intangibles contribute to market power through distinct mechanisms. Second, although this paper focuses on the composition of firm value and the dual roles of intangibles, the framework can be adapted to study other questions in finance. For example, differences in the risk exposures of value components may help explain cross-sectional variation in expected returns. Finally, extending the analysis beyond U.S. firms could reveal whether market power has also increased in other countries and whether its rise in the United States has come at the expense of foreign firms.

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Table 1: Parameter estimates

This table reports the model parameters. Panel A reports the calibrated parameters. Panel B lists the estimated parameters and their standard errors. The model is solved and simulated at annual frequency.

Variable	Notation	Value	SE
Panel A: Calibrated parameters			
Discount rate	r_f	0.04	
Scaling of profits	$\bar{Y}$	20	
Depreciation of physical capital	δ^K	0.12	
Depreciation of intangible capital	δ^N	0.2	
Panel B: Estimated parameters			
Group 1: Production technology			
Share of physical capital	α	0.8334	0.0579
Total share of capital	η	0.2570	0.0105
CES curvature	ρ	0.4974	0.1302
Fixed operating cost	f	0.3492	0.0403
Group 2: Investment and capital-specific productivity			
Adjustment cost of physical capital	b^K	2.4331	0.2274
Adjustment cost of intangible capital	b^N	1.8946	0.2982
Autocorrelation of z_t^K	ρ^K	0.8641	0.0297
Volatility of ζ_t^K shocks	σ_ζ^K	0.1804	0.0375
Autocorrelation of z_t^N	ρ^N	0.9161	0.0184
Volatility of ζ_t^N shocks	σ_ζ^N	0.3843	0.0897
Group 3: Markup-related			
Average effect of intangible capital on markup	$\bar{\mu}$	0.1280	0.0067
Curvature of the markup function	ϕ	0.8175	0.0190
Autocorrelation of ζ_t^μ	ρ^μ	0.7699	0.0115
Volatility of ζ_t^μ shocks	σ^μ	0.1547	0.0035
Group 4: SDF			
Volatility of ε_x^K	σ_x^K	0.0636	0.0131
Volatility of ε_x^N	σ_x^N	0.1398	0.0349
Risk price of ε_x^K	γ^K	1.1267	3.0054
Risk price of ε_x^N	γ^N	31.4845	7.9671

Table 2: Moments

This table reports moments used in the SMM. The model is solved and simulated at an annual frequency. I simulate the model for 100 independent samples, with each sample including 3500 firms over 200 years (the first 150 years form a burn-in period). I report the data moments and the means of the corresponding simulated moments. The SE column reports the sampling standard error of each empirical moment, calculated from the influence-function covariance matrix. Standard errors of firm-level moments are clustered by firm. Definitions of the variables are in Table IA.1.

Moment	Data	Simulated	SE
Panel A: Firm-level moments			
Average value-to-total-capital ratio	1.4022	1.4649	0.0133
Std. dev. of physical investment	0.1220	0.1228	0.0009
Std. dev. of intangible investment	0.1265	0.1239	0.0010
Average intangible share	0.4928	0.5019	0.0032
Std. dev. of intangible share	0.0848	0.0817	0.0011
Average profits-to-total-capital ratio	0.4212	0.4298	0.0027
AR(1) of profits-to-total-capital ratio	0.6807	0.6502	0.0046
Std. dev. of profits-to-physical-capital ratio	0.7314	0.7803	0.0111
AR(1) of profits-to-physical-capital ratio	0.6851	0.7004	0.0078
Std. dev. of profits-to-intangible-capital ratio	0.8860	0.8043	0.0147
AR(1) of profits-to-intangible-capital ratio	0.6430	0.6265	0.0091
Average SALE/COGS	1.7682	1.8158	0.0098
Std. dev. of SALE/COGS	0.4028	0.4185	0.0085
AR(1) of SALE/COGS	0.6425	0.7710	0.0131
Cross-sectional correlation between SALE/COGS and intangible share	0.2408	0.3388	0.0089
Panel B: Aggregate moments			
Average market excess return	0.0917	0.0797	0.0237
Average HML sorted by profitability	0.0397	0.0641	0.0275
Average HML sorted by investment	-0.0237	-0.0071	0.0260
Average HML sorted by markup	0.0277	0.0271	0.0193
Volatility of aggregate profits-to-total-capital	0.0353	0.0327	0.0030
AR(1) of aggregate profits-to-total-capital	0.8652	0.7564	0.0800
Volatility of aggregate physical investment	0.0230	0.0214	0.0025
Volatility of aggregate intangible investment	0.0149	0.0134	0.0013

Table 3: PNTR exposure and market-power value shares (1991–2007)

This table reports difference-in-differences estimates of the effect of PNTR-driven import competition on the two market power value shares. Section 4.6.2 describes the setting and the NTR gap measure of industry exposure. The Level columns use the value share itself, and the Growth columns use its annual change. Panel A interacts the NTR gap with a post-2000 indicator. Panel B uses separate interactions for the pre-period, the short run, and the medium run, omitting 1997–2000 as the reference period. All specifications include firm and year fixed effects. Standard errors, in parentheses, are clustered by NTR-gap industry family, the level at which Pierce and Schott (2016) define the tariff gap.

	Intangible market-power share		Non-intangible market-power share	
	Level	Growth	Level	Growth
Panel A: Static difference-in-differences				
Gap × Post	0.0276	0.0078	0.0049	-0.0225***
	(0.0198)	(0.0064)	(0.0468)	(0.0065)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	52,504	44,267	52,504	44,267
R^2	0.7436	0.2018	0.7895	0.1431
Within R^2	0.00054	0.00007	0.00001	0.00027
Panel B: Dynamic difference-in-differences				
Gap × Pre (1991–1996)	0.0046	-0.0089	-0.0081	-0.0105
	(0.0155)	(0.0080)	(0.0193)	(0.0085)
Gap × Short run (2001–2003)	0.0249	0.0044	0.0097	-0.0260**
	(0.0166)	(0.0092)	(0.0403)	(0.0105)
Gap × Medium run (2004–2007)	0.0343	0.0043	-0.0069	-0.0271***
	(0.0245)	(0.0078)	(0.0485)	(0.0094)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	52,504	44,267	52,504	44,267
R^2	0.7436	0.2019	0.7895	0.1432
Within R^2	0.00059	0.00014	0.00006	0.00031

Table 4: Aggregate decomposition

This table reports the level and covariance decomposition of V into terms attributed to physical capital, intangibles in production, intangible-driven market power, and non-intangible market power. Panel A presents the percentage contribution of each term to total market value. Each term is calculated as $\sum_i V_{it}^{\text{comp}} / \sum_i V_{it}^{\text{model}}$ and then averaged over time. Panel B reports the covariance decomposition following Equation (29) (average across years). The Late − Early row reports the difference between the late and early subperiod averages; standard errors are in parentheses.

		% $V^{\text{prod-phy}}$	% $V^{\text{prod-int}}$	% $V^{\text{mkp-int}}$	% $V^{\text{mkp-non}}$
Panel A: Level decomposition					
Full sample	(1976–2025)	55.74	14.14	29.60	0.52
Early	(1976–2000)	64.93	10.42	28.73	-4.08
Late	(2001–2025)	46.55	17.87	30.47	5.12
Late − Early		-18.38	7.45	1.74	9.20
		(1.13)	(0.83)	(0.68)	(0.83)
Panel B: Covariance decomposition					
Full sample	(1976–2025)	48.16	26.34	13.68	11.81
Early	(1976–2000)	57.32	21.86	11.52	9.30
Late	(2001–2025)	39.01	30.82	15.85	14.32
Late − Early		-18.31	8.96	4.33	5.02
		(1.45)	(1.17)	(0.50)	(0.87)

Table 5: The fitted value signal, book-to-market, and market-power value shares

This table regresses the log fundamental-to-market ratio (FE/ME) on the log book-to-market ratio and the model-implied value shares of intangible production, intangible market power, and non-intangible market power. All specifications include year fixed effects. Standard errors, two-way clustered by firm and year, are in parentheses.

	Full sample		1976–1999		2000–2025	
	(1)	(2)	(3)	(4)	(5)	(6)
$\log(BE^{\text{total}}/ME)$	0.0964***	0.0641**	0.2269***	0.1763***	-0.0346	-0.0487
	(0.0277)	(0.0280)	(0.0192)	(0.0365)	(0.0361)	(0.0307)
Intangible (production)		-0.5277***		-0.6637***		-0.4334***
		(0.0436)		(0.0506)		(0.0566)
Intangible (markup)		0.9884***		0.8854***		1.0400***
		(0.1092)		(0.1625)		(0.1355)
Non-intangible markup		0.8564***		0.6037***		1.0342***
		(0.0577)		(0.0827)		(0.0473)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.0696	0.1895	0.1195	0.2030	0.0268	0.1861
Within R^2	0.0086	0.1364	0.0573	0.1467	0.0010	0.1645
Observations	102,518	102,518	54,155	54,155	48,363	48,363

Table 6: Return spreads and trend regressions

Panel A reports the monthly value-weighted return spreads between value and growth stocks from annual NYSE-breakpoint decile sorts. The sample period is from July 1977 to June 2026 (588 months). Panel B reports estimates of $r_t = a + b(t/T) + e_t$ for the corresponding monthly return spreads. The intercept a is the fitted premium at the start of the sample, b is the total drift across the sample, both in % per month, and b/σ expresses the drift in standard deviations of the series. Newey–West t-statistics using automatic lag selection are in parentheses.

	Panel A: Return spread	Panel B: Trend regressions		
Sorting variable	% monthly	a	b	b/σ
Data ratios				
BM^{data} (original)	0.19	0.83	-1.27	-0.2290
	(0.72)	(1.82)	(-1.32)	
BM^{data} (incl. intangibles)	0.33	1.19	-1.72	-0.3311
	(1.27)	(2.60)	(-2.05)	
Model-implied ratios				
BM^{model} (ex. non-intangible markup)	0.18	1.01	-1.65	-0.4192
	(1.01)	(2.91)	(-2.98)	
BM^{model} (ex. all markup)	0.42	0.76	-0.69	-0.1858
	(2.60)	(2.30)	(-1.16)	
Fundamental-to-market				
Fitted FM	0.51	0.81	-0.60	-0.1532
	(3.20)	(2.48)	(-0.89)	
FM	0.45	1.06	-1.23	-0.2667
	(2.30)	(3.02)	(-1.88)	

Table 7: Properties of the q measures

This table reports the properties of the q measures used in the investment regressions: Panel A gives summary statistics and Panel B the pooled correlation matrix. Every measure divides a value component by end-of-period capital, where K is the physical-capital stock and N is the intangible-capital stock; V is the sum of market equity and the book value of debt, minus cash and short-term investments. $V^{\text{model}} = V^{\text{prod-phy}} + V^{\text{prod-int}} + V^{\text{mkp-int}} + V^{\text{mkp-non}}$ is the model-implied value, which removes the unexplained residual from market value. Skew is the sample skewness $g_1 = m_3/m_2^{3/2}$. All q measures are winsorized within each year at the 1st and 99th percentiles. The sample period is from 1976 to 2025.

Panel A: Summary statistics

Measure	Definition	Mean	SD	Median	Skew
Standard q	V/K	5.17	10.12	2.02	5.53
Total q	$V/(K+N)$	1.46	1.84	0.93	5.14
Model total q	$V^{\text{model}}/(K+N)$	1.41	0.99	1.11	1.52
ex. non-intangible markup	$(V^{\text{model}} - V^{\text{mkp-non}})/(K+N)$	1.33	0.86	1.01	1.41
ex. all markup	$(V^{\text{model}} - V^{\text{mkp-int}} - V^{\text{mkp-non}})/(K+N)$	0.97	0.77	0.71	1.34
Physical q	$V^{\text{prod-phy}}/K$	1.62	1.88	0.99	3.81
Intangible q (total)	$(V^{\text{prod-int}} + V^{\text{mkp-int}})/N$	1.40	0.95	1.08	2.52
Intangible q (production)	$V^{\text{prod-int}}/N$	0.53	0.65	0.22	1.75
Intangible q (markup)	$V^{\text{mkp-int}}/N$	0.87	0.55	0.77	4.05

Panel B: Pooled correlation matrix

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
(1) Standard q	1.00								
(2) Total q	0.72	1.00							
(3) Model total q	0.59	0.79	1.00						
(4) ex. non-intangible markup	0.56	0.79	0.87	1.00					
(5) ex. all markup	0.46	0.76	0.80	0.98	1.00				
(6) Physical q	0.73	0.68	0.69	0.79	0.75	1.00			
(7) Intangible q (total)	0.29	0.52	0.61	0.63	0.67	0.38	1.00		
(8) Intangible q (production)	0.40	0.60	0.62	0.78	0.77	0.50	0.83	1.00	
(9) Intangible q (markup)	0.02	0.19	0.32	0.18	0.25	0.08	0.75	0.25	1.00

Table 8: Investment and q: horse race

This table reports the investment-q regressions for different total q measures. Each column is a univariate regression of the investment rate on a single q. Panel A uses physical investment I^K/K and Panel B intangible investment I^N/N. Both investment rates are measured one year ahead. Model total q is the model-implied value over $K+N$; the last two columns strip the non-intangible market-power component $V^{\text{mkp-non}}$, and then both market-power components $V^{\text{mkp-int}}$ and $V^{\text{mkp-non}}$, out of that value. All q measures and investment rates are winsorized within each year at the 1st and 99th percentiles. All specifications include firm and year fixed effects, with standard errors two-way clustered by firm and year in parentheses.

	Standard q	Total q	Model total q	Model total q (ex. non-intangible markup)	Model total q (ex. all markup)
	(1)	(2)	(3)	(4)	(5)
Panel A: Physical investment, I^K/K					
q	0.0058***	0.0312***	0.0660***	0.0703***	0.0744***
	(0.0006)	(0.0027)	(0.0033)	(0.0034)	(0.0036)
Observations	114,668	114,668	114,668	114,668	114,668
R^2	0.4500	0.4722	0.4727	0.4722	0.4675
Within R^2	0.0842	0.1210	0.1219	0.1212	0.1133
Panel B: Intangible investment, I^N/N					
q	0.0035***	0.0281***	0.0568***	0.0571***	0.0652***
	(0.0004)	(0.0020)	(0.0028)	(0.0028)	(0.0031)
Observations	114,668	114,668	114,668	114,668	114,668
R^2	0.5962	0.6365	0.6316	0.6256	0.6298
Within R^2	0.0429	0.1386	0.1270	0.1127	0.1226

Table 9: Investment and q: value components

Panel A reports investment-q regressions using the value components. Columns (1)–(3) explain physical investment I^K/K and columns (4)–(6) intangible investment I^N/N; both dependent variables are measured one year ahead. Components are scaled by K in columns (1)–(3) and by N in columns (4)–(6). Columns (1) and (4) use the components generated by the corresponding capital stock, columns (2) and (5) use all four components, and columns (3) and (6) add total q. Panel B decomposes the within-R^2 of columns (2) and (5) into Shapley shares, which add to 100. All q measures and investment rates are winsorized within year at the 1st and 99th percentiles. All specifications include firm and year fixed effects; standard errors in Panel A are two-way clustered by firm and year.

	Physical investment, I^K/K			Intangible investment, I^N/N		
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Investment regressions						
$V^{\text{prod-phy}}$	0.0257***	0.0227***	0.0067***		0.0001	0.0006
	(0.0027)	(0.0026)	(0.0017)		(0.0004)	(0.0004)
$V^{\text{prod-int}}$		0.0070***	0.0010	0.0264***	0.0268***	-0.0033
		(0.0010)	(0.0007)	(0.0023)	(0.0024)	(0.0027)
$V^{\text{mkp-int}}$		-0.0009	0.0124***	0.0704***	0.0673***	0.0457***
		(0.0015)	(0.0018)	(0.0068)	(0.0067)	(0.0061)
$V^{\text{mkp-non}}$		0.0042***	-0.0026***		0.0002	0.0008**
		(0.0010)	(0.0008)		(0.0004)	(0.0004)
Total q			0.0263***			0.0254***
			(0.0025)			(0.0021)
Observations	114,668	114,668	114,668	114,668	114,668	114,668
R^2	0.4466	0.4519	0.4832	0.6129	0.6130	0.6482
Within R^2	0.0785	0.0873	0.1394	0.0827	0.0827	0.1661

Panel B: Shapley decomposition of within-R^2 (%)		
	Physical investment, I^K/K	Intangible investment, I^N/N
$V^{\text{prod-phy}}$	70.50	16.61
$V^{\text{prod-int}}$	19.01	32.51
$V^{\text{mkp-int}}$	9.06	45.45
$V^{\text{mkp-non}}$	1.42	5.43
Total within R^2	0.0873	0.0827
Observations	114,668	114,668

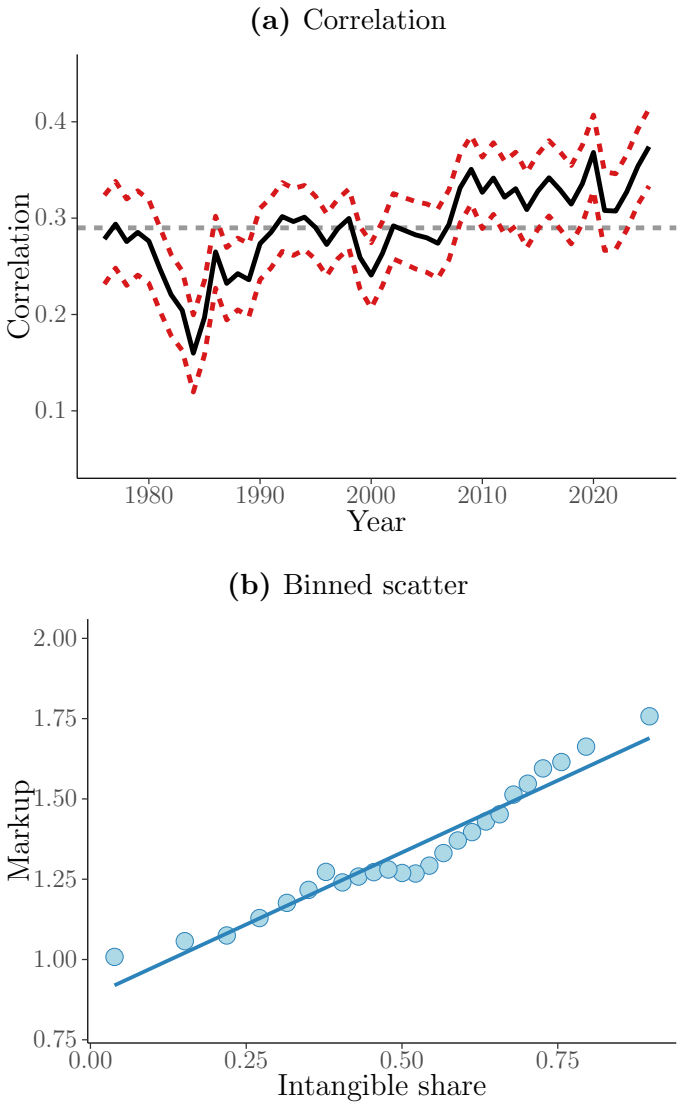


Figure 1: Cross-sectional correlation between intangible share and markups (1976–2025)

This figure plots the relationship between intangible capital share and markups. Markups are estimated under Cobb-Douglas production technology that includes intangible capital as an input. Panel (a) plots the annual cross-sectional correlation, demeaning both variables within each two-digit industry-year, with the 95% confidence interval in red and the sample average in gray. Panel (b) plots a binned scatter: both variables are residualized on two-digit industry and year fixed effects, sorted into 25 equal-sized bins on the residualized intangible share, and plotted as bin means with the sample means added back, with an OLS fitted line. Markups are winsorized within year at the 2.5th and 97.5th percentiles. Intangible share is intangible capital divided by the sum of physical and intangible capital, with intangible capital from Peters and Taylor (2017) and physical capital measured by Compustat item PPEGT.

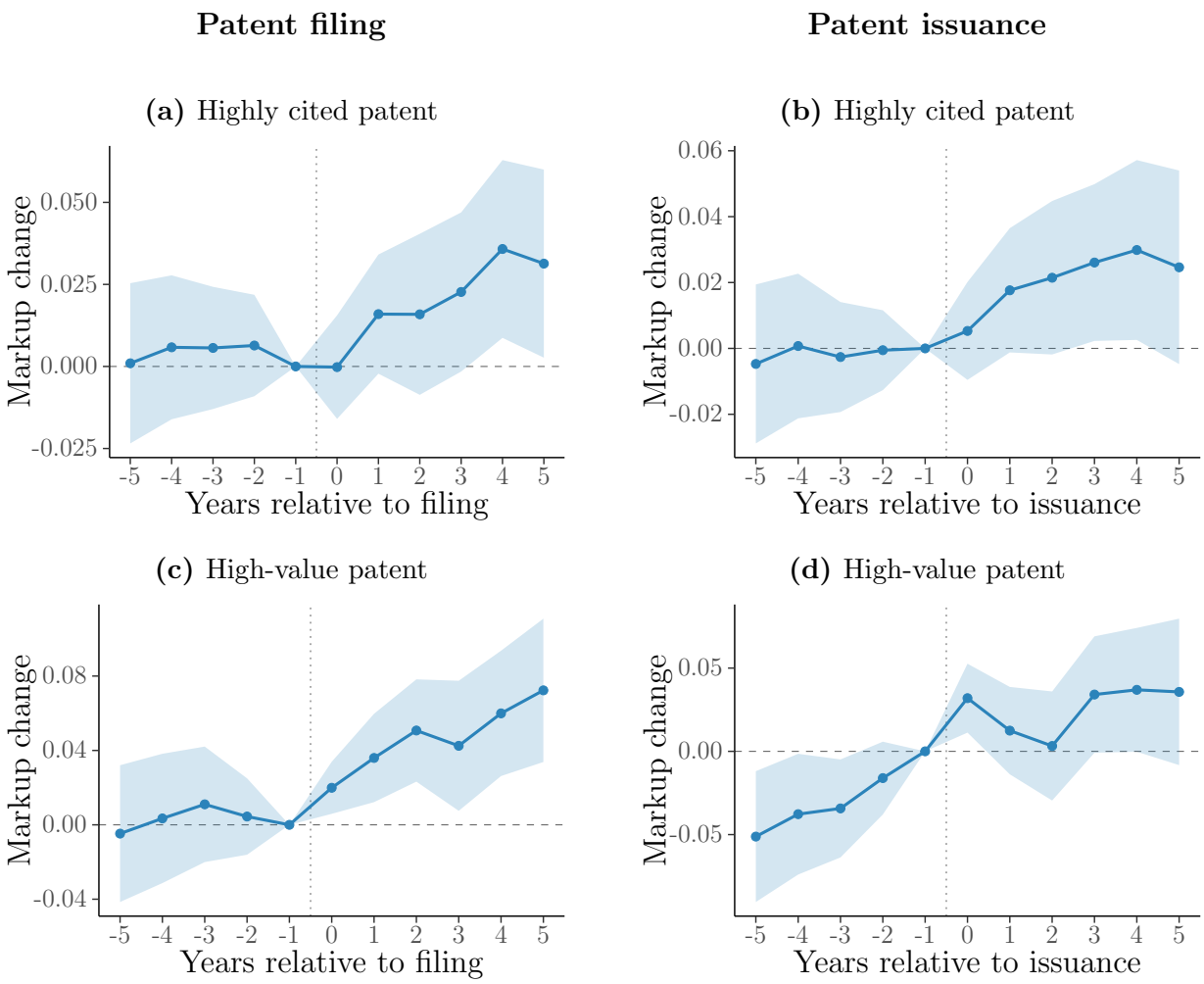


Figure 2: Event studies under Cobb-Douglas with intangibles

This figure plots dynamic treatment effects of important patent filings (left column) and issuances (right column) on markups, using the Cobb-Douglas production-function estimate that includes intangible capital as a production input. Panels (a) and (b) classify a patent as important if its forward-citation count lies in the top 10% within its Fama-French 48 industry in the filing or issuance year; Panels (c) and (d) instead classify a patent as important if its stock-market value, as measured by Kogan et al. (2017), lies in the top 10% within the same industry-year cell. A firm is treated in the year of its first important patent, and firms that never have an important patent serve as the never-treated control group. Coefficients are estimated with the Sun and Abraham (2021) interaction-weighted estimator. The specification includes firm and year fixed effects. The estimation window is $\tau \in [-5, +5]$. Shaded areas are 95% confidence intervals, with standard errors clustered by firm.

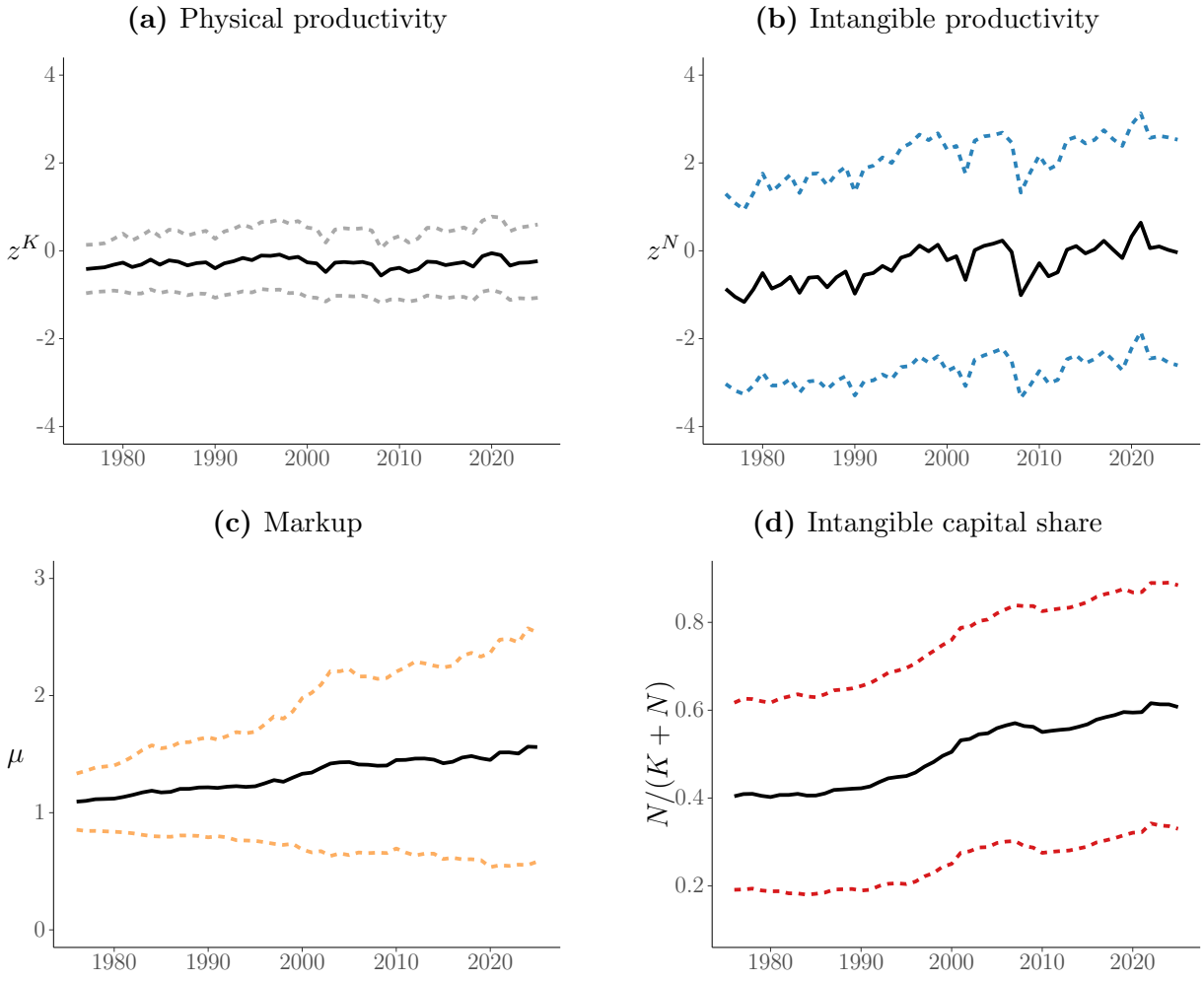
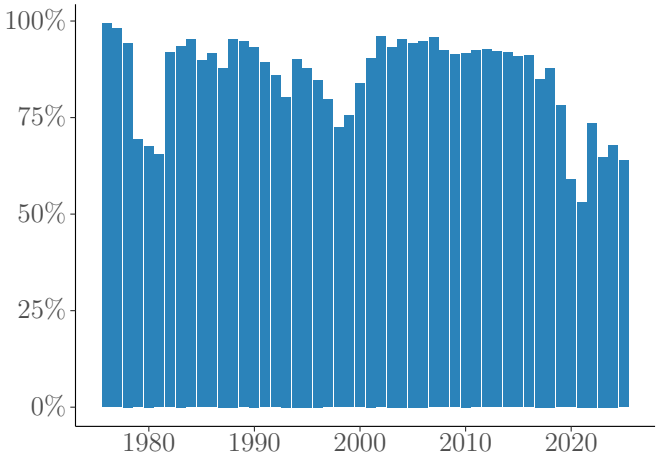


Figure 3: Time series of estimated state variables (1976–2025)

This figure plots the estimated state variables, averaged over firms in each year. Panels (a) and (b) show the cross-sectional average productivity for physical capital, $\bar{z}_t^K = x_t^K$, and intangible capital, $\bar{z}_t^N = x_t^N$, respectively. Panels (c) and (d) show the cross-sectional average markup, $\bar{\mu}_t$, and intangible capital share, $\overline{N/(K+N)}_t$, respectively. In each panel, the solid line is the cross-sectional mean and the dashed lines represent the mean $\pm$ one cross-sectional standard deviation, reflecting the dispersion across firms.

(a) R-squared for V



(b) R-squared for $V/(K+N)$

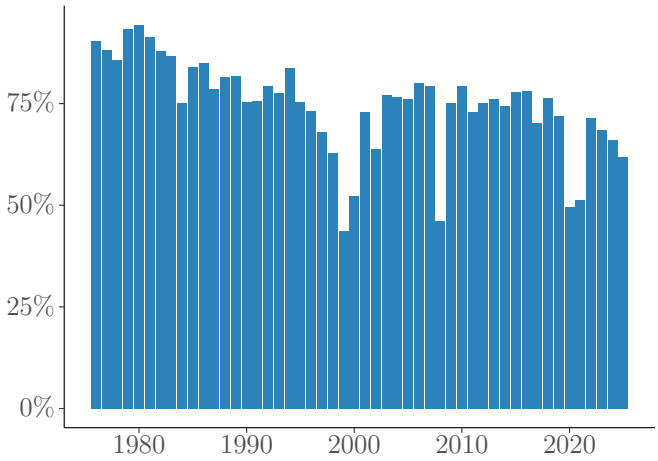


Figure 4: Cross-sectional R-squared over time (1976–2025)

This figure plots the cross-sectional R-squared defined as $\hat{R}^2_t = 1 - \frac{\sum_i \left(V_{it} - V_{it}^{\text{model}}\right)^2}{\sum_i \left(V_{it} - \bar{V}_t\right)^2}$, which restricts both intercept to be zero and slope to be one. Panel (a) uses firm value V_{it} and panel (b) uses V_{it} scaled by total capital. V_{it}^{model} is the model-implied counterpart and $\bar{V}_t$ is the cross-sectional mean.

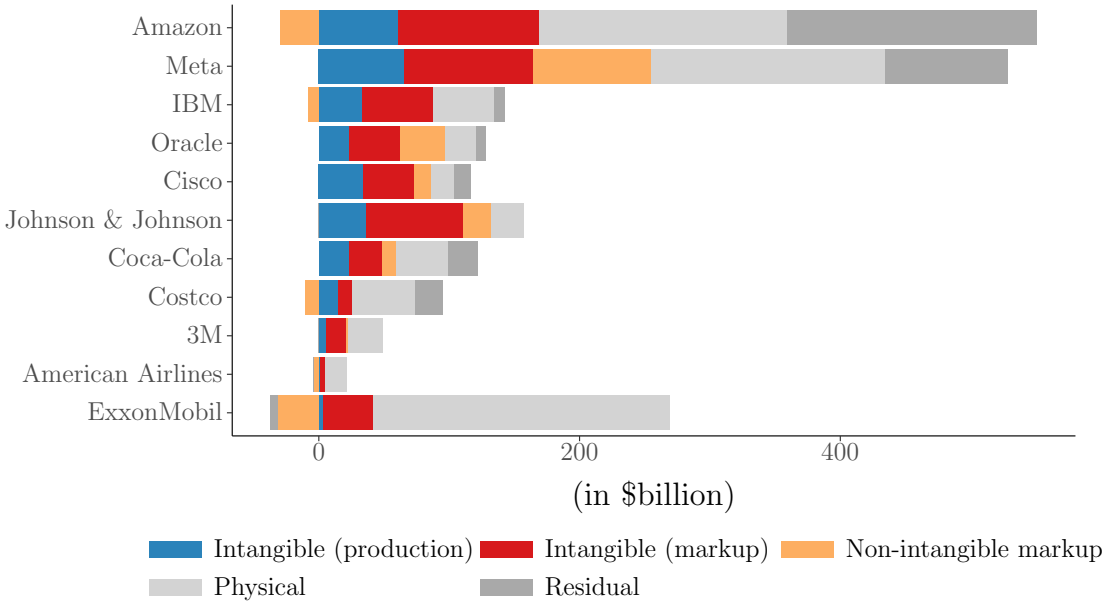


Figure 5: Decomposition for individual firms

This figure plots the value decomposition for a set of example firms based on Equation (18). Each bar represents a firm's average value across sample years.

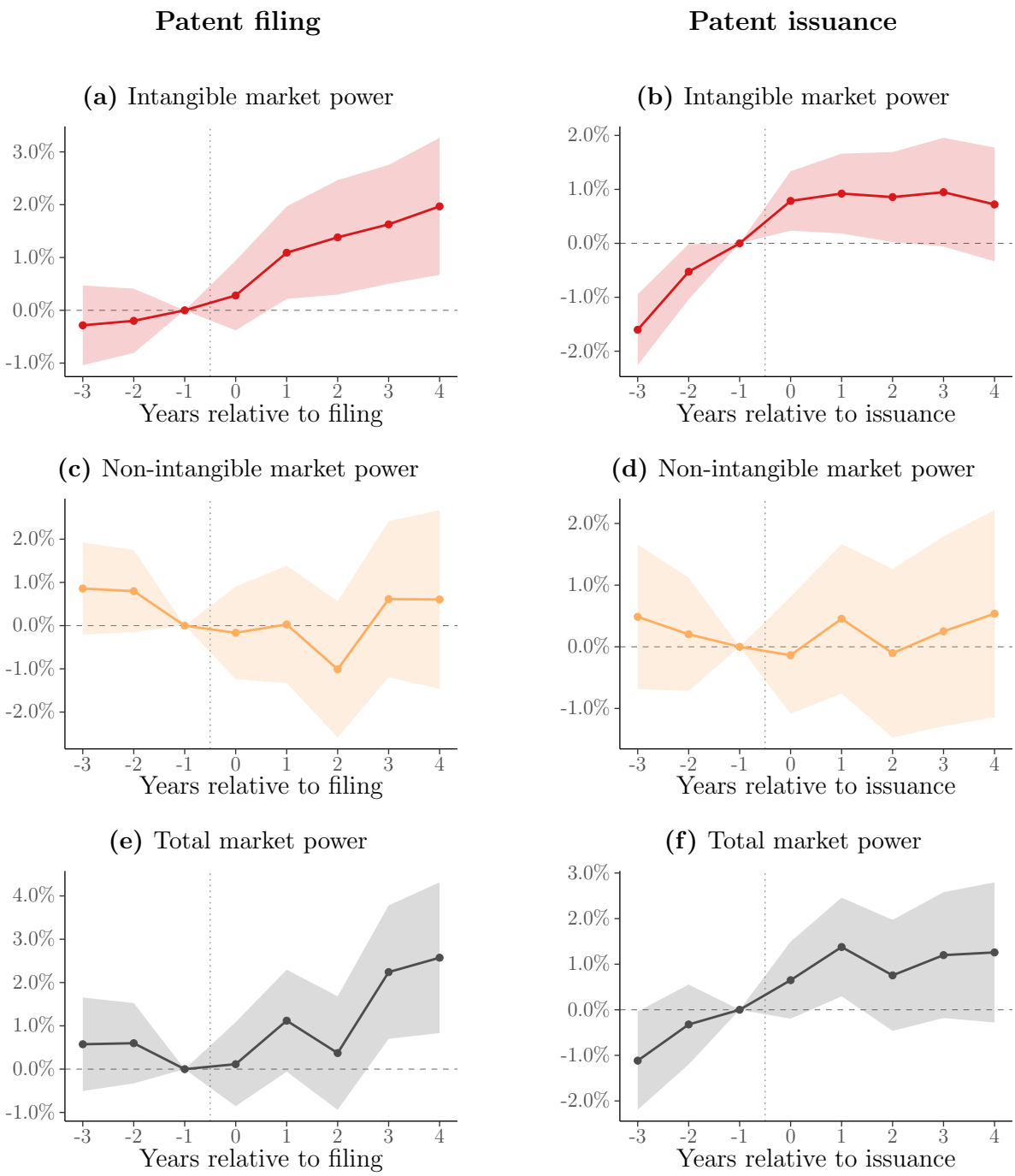


Figure 6: Change in the share of market power around patent filing and issuance

This figure plots dynamic treatment effects of important patent filings (left column) and issuances (right column) on the share of firm value attributable to market power. The figure shows the results for the value share of intangible-driven market power, non-intangible market power, and total market power. Each share is defined as the corresponding component divided by V^{model}. The event-study design, patent-importance definition, and estimation follow Figure 2. Shaded areas are 95% confidence intervals, with standard errors clustered by firm.

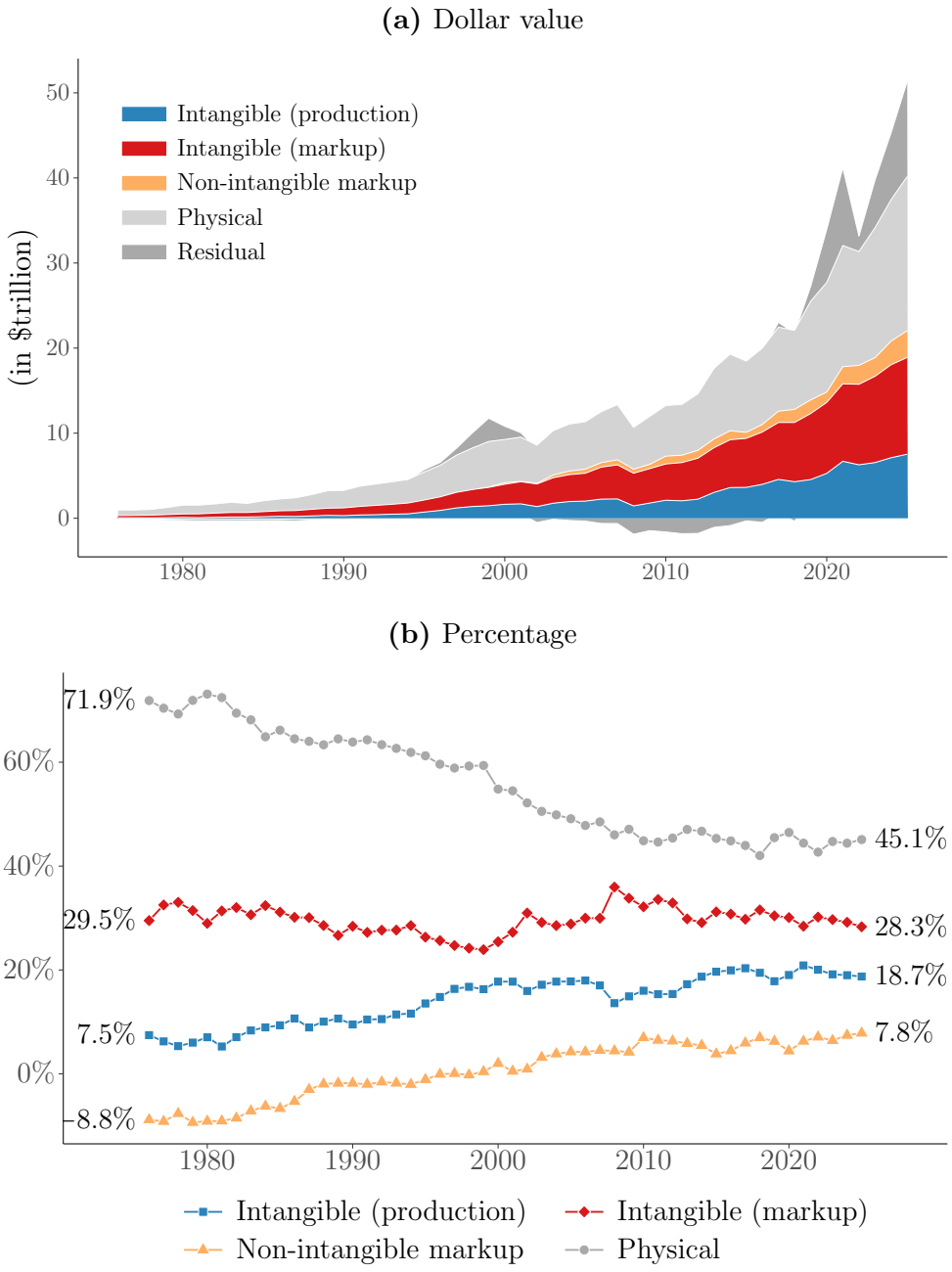


Figure 7: Level decomposition over time (1976–2025)

This figure plots the aggregate value decomposition based on Equation (18). Panel (a) sums each value component across all firms in the cross section for each year and reports the totals in trillions of dollars as a stacked area chart. Panel (b) shows the percentage decomposition, where each component's share is computed as its aggregate dollar value divided by total model-implied firm value.

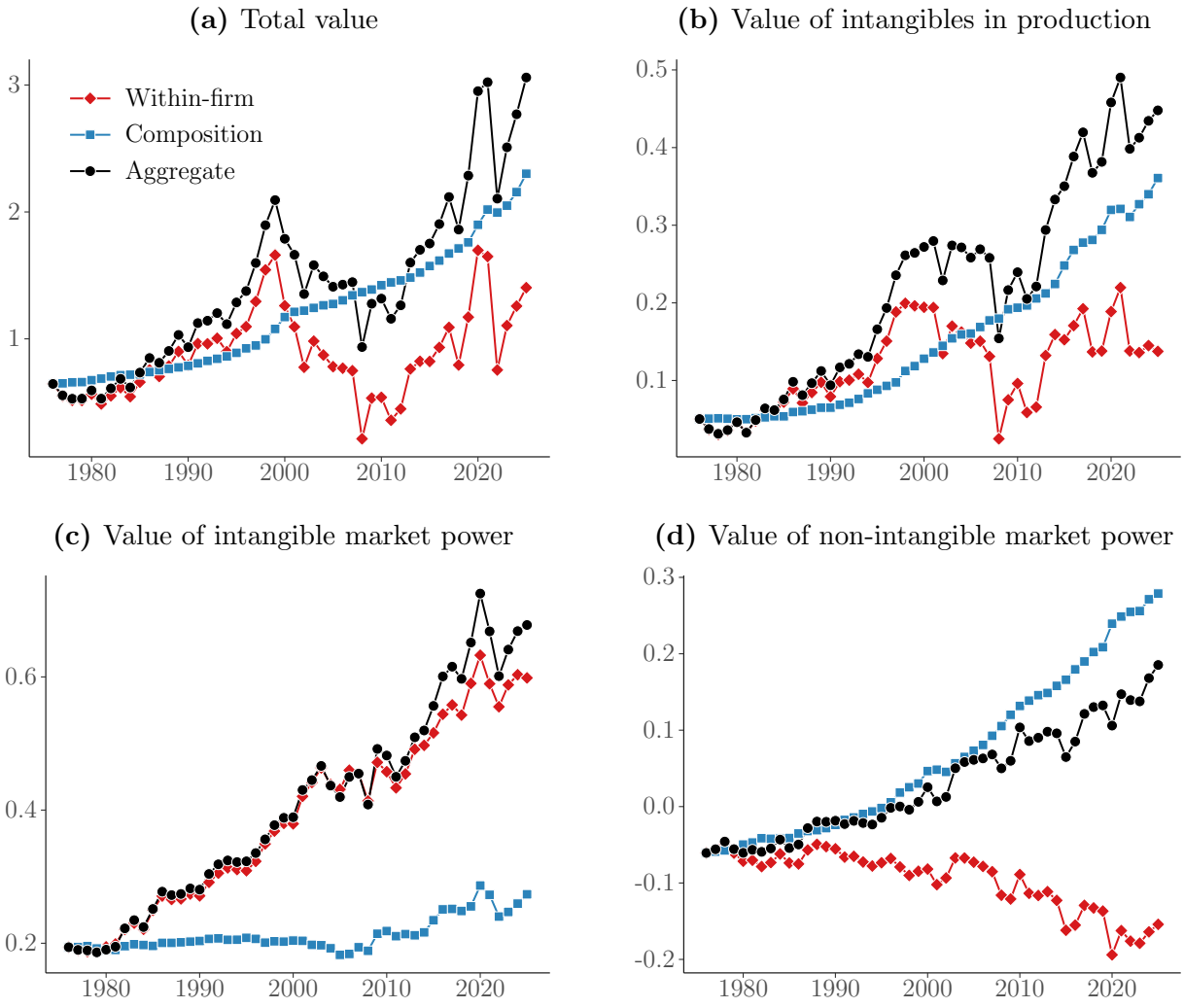


Figure 8: Compositional vs. within-firm change

This figure plots the decomposition of changes in aggregate value into within-firm and compositional components. The decomposition is defined in Equation (26). Panel (a) shows the decomposition of the total value-to-sales ratio. Panel (b) shows the decomposition of the value of intangibles in production scaled by sales. Panels (c) and (d) show the corresponding decompositions of intangible and non-intangible market power value scaled by sales.

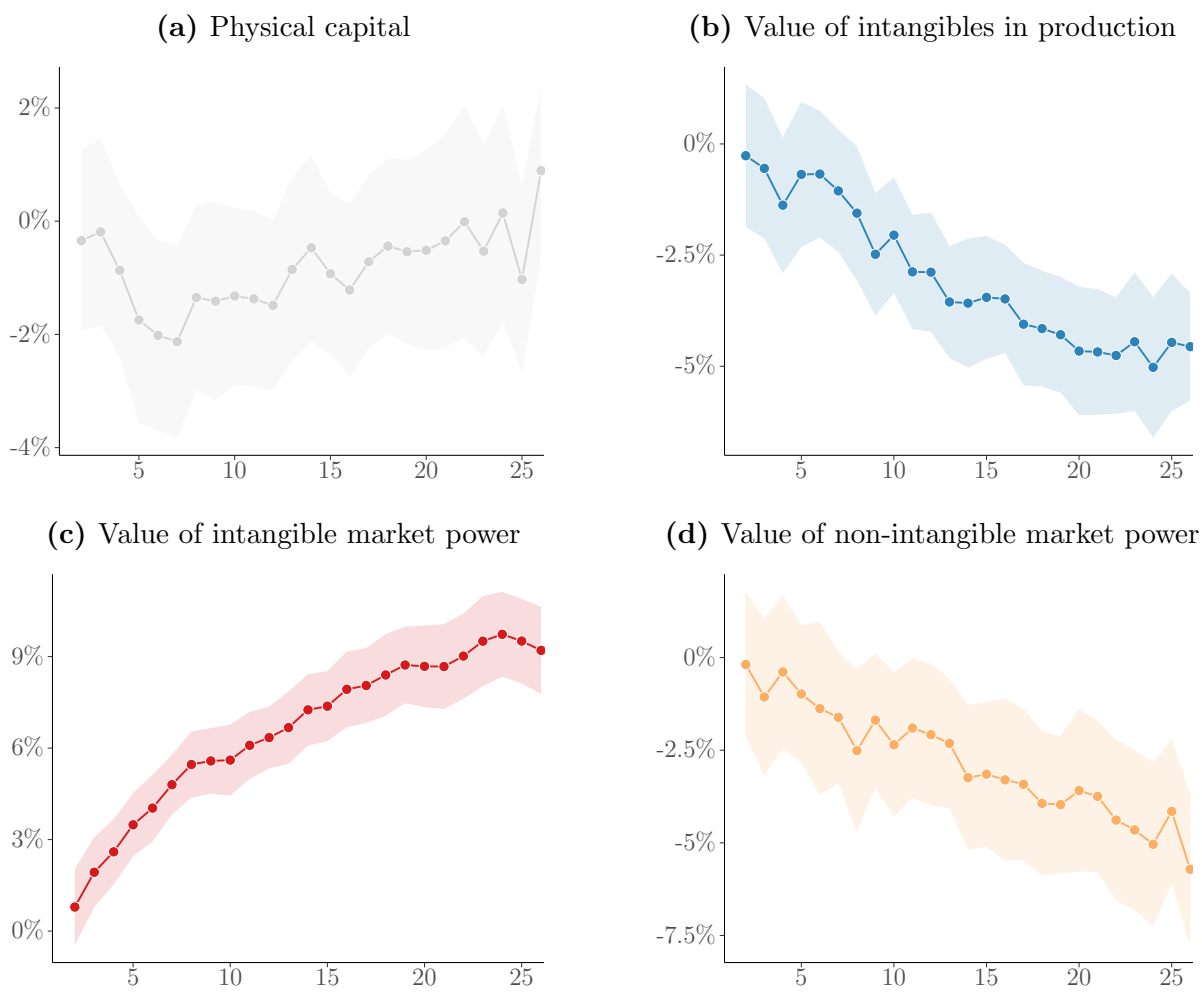


Figure 9: Life cycle changes

This figure plots the evolution of each value component's share over a firm's life cycle. For each component, I regress the ratio of the component to total model-implied value on firm age dummies with year and cohort fixed effects, restricting the sample to firms with at least 15 years of data. The solid line shows the estimated coefficients and the shaded area represents the 95% confidence interval.

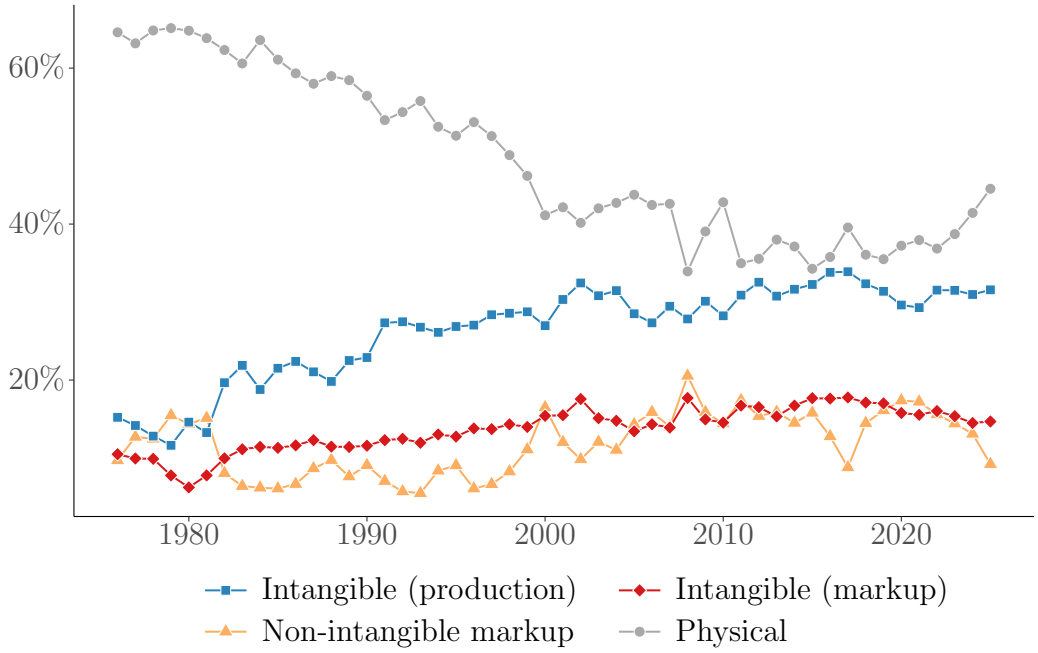


Figure 10: Covariance decomposition over time (1976–2025)

This figure plots the aggregate covariance decomposition for each year based on Equation (29).

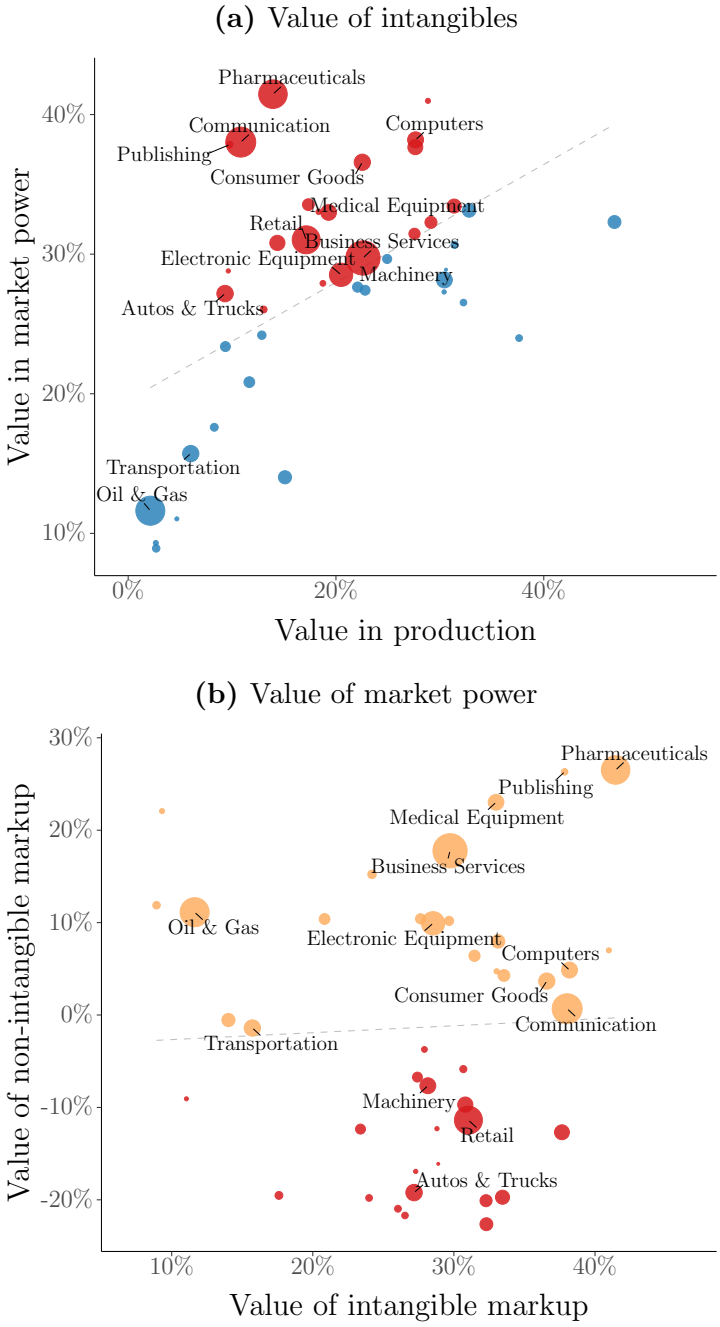


Figure 11: Sector-level relationships between value components

This figure plots industry-level scatter plots using the Fama-French 48 industry classification (excluding banking, insurance, real estate, trading, utilities, and other). Panel (a) plots the share of firm value attributable to intangible market power against the share from intangibles in production. Panel (b) plots the share from non-intangible market power against the share from intangible market power. Each point represents an industry average, with its size proportional to the industry's share of total model-implied value. The dashed line shows the linear fit. Labels are shown for the twelve largest industries and for the two industries farthest from the fitted line in each panel.

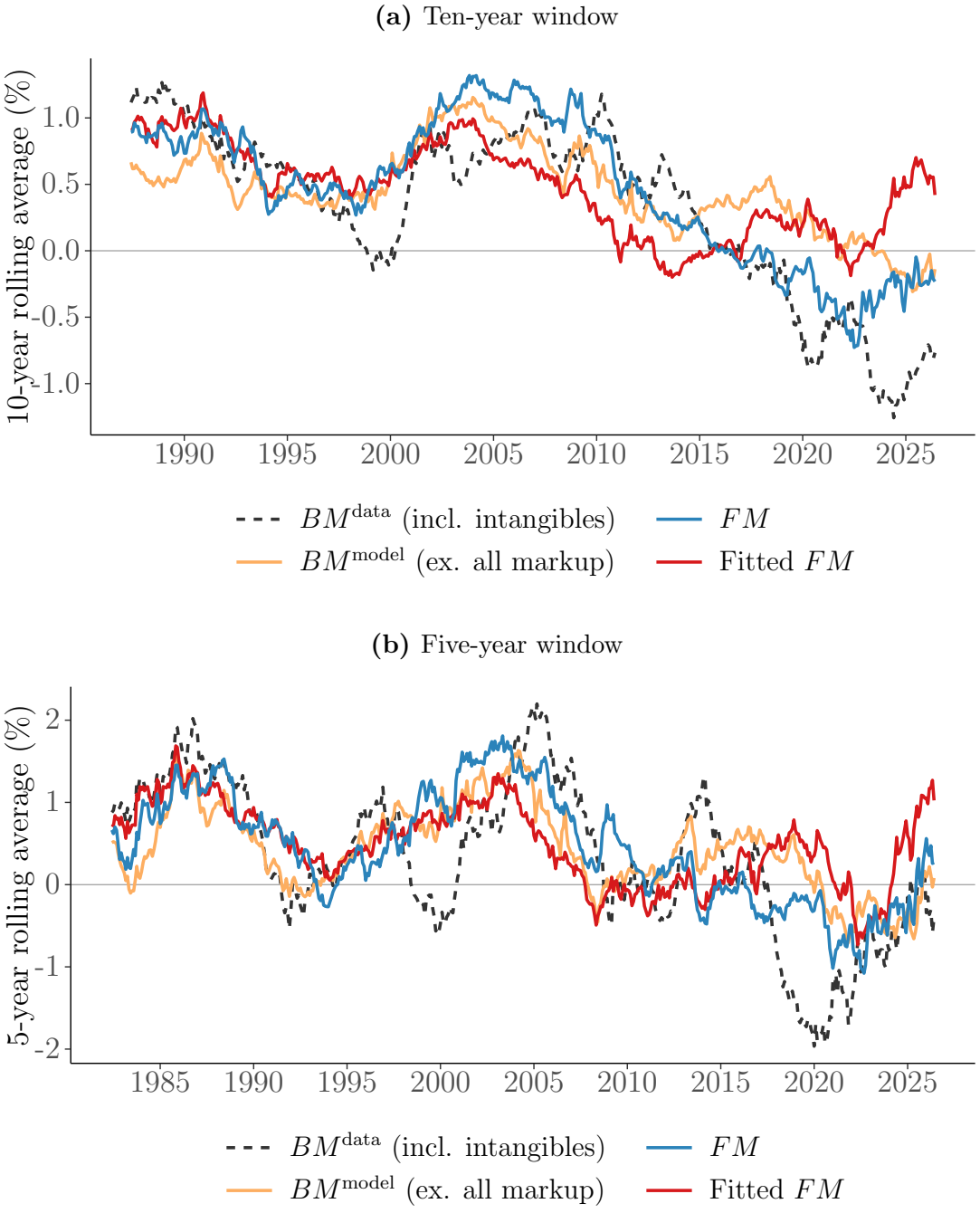


Figure 12: Trailing average of the value premium

This figure plots trailing averages of the monthly value premium, value minus growth in % per month, for four sorting variables: the intangible-adjusted book-to-market ratio, the model-implied ratio that removes the value of all market power, fitted FM, and FM. Portfolios are annual NYSE-breakpoint deciles, value weighted by lagged market equity, over July 1977 to June 2026. Panel (a) uses a ten-year window and Panel (b) a five-year window. Windows are right-aligned, so the value plotted at June 2026 in Panel (a) is the average over July 2016 to June 2026.

Appendix: Derivation of the Operating Profits

A.1 Consumer Demand

Assume each firm operates within its own distinct market segment, producing a single, differentiated variety. The demand function faced by firm i is $Y(P_{it})$, where P_{it} is the price of product i. I parameterize $Y(\cdot)$ as

$$Y(P_{it}) = \left(\frac{P_{it}}{P_t}\right)^{-\frac{\mu_{it}}{\mu_{it}-1}} \frac{\bar{Y}}{P_t}, \quad \mu_{it} > 1, \tag{A.1}$$

where μ_{it} represents the firm's pricing power for product i, $\bar{Y}$ is total market size for this specific product, and P_t is an exogenous price index that is absorbed by the productivity shocks (see the next section). Intuitively, when μ_{it} is high, consumers become less sensitive to prices and firms can set higher prices to extract rents. In the next subsection, I show that firms optimally choose the markup to be μ_{it} in their profit maximization.

The total expenditure $\bar{Y}$ is held constant and serves as the numeraire in this market. This demand function implicitly assumes that all firms face the same demand curve, and demand variation comes from movement along the curve and variation of demand elasticity (through changing market power), while the total expenditure $\bar{Y}$ is constant across firms. This assumption is not too restrictive because I scale the firm size in the data to match the distribution in the model simulation and I match the valuation ratio to extract productivity states. However, the remaining unobserved demand-scale heterogeneity may still be absorbed by the recovered productivity states.

This demand function is consistent with a rational consumer preference system as shown in Matsuyama and Ushchev (2017) and Baqaee, Farhi and Sangani (2024), and it accommodates heterogeneous markups across firms. In this partial equilibrium framework, market power is captured through the shape of the demand function, which governs consumers' price sensitivity. This modeling choice does not imply that consumer preferences are the

sole primitive source of market power. In a general equilibrium setting, markups are equilibrium outcomes jointly determined by demand and supply forces: a firm's ability to charge high prices reflects not only consumers' willingness to pay but also supply-side factors such as product differentiation, proprietary technology, or competitive structure. The demand function in my model should therefore be interpreted as a reduced-form representation of these combined forces.

A.1.1 Special Cases of the Demand Function

The markup is less than or equal to 1. When $\mu_{it} \leq 1$, I assume the firm operates under perfect competition and takes the price P_t as given. This assumption corresponds to the limiting case of the demand function (A.1) as μ_{it} approaches 1 from above.

P_{it} is below the price index P_t. If the firm has the ability to produce with low marginal cost and undercut the price index, I assume the firm captures the entire market and earns total revenue $\bar{Y}$. This case corresponds to a sector monopoly as in De Ridder (2024) and Aghion et al. (2025), where the most efficient producer dominates the entire sector and earns constant revenue. The price this firm can set is constrained by the second-best competitor (potential entrant), and μ_{it} corresponds to the productivity gap between the incumbent and its closest rival. It is consistent with the relation between μ_{it} and intangible capital, as investments in intangibles enhance productivity and widen this competitive advantage.

A.2 Profit Maximization

At each time t, given the demand equation (A.1), firm i solves the following problem:

$$\begin{aligned} \Pi(z_{it}^K, z_{it}^N, \zeta_{it}^\mu, K_{it}, N_{it}) = \max_{P_{it}, L_{it}} \; & P_{it}Y_{it} - P_{Lit}L_{it} \\ \text{subject to} \quad & \mathbf{K}(z_{it}^K, z_{it}^N, K_{it}, N_{it})^\eta L_{it}^{1-\eta} = Y_{it}, \end{aligned} \tag{A.2}$$

where P_{Lit} is the price of the costlessly adjustable input and firms take it as given. Let λ_{it} be the Lagrangian multiplier for the production constraint. λ_{it} represents the marginal cost of production (given the capital inputs).

Taking the first-order condition with respect to P_{it} yields the optimal markup, $P_{it}/\lambda_{it} = \mu_{it}$. I then substitute P_{it} out using the constraint and rewrite the problem as

$$\max_{L_{it}} \left[\mathbf{K}_{it}^{\eta} L_{it}^{1-\eta}\right]^{\frac{1}{\mu_{it}}} \bar{Y}^{\frac{\mu_{it}-1}{\mu_{it}}} P_t^{\frac{1}{\mu_{it}}} - P_{Lit} L_{it}. \tag{A.3}$$

The first-order condition with respect to L_{it} is $P_{Lit} = \frac{1-\eta}{\mu_{it}} \bar{Y}^{\frac{\mu_{it}-1}{\mu_{it}}} P_t^{\frac{1}{\mu_{it}}} Y_{it}^{\frac{1}{\mu_{it}}} / L_{it}$. Solving for L_{it} yields

$$L_{it} = \left(\frac{(1-\eta) \bar{Y}^{\frac{\mu_{it}-1}{\mu_{it}}} P_t^{\frac{1}{\mu_{it}}}}{\mu_{it} P_{Lit}} \right)^{\frac{\mu_{it}}{\mu_{it}-1+\eta}} \mathbf{K}_{it}^{\frac{\eta}{\mu_{it}-1+\eta}}. \tag{A.4}$$

Substituting the first-order condition back into Π gives

$$\Pi_{it} = \left(1 - \frac{1-\eta}{\mu_{it}}\right) P_{it} Y_{it}. \tag{A.5}$$

Plugging the optimal L_{it} into the profit function,

$$\begin{aligned}
\Pi_{it} &= \left[1 - \frac{1-\eta}{\mu_{it}}\right] \left(\frac{1-\eta}{\mu_{it}}\right)^{\frac{1-\eta}{\mu_{it}-1+\eta}} P_t^{\frac{1}{\mu_{it}-1+\eta}} \bar{Y}^{\frac{\mu_{it}-1}{\mu_{it}-1+\eta}} P_{Lit}^{-\frac{1-\eta}{\mu_{it}-1+\eta}} \mathbf{K}_{it}^{\frac{\eta}{\mu_{it}-1+\eta}} \\
&= \left[1 - \frac{1-\eta}{\mu_{it}}\right] \left(\frac{1-\eta}{\mu_{it}}\right)^{\frac{1-\eta}{\mu_{it}-1+\eta}} \bar{Y} \left(\frac{\mathbf{K}_{it}^{\eta} P_t}{P_{Lit}^{1-\eta} \bar{Y}^{\eta}}\right)^{\frac{1}{\mu_{it}-1+\eta}} \\
&= \left[1 - \frac{1-\eta}{\mu_{it}}\right] \left(\frac{1-\eta}{\mu_{it}}\right)^{\frac{1-\eta}{\mu_{it}-1+\eta}} \bar{Y} \cdot \tilde{\mathbf{K}}_{it}^{\frac{\eta}{\mu_{it}-1+\eta}}.
\end{aligned} \tag{A.6}$$

The last equality is due to the fact that the term $\frac{P_t}{P_{Lit}^{1-\eta} \bar{Y}^{\eta}}$ can be absorbed by shocks z_{it}^K and z_{it}^N within the capital aggregator. In the main text, I drop the tilde above $\mathbf{K}$ to be more concise.

The demand function (A.1) is well-defined for $\mu_{it} > 1$. When $\mu_{it} = 1$ with price equal to

the marginal cost, the firm operates under perfect competition and is a price taker. In this case, the firm solves

$$\Pi(z_{it}^K, z_{it}^N, \zeta_{it}^{\mu}, K_{it}, N_{it}) = \max_{L_{it}} P_t \mathbf{K}_{it}^{\eta} L_{it}^{1-\eta} - P_{Lit} L_{it}. \tag{A.7}$$

The first-order condition is given by $P_{Lit} = (1-\eta) P_t \mathbf{K}_{it}^{\eta} L_{it}^{-\eta}$, which implies $L_{it} = \left(\frac{(1-\eta)P_t}{P_{Lit}}\right)^{\frac{1}{\eta}} \mathbf{K}_{it}$. Therefore, the profit function is

$$\begin{aligned} \Pi_{it} &= P_t \mathbf{K}_{it}^{\eta} L_{it}^{1-\eta} - (1-\eta) P_t \mathbf{K}_{it}^{\eta} L_{it}^{1-\eta} \\ &= \eta P_t \mathbf{K}_{it}^{\eta} L_{it}^{1-\eta} \\ &= \eta (1-\eta)^{\frac{1-\eta}{\eta}} P_t^{\frac{1}{\eta}} P_{Lit}^{-\frac{1-\eta}{\eta}} \mathbf{K}_{it}, \end{aligned} \tag{A.8}$$

which is equivalent to Equation (A.6) when setting $\mu_{it} = 1$. Therefore, the profit function (A.6) nests the perfect competition case.

When the firm produces at very low cost, it captures the entire product market niche and the profit is then given by (following Equation (A.5) and the fact that total sales equal the market size)

$$\Pi_{it} = \bar{Y} \cdot \left(1 - \frac{1-\eta}{\mu_{it}}\right). \tag{A.9}$$

Internet Appendix

"What Are Intangibles Worth? Production versus Market Power"

By Yicheng Liu

Contents

A Data, Measurement, and Methodology Details IA.1

A.1 Measure of Markups IA.1

A.2 Estimation Details IA.3

A.3 Shapley Decomposition IA.4

B Additional Results IA.7

B.1 Alternative Specifications of Estimation IA.7

B.2 Additional Evidence on the Relation between Intangibles and Markups IA.9

B.3 Additional Validation IA.10

B.4 R&D, SG&A, and Acquired Intangible Capital IA.10

B.5 Alternative Specification for Life Cycle Regressions IA.11

B.6 Additional Evidence on the Value Premium Decline IA.11

B.7 Alternative Definitions of Investment Rates and q IA.11

A Data, Measurement, and Methodology Details

A.1 Measure of Markups

Estimation of firm-level markups typically relies on the production function approach in the IO literature (e.g., Hall, 1988; De Loecker, Eeckhout and Unger, 2020). This approach exploits the cost-minimization condition that ties marginal cost to input demand. Markups are then estimated as the wedge between the output elasticity of a variable input and that input's observed revenue share. The measure is given by

$$\mu_{it} = \eta_{it} \frac{\text{SALE}_{it}}{\text{VarCost}_{it}}, \tag{A.1}$$

where VarCost_{it} is the cost of a variable input that can be freely adjusted, and η_{it} is the output elasticity of that input.

While this measure provides markup estimates for nearly all public firms in Compustat, it is criticized for relying on strong assumptions: the production function can be identified from revenue-based data when quantity is not observed, and COGS from Compustat fully represents the variable inputs.[24] However, De Ridder, Grassi and Morzenti (2026) argue that even though the level of markup can be biased under the production approach, the dispersion among firms is accurately captured by such estimates, which is the focus of this paper. To further ensure the validity of this measure, I include 6 estimates based on different specifications.

Model-free markup proxies. As shown in De Ridder, Grassi and Morzenti (2026), firm-level variation in markups is primarily driven by the variation in the revenue-to-cost ratio $\text{SALE}_{it}/\text{VarCost}_{it}$. I first document the correlation between intangible capital share and this ratio, which is free of production function estimation. I consider two versions of VarCost_{it}.

[24] See Traina (2018), Bond et al. (2021), Doraszelski and Jaumandreu (2022), Raval (2023), and Demirer (2022).

The first uses COGS following De Loecker, Eeckhout and Unger (2020). The second includes both COGS and some of the SG&A expenses. Traina (2018) suggests that at least some of the selling and management expenses are also variable costs. I follow Ayyagari, Demirgüç-Kunt and Maksimovic (2024) to measure variable costs as:

$$\text{COGS} + \text{XSGA}^* = \text{COGS} + (\text{XSGA} - \text{XRD} - \text{RDIP} - 0.3 \times \text{XSGA}), \tag{A.2}$$

where XSGA is the Compustat item selling, general, and administrative expenses. XRD and RDIP measure R&D and in-process R&D expenses in Compustat. In other words, this measure of variable costs recognizes R&D and some SG&A expenses as capital investment, while the rest is variable.

Markup based on different production functions. I further estimate markups under four production function specifications, varying along two dimensions: production technology (translog vs. Cobb-Douglas) and whether intangible capital is included as a production input. The translog production function approximates arbitrary production technologies via a second-order Taylor expansion in logarithms. In general, a translog function with n inputs $X_1, X_2, \ldots, X_n$ can be written as:

$$\log Y = \alpha_0 + \sum_{i=1}^{n} \alpha_i \log X_i + \sum_{i=1}^{n} \sum_{j=1}^{n} \beta_{ij} \log X_i \log X_j. \tag{A.3}$$

Under the translog specification, the output elasticity of the variable input, $\eta_i = \partial \log Y / \partial \log X_i$, is firm-specific and depends on the levels of all production inputs. Cobb-Douglas is a nested special case with $\beta_{ij} = 0$ for all i, j, yielding a constant output elasticity. For each production technology, I consider two input configurations: one with physical capital and the variable input only, and one that additionally includes intangible capital. In all specifications, I use cost of goods sold (COGS) as the variable input expenditure.

A.2 Estimation Details

The SMM estimation minimizes the distance between data moments and simulated moments. The estimation is implemented in the following steps. For an arbitrary set of parameters Θ, I solve the model for the value and policy functions. Then I simulate K panels of N firms $\times$ $T + 150$ years and drop the first 150 years as a burn-in period to remove the impact of initial states. In the simulation, if a firm's value drops to 0, I assume the firm exits and is immediately replaced by a new firm with the same capital and redrawn shocks.

Next, I compute the model-implied moments in the simulation. Let x_{it} be the actual data vector, $i \in \{1, \ldots, N\}$, $t \in \{1, \ldots, T\}$, and let $y_{itk}(\Theta)$ be the simulated vector from simulation k, $i \in \{1, \ldots, N\}$, $t \in \{1, \ldots, T\}$, and $k \in \{1, \ldots, K\}$. The simulated data vector, $y_{itk}(\Theta)$, depends on a vector of structural parameters, Θ. Define the moment conditions as

$$\frac{1}{NT} \sum_{i=1}^{N} \sum_{t=1}^{T} \left[h(x_{it}) - \frac{1}{K} \sum_{k=1}^{K} h(y_{itk}(\Theta)) \right] \equiv \Psi^A - \Psi^S(\Theta), \tag{A.4}$$

where $h(y_{itk}(\Theta))$ is a vector of simulated moments and $h(x_{it})$ is the vector of moments from the actual data. $\Psi^A = \frac{1}{NT} \sum_{i=1}^{N} \sum_{t=1}^{T} h(x_{it})$, $\Psi^S(\Theta) = \frac{1}{NTK} \sum_{i=1}^{N} \sum_{t=1}^{T} \sum_{k=1}^{K} h(y_{itk}(\Theta))$.

The simulated moments estimator is defined as the solution to the minimization of

$$\hat{\Theta} = \arg\min_{\Theta} \left[\Psi^A - \Psi^S(\Theta) \right]' \hat{W} \left[\Psi^A - \Psi^S(\Theta) \right], \tag{A.5}$$

where $\hat{W}$ is a positive definite matrix that converges in probability to a deterministic positive definite matrix W. It is constructed by calculating the inverse of the variance-covariance matrix of the data moments. I construct the variance-covariance matrix using the influence function method following Erickson and Whited (2002). I demean each variable at the industry-year level and then calculate the influence functions for each moment using the demeaned data. I then covary these influence functions by summing in both the cross-sectional and time series dimensions.

I use the simulated annealing algorithm to solve the minimization problem. The algorithm starts with an initial guess of parameters and iteratively updates the parameters by adding a random perturbation. The new parameters are accepted with a probability that depends on the change in the objective function and a temperature parameter that gradually decreases over iterations. This allows the algorithm to explore the parameter space and avoid local minima. I set the initial temperature to be sufficiently high to allow for exploration and also start with different initial guesses to ensure the robustness of the estimation results.

The simulated moments estimator is asymptotically normal for fixed K. Denote $g(\Theta) \equiv \Psi^A - \Psi^S(\Theta)$, let Θ_0 be the true parameter vector, and define $G \equiv \partial g(\Theta_0)/\partial \Theta'$. The asymptotic distribution of $\hat{\Theta}$ is given by:

$$\sqrt{n}(\hat{\Theta} - \Theta_0) \xrightarrow{d} \mathcal{N}\left(0, \text{avar}(\hat{\Theta})\right), \tag{A.6}$$

where

$$\text{avar}(\hat{\Theta}) = \left(1 + \frac{1}{K}\right)(G'WG)^{-1}G'W\Omega WG(G'WG)^{-1}, \tag{A.7}$$

where Ω is the asymptotic covariance matrix of the moment vector. I estimate this covariance matrix using the influence function of the moment vector, clustered at the firm level, following Erickson and Whited (2002).

A.3 Shapley Decomposition

As described in Section 4.3, separating the production and markup channels of intangible capital requires treating N^{prod} and N^{mkp} as distinct variables. Since the original value function V has only a single state variable N, I solve an auxiliary value function $\tilde{V}$ from a model with the same parameters but two separate intangible capital stocks: N^{prod} enters only the production function and N^{mkp} enters only the markup function. One concern is that in the auxiliary problem, firms need to invest in two copies of intangible capital to achieve the same level of production and market power as in the original problem. To ensure that the two

problems have a comparable cost structure, I set the adjustment cost as a function of total investment rates in two intangibles, which is given by $\frac{b^N}{2}\left(\frac{I_{it}^{N,\text{prod}}+I_{it}^{N,\text{mkp}}}{N_{it}^{\text{prod}}+N_{it}^{\text{mkp}}}\right)^2(N_{it}^{\text{prod}}+N_{it}^{\text{mkp}})$, where $I_{it}^{N,\text{prod}}$ and $I_{it}^{N,\text{mkp}}$ are the investments in the two intangible capital stocks.

I then define a composite evaluation rule $\hat{V}$ that governs how coalitions are evaluated throughout the Shapley decomposition:

$$\hat{V}(\cdot)=\begin{cases}V(\cdot) & \text{if } N^{\text{prod}}=N^{\text{mkp}}\\ \tilde{V}(\cdot) & \text{if } N^{\text{prod}}\neq N^{\text{mkp}}.\end{cases} \tag{A.8}$$

That is, when N^{prod} and N^{mkp} are at the same value—both at benchmark or both at their firm-level value—the coalition is evaluated using the original V. The auxiliary $\tilde{V}$ is used only when exactly one of the two has been switched on. This ensures that the decomposition recovers the original V_{it} at the grand coalition while allowing the production and markup channels to be separated.

I then run a Shapley decomposition over six variables $\mathcal{N}=\{z^K,z^N,\zeta^\mu,K,N^{\text{prod}},N^{\text{mkp}}\}$. For a given variable $j\in\mathcal{N}$ and a subset $S\subseteq\mathcal{N}\setminus\{j\}$, define $\hat{V}(S\cup\{j\})$ as the composite value function evaluated with variable j and every variable in S set to their firm-level values, and all remaining variables set to their benchmark values. Similarly, $\hat{V}(S)$ sets every variable in S to its firm-level value and all other variables (including j) to their benchmark values. The Shapley value of variable j is:

$$V_{it}^j=\sum_{S\subseteq\mathcal{N}\setminus\{j\}}\frac{|S|!\,(5-|S|)!}{6!}\left[\hat{V}(S\cup\{j\})-\hat{V}(S)\right], \tag{A.9}$$

where $|S|$ denotes the cardinality of S. The sum ranges over all $2^5=32$ subsets of the five variables other than j, weighting each marginal contribution by the Shapley coefficients. The empty coalition corresponds to all variables at their benchmark values, yielding $\hat{V}(\emptyset)=\bar{V}$, and the grand coalition corresponds to all variables at their firm-level values, yielding $\hat{V}(\mathcal{N})=V_{it}$, since $N^{\text{prod}}=N^{\text{mkp}}=N_{it}$ at the grand coalition.

Intuitively, the Shapley decomposition calculates the average change in firm value when a variable moves from its benchmark to its firm-level value, across all possible orderings in which the six variables are added one by one. I set the benchmark productivity and exogenous markup shocks to zero and the benchmark capital stocks to their minimum grid values. The decomposition term $V_{it}^{\zeta^{\mu}}$ captures the marginal contribution of the exogenous markup shock relative to its zero-shock benchmark, averaged across all possible orderings of the other variables. The term $V_{it}^{N,\text{prod}}$ measures the value of adding intangible capital to the production process only, without its effect on markup, since N^{mkp} is treated as a separate variable. Similarly, $V_{it}^{N,\text{mkp}}$ captures the value of intangible capital's contribution to market power, holding the production channel fixed.

B Additional Results

This section organizes the supplemental empirical results by topic. Table IA.1 collects the variable definitions used across these analyses.

B.1 Alternative Specifications of Estimation

Table IA.2 re-estimates the model under five alternative specifications and reports the parameter estimates. Figure IA.1 reports the relative importance of intangibles' value in production and market power for each specification, including the baseline version in the main text. Table IA.3 shows other main results: Panel A reports the model fit, Panel B reports the covariance decomposition, Panel C reports the within-firm and compositional decomposition, and Panel D reports the life-cycle patterns. The main findings hold across specifications.

B.1.1 Two-Stage Estimation

The main specification estimates all 18 parameters jointly by targeting the 23 moments in Table 2. Since return moments are measured with larger standard errors in the data, the joint estimation allocates smaller weights to these moments. Column (2) of Table IA.2 instead estimates the model in two stages, so return moments can be matched more precisely. The first stage estimates the 16 parameters by targeting the 19 moments on firm characteristics and aggregate quantities. The second stage holds these parameters fixed and estimates the risk prices γ^K and γ^N from return moments.

B.1.2 Physical Investment Measured with PPENT

The main specification measures physical investment with capital expenditures (Compustat item CAPX), which matches the definition in the investment-q regressions in Section 6.2. Column (3) of Table IA.2 instead measures physical investment as the change in net property, plant, and equipment plus depreciation, $\text{PPENT}_t - \text{PPENT}_{t-1} + (\text{DP} - \text{AM})$. Production

value accounts for a larger share of the cross-sectional variation in firm value. The results are otherwise broadly similar to the main specification.

B.1.3 Reversible Intangible Investment

The main specification does not allow disinvestment in intangible capital (Section 3). Column (4) of Table IA.2 relaxes this assumption: intangible investment can be negative and incurs the same quadratic adjustment cost b^N in both directions. The estimated adjustment cost of intangible capital is higher than in the main specification, which partly replaces the friction previously imposed by irreversibility. The main results are similar. Intangible-driven market power accounts for a slightly larger share of firm value and of its cross-sectional variation.

B.1.4 Alternative Intangible Capital Measure

The main specification measures intangible capital following Peters and Taylor (2017), who capitalize R&D and a fixed 30% of SG&A expenditures. Column (5) of Table IA.2 instead uses the measure of Ewens, Peters and Wang (2024), which allows the capitalization share of SG&A and depreciation rates to vary across industries based on estimation from the market prices of acquired intangibles. The resulting measure shows a smaller intangible capital stock relative to the main specification. Under this alternative measure, intangible-driven market power accounts for a smaller share of firm value, but its patterns over time, across the firm life cycle, and in the cross section remain similar to those in the main specification.

B.1.5 Including SG&A in Markups

The main specification uses SALE/COGS as a proxy for markups. Column (6) of Table IA.2 instead includes part of SG&A in variable costs, following Ayyagari, Demirgüç-Kunt and Maksimovic (2024), and measures markups as SALE/(COGS + XSGA*), where XSGA* includes 70% of SG&A expenditures. All markup-based moments are recomputed using this measure, and firm-years with missing SG&A are excluded. This specification provides an

aggressive treatment of SG&A as variable costs: 30% is capitalized as intangible investment, while all remaining SG&A is treated as variable. Moreover, because SG&A enters both the denominator of the markup measure and the construction of intangible capital, this specification mechanically weakens the correlation between markups and intangible intensity. The resulting estimate of intangible-driven market power can therefore be viewed as a lower bound on its contribution to firm value.

The parameter estimates are consistent with this interpretation. The estimated effect of intangible capital on markups is smaller under this alternative measure. The capital share, η, also declines substantially because including SG&A raises measured variable costs and therefore increases the production share of variable inputs. Nonetheless, the main findings remain similar.

B.2 Additional Evidence on Intangibles and Markups

Figures IA.2 and IA.3 test whether the positive relation between intangible intensity and markups is sensitive to the markup measure. They repeat the annual correlations and binned-scatter plots for two model-free sales-to-cost ratios and four production-function estimates that vary the functional form and inclusion of intangible capital. Table IA.4 reports corresponding regressions with firm controls and industry and year fixed effects. The positive association remains across all six measures.

Figure IA.4 uses a stacked difference-in-differences design that treats every important patent as an event. Figures IA.5–IA.8 vary the markup measure, patent definition, and event date, covering filings and issuances for highly cited and high-value patents under all six markup measures. Because the stacked estimates display stronger pre-trends, I treat them as robustness evidence rather than the preferred design.

B.3 Additional Validation

Figure IA.9(a) plots the relation between the intangible share and the total value of intangibles as a percentage of total value across firms. The intangible intensity is measured as the ratio of intangible capital to total capital, and the value of intangibles is the sum of the value of intangible capital in production and intangible-driven market power. As expected, firms with higher intangible intensity tend to have a larger share of their value attributed to intangibles. Similarly, Figure IA.9(b) plots the relation between the markup and the value of market power as a percentage of total value across firms. Firms with higher markups tend to derive more value from market power, and this relation is more concave.

Table IA.5 relates the intangible and non-intangible market-power value shares to standard measures of product-market dynamics. Panel A aggregates the value shares to NAICS-3 industry-year cells and considers concentration and establishment net entry, while Panel B uses firm-level TNIC concentration, product similarity, and fluidity within NAICS-3 industry-year cells. The components have different associations with these measures, but the evidence is descriptive rather than causal.

B.4 R&D, SG&A, and Acquired Intangible Capital

Different types of intangible capital may contribute differently to production and market power. For tractability and identification, I model intangible capital as a single state variable. Adding separate intangible states would substantially increase the computational burden and make it difficult to separately identify their effects from the data. Instead, this section examines whether the relative importance of the two channels varies with the composition of firms' intangible capital.

Figure IA.10 plots the ratio of intangible value in production to intangible value in market power against the share of each type of intangible capital. I consider knowledge capital (capitalized R&D), organization capital (capitalized SG&A), and acquired intangible capital (Compustat item INTAN), which together comprise the intangible stock used in the paper.

Knowledge capital and acquired intangibles both exhibit a steeper positive relation between the value ratio and their capital shares, suggesting that they contribute relatively more to production as their importance increases. By contrast, organization capital is associated with relatively more market power value and a much flatter relation. Because SG&A includes advertising and marketing expenditures, organization capital likely captures some brand capital, which can reduce consumers' price sensitivity.

B.5 Alternative Specification for Life Cycle Regressions

Figure IA.11 replaces the age indicators in the baseline life-cycle regressions with natural cubic splines, while retaining the sample restriction to firms observed for at least 15 years and the year and cohort fixed effects. The smoother specification yields the same qualitative reallocation within intangible value: the production component declines with firm age, whereas intangible-driven market power rises.

B.6 Additional Evidence on the Value Premium Decline

Table IA.6 replaces the continuous time-trend specification with early- and late-sample portfolio returns based on three alternative split dates: July 1996, January 2002, and January 2010. The magnitude and significance of the early-late comparison vary with the selected date, illustrating why the main analysis uses the full monthly series rather than a single sample split. Across the alternatives, the fitted fundamental-to-market signal continues to deliver the strongest value premium in the second subsample.

B.7 Alternative Definitions of Investment Rates and q

Table IA.7 uses the exact Peters–Taylor definition of total q, rather than the paper's cash-adjusted measure, and tests whether it retains explanatory power after the four model-implied value components are included. This specification checks that the component results

do not depend on the paper's particular construction of observed total q.

Tables IA.8 and IA.9 instead scale total, physical, and intangible investment—and the corresponding value measures—by total capital $K + N$. This common denominator makes coefficients comparable across investment margins but also introduces variation in capital composition. The horse races, component regressions, and Shapley decomposition remain qualitatively similar to the baseline results that scale each investment rate by its corresponding capital stock.

Table IA.1: Variable definitions

Variable	Definition
Intangible capital	Accumulated R&D and capitalized SG&A using the perpetual-inventory method, plus balance-sheet goodwill, from Peters and Taylor (2017)
Physical capital	PPEGT
Market equity	Price (CRSP item PRC) × shares outstanding (SHROUT)
Total firm value	Market equity + DLC + DLTT − CHE
Profitability	(REVT − COGS) / (PPEGT + intangible capital)
Physical investment rate	CAPX / PPEGT
Intangible investment rate	(XRD + capitalized share of XSGA) / intangible capital
Intangible share	Intangible capital / (PPEGT + intangible capital)

Table IA.2: Parameter estimates across estimation specifications

This table reports the estimated parameters for the specifications in Table IA.3. Column (1) is the main specification in Panel B of Table 1. The calibrated parameters are the same as in Panel A of Table 1 in all specifications. The model is solved and simulated at annual frequency.

Variable	Notation	Main (1)	Two-stage estimation (2)	PPENT investment (3)	Reversible intangibles (4)	EPW intangibles (5)	SG&A in markup (6)
Group 1: Production technology							
Share of physical capital	α	0.8334	0.8696	0.8440	0.8365	0.9010	0.9118
Total share of capital	η	0.2570	0.2607	0.2468	0.2326	0.2630	0.1218
CES curvature	ρ	0.4974	0.4907	0.2541	0.6392	0.3714	0.4015
Fixed operating cost	f	0.3492	0.3377	0.2729	0.5276	0.6429	0.3577
Group 2: Investment and capital-specific productivity							
Adjustment cost of physical capital	b^K	2.4331	2.6202	0.8821	1.9949	2.9172	4.0489
Adjustment cost of intangible capital	b^N	1.8946	1.8992	1.7602	2.7711	0.8069	1.7331
Autocorrelation of z_t^K	ρ^K	0.8641	0.8894	0.6911	0.8861	0.8968	0.8936
Volatility of ζ_t^K shocks	σ_ζ^K	0.1804	0.1525	0.1460	0.2164	0.1337	0.1750
Autocorrelation of z_t^N	ρ^N	0.9161	0.9327	0.9356	0.9403	0.9869	0.9477
Volatility of ζ_t^N shocks	σ_ζ^N	0.3843	0.3694	0.3384	0.3713	0.1104	0.3149
Group 3: Markup-related							
Average effect of intangible capital on markup	$\bar{\mu}$	0.1280	0.1273	0.1240	0.1344	0.1245	0.1299
Curvature of the markup function	ϕ	0.8175	0.8021	0.8236	0.8086	0.7137	0.5407
Autocorrelation of ζ_t^μ	ρ^μ	0.7699	0.7795	0.7698	0.7593	0.8014	0.7744
Volatility of ζ_t^μ shocks	σ^μ	0.1547	0.1504	0.1502	0.1502	0.1581	0.0676
Group 4: SDF							
Volatility of ε_x^K	σ_x^K	0.0636	0.0442	0.0377	0.0896	0.0504	0.0620
Volatility of ε_x^N	σ_x^N	0.1398	0.1322	0.0932	0.1780	0.1072	0.0489
Risk price of ε_x^K	γ^K	1.1267	16.4510	0.0347	0.0246	0.6408	0.1432
Risk price of ε_x^N	γ^N	31.4845	31.7192	47.0380	23.5776	40.6879	37.7244

Table IA.3: Main results across estimation specifications

Column (1) is the main specification, and Columns (2)–(6) are the alternative specifications in Section B.1. Figure IA.1 reports the level decomposition. Panel A reports the cross-sectional R^2 of V (slope one, zero intercept), averaged across years. Panel B reports the covariance decomposition (Equation (29)), averaged over 1976–2025. Panel C reports the cumulative compositional and within-firm changes (Equation (26)) as a percentage of their sum; values above 100 indicate that the other channel moves in the opposite direction. Panel D reports the firm-age coefficients from the life-cycle regression, with age 1 omitted.

	Main (1)	Two-stage estimation (2)	PPENT investment (3)	Reversible intangibles (4)	EPW intangibles (5)	SG&A in markup (6)
Panel A: Model fit (%)						
Average cross-sectional R^2	85.52	85.74	80.28	87.13	90.70	96.35
Panel B: Covariance decomposition, 1976–2025 (%)						
Intangibles in production	26.34	28.90	53.55	23.77	23.25	29.00
Intangible market power	13.68	13.94	13.73	16.34	9.37	13.75
Non-intangible market power	11.81	10.94	10.92	9.60	10.18	12.46
Panel C: Share of cumulative change (%)						
Intangibles in production						
Compositional	78.08	79.44	93.03	70.86	72.03	69.87
Within-firm	21.92	20.56	6.97	29.14	27.97	30.13
Intangible market power						
Compositional	16.42	19.08	11.71	20.21	29.19	32.38
Within-firm	83.58	80.92	88.29	79.79	70.81	67.62
Non-intangible market power						
Compositional	137.99	139.76	142.54	149.52	119.46	107.76
Within-firm	-37.99	-39.76	-42.54	-49.52	-19.46	-7.76
Panel D: Life-cycle coefficients (%)						
Intangibles in production						
Age 15	-3.45	-3.76	-9.54	-2.91	-2.85	-1.32
Age 25	-4.46	-4.70	-12.31	-3.87	-3.75	-0.83
Intangible market power						
Age 15	7.37	7.20	7.52	8.70	4.50	3.73
Age 25	9.51	9.29	9.45	11.09	5.76	5.06
Non-intangible market power						
Age 15	-3.15	-3.30	-3.04	-5.67	-1.17	-5.81
Age 25	-4.15	-4.35	-4.03	-7.84	-1.96	-7.90

Table IA.4: Relation between intangible share and markups

This table reports OLS regressions of markups on the intangible capital share and firm-level controls. The dependent variable in each column is a different markup measure: column (1) uses SALE/COGS; column (2) uses SALE/(COGS + XSGA*); columns (3) and (4) use Cobb-Douglas production function estimates without and with intangible capital as a production input; columns (5) and (6) use translog production function estimates without and with intangible capital. Intangible share is intangible capital divided by the sum of physical and intangible capital. Leverage is total debt divided by total assets. ROA is return on assets. The sample is the model decomposition sample used throughout the paper. All specifications include two-digit industry and year fixed effects. Standard errors in parentheses are two-way clustered by firm and year.

	(1) $\frac{\text{SALE}}{\text{COGS}}$	(2) $\frac{\text{SALE}}{\text{COGS+XSGA}^*}$	(3) CD	(4) CD (intan.)	(5) TL	(6) TL (intan.)
Intan. share	1.1880***	0.2435***	1.0897***	0.8976***	1.0862***	0.9383***
	(0.0730)	(0.0255)	(0.0615)	(0.0497)	(0.0620)	(0.0549)
Log age	-0.1227***	-0.0329***	-0.1112***	-0.0923***	-0.1120***	-0.0940***
	(0.0134)	(0.0047)	(0.0109)	(0.0090)	(0.0110)	(0.0103)
ROA	1.0849***	0.7200***	0.7378***	0.6468***	0.7306***	0.6677***
	(0.1101)	(0.0765)	(0.0925)	(0.0814)	(0.0899)	(0.0828)
Leverage	-0.1295**	0.0083	-0.1364***	-0.1089***	-0.1347***	-0.1198***
	(0.0500)	(0.0266)	(0.0390)	(0.0334)	(0.0386)	(0.0360)
Log assets	0.0047	0.0208***	0.0045	0.0037	-0.0105**	-0.0192***
	(0.0063)	(0.0027)	(0.0052)	(0.0043)	(0.0051)	(0.0050)
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	122,670	115,089	109,851	109,850	109,851	109,850
Adj. R^2	0.2481	0.3311	0.2362	0.2229	0.2433	0.1912
Within R^2	0.1105	0.2041	0.1218	0.1225	0.1205	0.1103

Table IA.5: Market-power value shares and industry dynamics

This table reports panel regressions of the intangible and non-intangible market-power value shares on product-market competition measures. Each column is a separate univariate regression of a value share on one competition measure. Panel A is estimated at the industry level: the value shares are aggregated to NAICS-3 industry-year cells (value-weighted, the ratio of aggregate markup value to aggregate firm value in the industry) and regressed on industry-level competition measures with year fixed effects. The measures are the sales Herfindahl (HHI) and the four-firm concentration ratio (CR4), both computed at the NAICS-3 level, and the Census Business Dynamics Statistics establishment net entry rate (entry minus exit). Panel B is estimated at the firm level with NAICS-3 industry-by-year fixed effects. The measures are the Hoberg and Phillips (2016) TNIC Herfindahl (TNIC HHI), TNIC similarity, and product-market fluidity (the change in rivals' product vocabulary relative to the firm's own). All Panel B regressions control for firm size (log sales). Standard errors in parentheses are two-way clustered by industry and year in Panel A and by firm and year in Panel B.

Panel A: Industry-level product market measures (1976–2025)

	Intangible market-power share			Non-intangible market-power share		
	HHI	CR4	Net entry	HHI	CR4	Net entry
Coefficient	0.0025	-0.0227	-0.0057***	-0.0253	-0.0879	0.0052*
	(0.0318)	(0.0355)	(0.0019)	(0.0635)	(0.0592)	(0.0030)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	3,899	3,899	3,179	3,899	3,899	3,179
R^2	0.0666	0.0683	0.0868	0.0208	0.0307	0.0189
Within R^2	0.0000	0.0018	0.0193	0.0009	0.0110	0.0061

Panel B: Firm-level product market measures (1988–2023)

	Intangible market-power share			Non-intangible market-power share		
	TNIC HHI	Similarity	Fluidity	TNIC HHI	Similarity	Fluidity
Coefficient	0.0233***	0.0008**	-0.0014**	-0.0352***	-0.0038***	-0.0013
	(0.0068)	(0.0003)	(0.0007)	(0.0106)	(0.0007)	(0.0013)
Industry × Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Firm size control	Yes	Yes	Yes	Yes	Yes	Yes
Observations	93,473	93,473	91,054	93,473	93,473	91,054
R^2	0.2912	0.2912	0.2909	0.2802	0.2855	0.2800
Within R^2	0.0172	0.0172	0.0170	0.0109	0.0183	0.0104

Table IA.6: Value premiums by sample split date

This table reports the value premium, value minus growth in % per month, estimated separately in the first and second subperiods of the sample for three candidate split dates. The sorts are identical to those in Panel A of Table 6: yearly NYSE-breakpoint decile sorts, value weighted by lagged market equity. The full return sample runs from July 1977 to June 2026, so for each split date the first subperiod runs from July 1977 to the month before the split and the second subperiod from the split month to June 2026; the two subperiods for each split date therefore partition the full sample exactly. Newey–West t-statistics using automatic lag selection are in parentheses beneath the estimates. For the BM ratios, the value portfolio is the low market-to-book decile; for fitted FM and FM, it is the high decile.

Split date	July 1996		January 2002		January 2010	
	First	Second	First	Second	First	Second
Data ratios						
BM^{data} (original)	0.49	0.01	0.43	-0.04	0.51	-0.44
	(1.40)	(0.02)	(1.40)	(-0.09)	(1.86)	(-0.81)
BM^{data} (incl. intangibles)	0.74	0.06	0.71	-0.05	0.68	-0.37
	(2.03)	(0.19)	(2.09)	(-0.15)	(2.31)	(-0.83)
Model-implied ratios						
BM^{model} (ex. exogenous markup)	0.54	-0.05	0.55	-0.19	0.46	-0.37
	(1.96)	(-0.23)	(1.99)	(-0.94)	(2.09)	(-1.55)
BM^{model} (ex. all markup)	0.50	0.36	0.67	0.16	0.59	0.08
	(2.11)	(1.65)	(2.95)	(0.73)	(2.99)	(0.28)
Fundamental-to-market						
Fitted FM	0.72	0.39	0.78	0.24	0.61	0.33
	(3.32)	(1.78)	(4.79)	(0.98)	(3.81)	(0.97)
FM	0.61	0.34	0.83	0.06	0.77	-0.19
	(2.25)	(1.30)	(3.37)	(0.23)	(3.81)	(-0.61)

Table IA.7: Investment and q: Peters-Taylor q and the value components

This table asks whether Peters-Taylor q still explains investment after controlling for the model's four value components. Columns (1)–(2) explain physical investment I^K/K and columns (3)–(4) intangible investment I^N/N; both dependent variables are measured one year ahead, so the regression is of investment in $t+1$ on q in t. Columns (1) and (3) use Peters-Taylor q on its own, and columns (2) and (4) add the four value components alongside it. The components are scaled by the capital stock of the margin being explained: by K in column (2) and by N in column (4). The sample is restricted to the firm-years for which Peters-Taylor q is available, and is therefore slightly smaller than in the other investment tables. All specifications include firm and year fixed effects, with standard errors two-way clustered by firm and year in parentheses.

	Physical, I^K/K		Intangible, I^N/N	
	(1)	(2)	(3)	(4)
Peters-Taylor q	0.0324***	0.0257***	0.0289***	0.0247***
	(0.0030)	(0.0025)	(0.0020)	(0.0019)
$V^{\text{prod-phy}}$		0.0094***		0.0005
		(0.0017)		(0.0004)
$V^{\text{prod-int}}$		0.0018**		0.0033
		(0.0007)		(0.0022)
$V^{\text{mkp-int}}$		0.0105***		0.0477***
		(0.0019)		(0.0059)
$V^{\text{mkp-non}}$		-0.0018*		0.0007*
		(0.0009)		(0.0004)
Observations	114,402	114,402	114,402	114,402
R^2	0.4656	0.4793	0.6295	0.6424
Within R^2	0.1099	0.1329	0.1227	0.1533

Table IA.8: Investment and q: horse race, scaled by total capital

This table repeats Table 8 with every investment rate scaled by total capital $K+N$ rather than by the capital stock of its own margin, which makes the three margins comparable on a common base. Each column is a separate regression of the investment rate on a single q, so the coefficients in a panel are directly comparable across columns. Panel A uses total investment $I/(K+N)$, Panel B uses physical investment $I^K/(K+N)$, and Panel C uses intangible investment $I^N/(K+N)$; all three are led one year, so the regression is of investment in $t+1$ on q in t. Model total q is the model-implied value over $K+N$; the last two columns strip the non-intangible market-power component $V^{\text{mkp-non}}$, and then both market-power components $V^{\text{mkp-int}}$ and $V^{\text{mkp-non}}$, out of that value. All specifications include firm and year fixed effects, with standard errors two-way clustered by firm and year in parentheses.

	Standard q	Total q	Model total q	Model total q (ex. non-intangible markup)	Model total q (ex. all markup)
	(1)	(2)	(3)	(4)	(5)
Panel A: Total investment, $I/(K+N)$					
q	0.0041***	0.0297***	0.0616***	0.0644***	0.0700***
	(0.0005)	(0.0024)	(0.0029)	(0.0029)	(0.0031)
Observations	114,668	114,668	114,668	114,668	114,668
R^2	0.5696	0.6211	0.6184	0.6150	0.6139
Within R^2	0.0715	0.1827	0.1769	0.1695	0.1672
Panel B: Physical investment, $I^K/(K+N)$					
q	0.0015***	0.0139***	0.0320***	0.0328***	0.0368***
	(0.0002)	(0.0016)	(0.0020)	(0.0021)	(0.0022)
Observations	114,668	114,668	114,668	114,668	114,668
R^2	0.5058	0.5385	0.5466	0.5427	0.5449
Within R^2	0.0187	0.0837	0.0998	0.0920	0.0965
Panel C: Intangible investment, $I^N/(K+N)$					
q	0.0026***	0.0151***	0.0283***	0.0303***	0.0317***
	(0.0003)	(0.0008)	(0.0013)	(0.0014)	(0.0015)
Observations	114,668	114,668	114,668	114,668	114,668
R^2	0.7847	0.8026	0.7933	0.7933	0.7905
Within R^2	0.1097	0.1838	0.1451	0.1454	0.1335

Table IA.9: Investment and q: value components, scaled by total capital

This table repeats Table 9 with every investment rate and value component scaled by total capital $K+N$ rather than by the capital stock of the margin being explained, and adds the total-investment margin. Panel A reports the investment regressions. Columns (1)–(2) explain total investment $I/(K+N)$, columns (3)–(5) physical investment $I^K/(K+N)$, and columns (6)–(8) intangible investment $I^N/(K+N)$; all rates are measured one year ahead. Columns (3) and (6) use the components generated by the corresponding capital stock, columns (1), (4), and (7) use all four components, and columns (2), (5), and (8) add total q. Panel B decomposes the within-R^2 of columns (1), (4), and (7) into Shapley shares, which add to 100. All specifications include firm and year fixed effects; standard errors in Panel A are two-way clustered by firm and year.

	Total, $I/(K+N)$		Physical, $I^K/(K+N)$			Intangible, $I^N/(K+N)$		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Investment regressions								
$V^{\text{prod-phy}}$	0.0747***	0.0447***	0.0528***	0.0519***	0.0419***		0.0210***	0.0018
	(0.0032)	(0.0031)	(0.0027)	(0.0026)	(0.0025)		(0.0016)	(0.0014)
$V^{\text{prod-int}}$	0.0478***	0.0150***		0.0133***	0.0024	0.0348***	0.0332***	0.0122***
	(0.0038)	(0.0034)		(0.0019)	(0.0019)	(0.0029)	(0.0026)	(0.0022)
$V^{\text{mkp-int}}$	0.0529***	0.0468***		-0.0252***	-0.0272***	0.1171***	0.0800***	0.0761***
	(0.0110)	(0.0105)		(0.0067)	(0.0064)	(0.0075)	(0.0072)	(0.0071)
$V^{\text{mkp-non}}$	0.0412***	0.0274***		0.0298***	0.0252***		0.0101***	0.0012
	(0.0042)	(0.0041)		(0.0028)	(0.0027)		(0.0023)	(0.0021)
Total q		0.0178***			0.0059***			0.0114***
		(0.0016)			(0.0008)			(0.0008)
Observations	114,668	114,668	114,668	114,668	114,668	114,668	114,668	114,668
R^2	0.6211	0.6355	0.5532	0.5579	0.5615	0.7896	0.7970	0.8090
Within R^2	0.1826	0.2138	0.1130	0.1222	0.1295	0.1301	0.1604	0.2100

Panel B: Shapley decomposition of within-R^2 (%)			
	Total, $I/(K+N)$	Physical, $I^K/(K+N)$	Intangible, $I^N/(K+N)$
$V^{\text{prod-phy}}$	62.87	83.03	30.39
$V^{\text{prod-int}}$	16.55	5.14	33.77
$V^{\text{mkp-int}}$	15.89	5.33	33.58
$V^{\text{mkp-non}}$	4.70	6.51	2.26
Total within R^2	0.1826	0.1222	0.1604
Observations	114,668	114,668	114,668

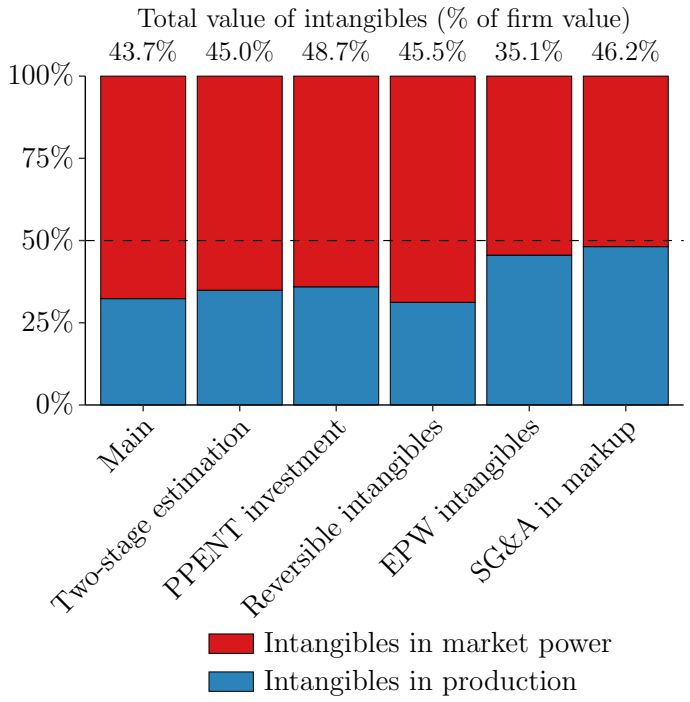


Figure IA.1: Intangible value across estimation specifications

This figure splits intangible value into intangibles in production and intangible market power for each specification in Table IA.3. Each bar reports the share of each component in their sum, based on the level decomposition averaged over 1976–2025. The number above each bar is intangible value as a percentage of total model-implied firm value. The dashed line marks 50%.

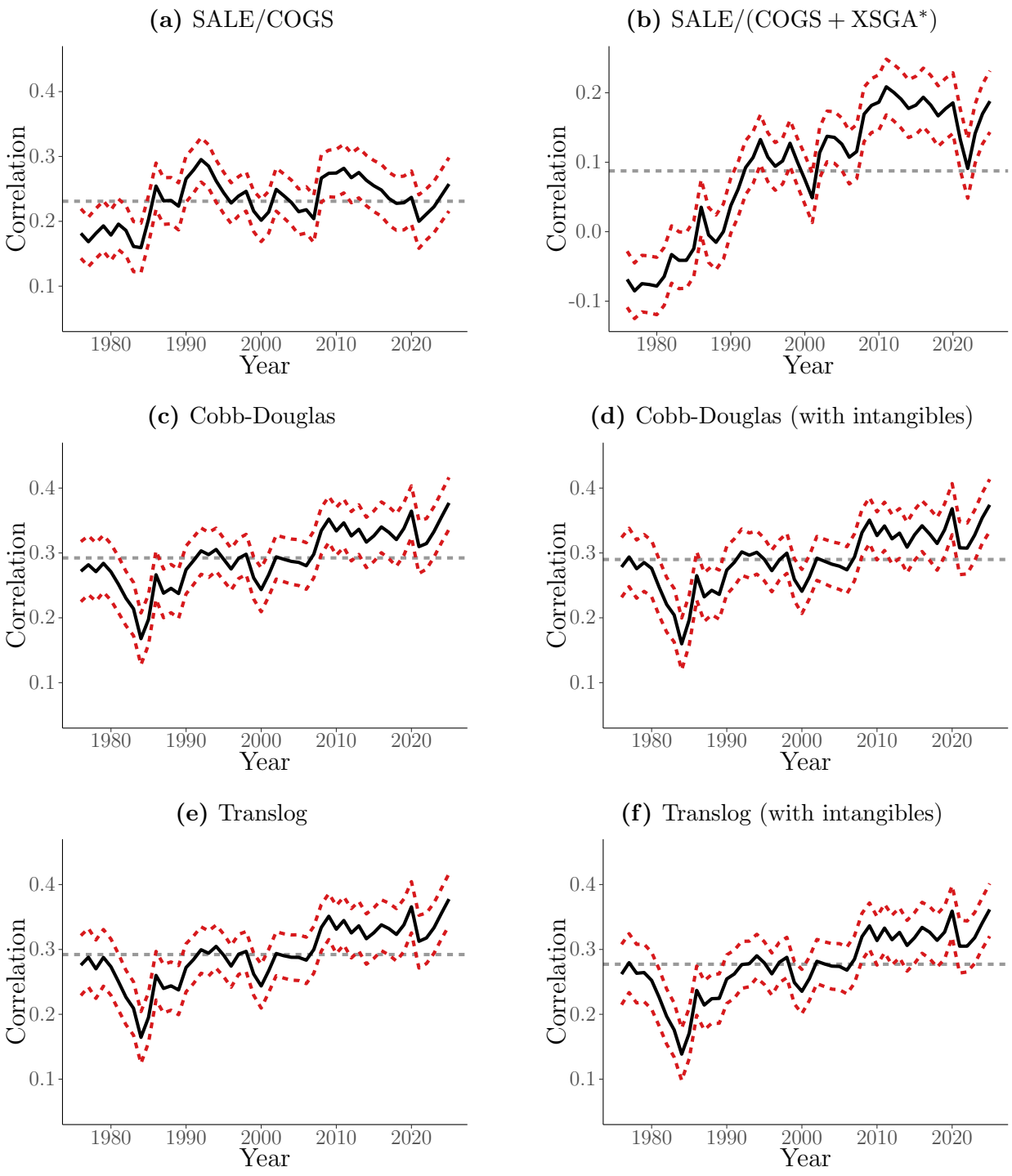


Figure IA.2: Correlation between intangibles and markup: alternative specifications (1976–2025)

This figure repeats Panel (a) of Figure 1 for all six markup measures. The specification is identical; see the note to Figure 1.

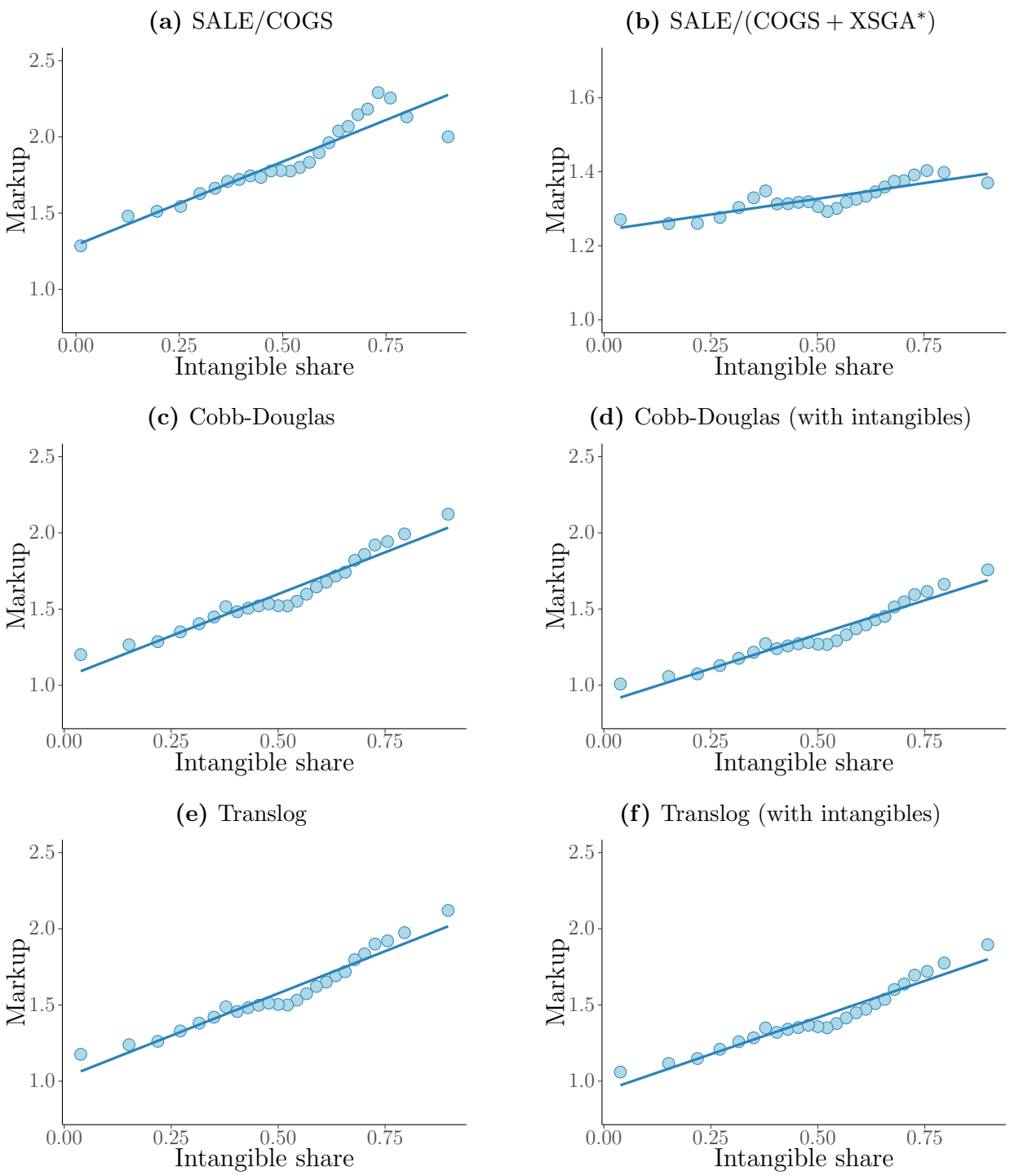


Figure IA.3: Binned scatter plots of markup against intangible share: alternative specifications (1976–2025)

This figure repeats Panel (b) of Figure 1 for all six markup measures. The specification is identical; see the note to Figure 1.

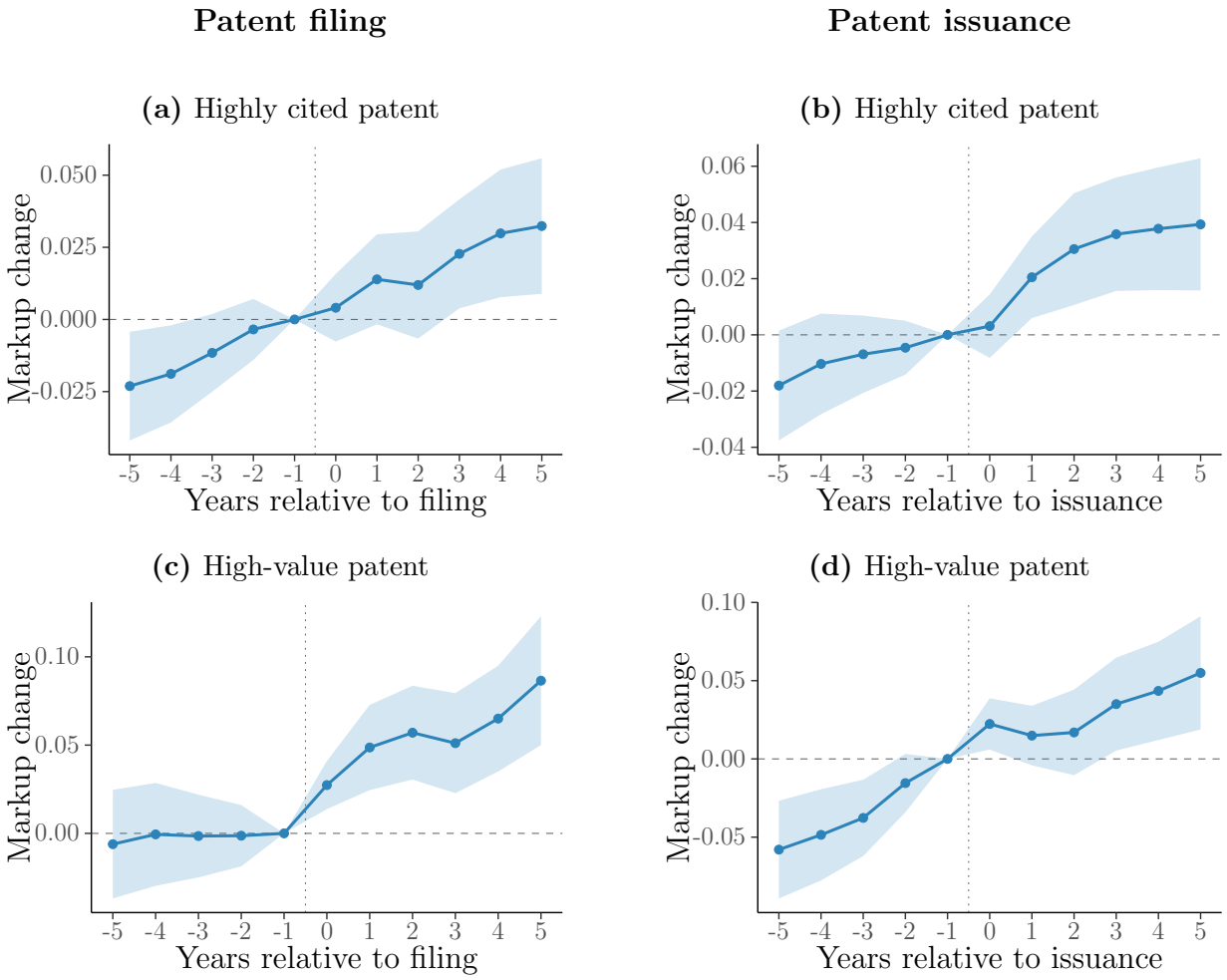


Figure IA.4: Event studies under Cobb-Douglas with intangibles: stacked difference-in-differences

This figure reproduces Figure 2 using a stacked difference-in-differences design in place of the first-event specification for important patent filings (left column) and issuances (right column). Panels (a) and (b) define important patents by forward citations in the top 10% within their Fama-French 48 industry and year; Panels (c) and (d) instead use the Kogan et al. (2017) stock market value of the patent. Each firm-year with an important patent forms a separate event stack: the treated firm is followed over $\tau \in [-5, +5]$, and controls are firms in the same Fama-French 48 industry with no important patent within that window. Stacks are pooled with event-specific firm and year fixed effects, and treated firms are required to have markup data in every pre-event year $\tau \in \{-5, \ldots, -1\}$. Shaded areas are 95% confidence intervals, with standard errors two-way clustered by event and firm.

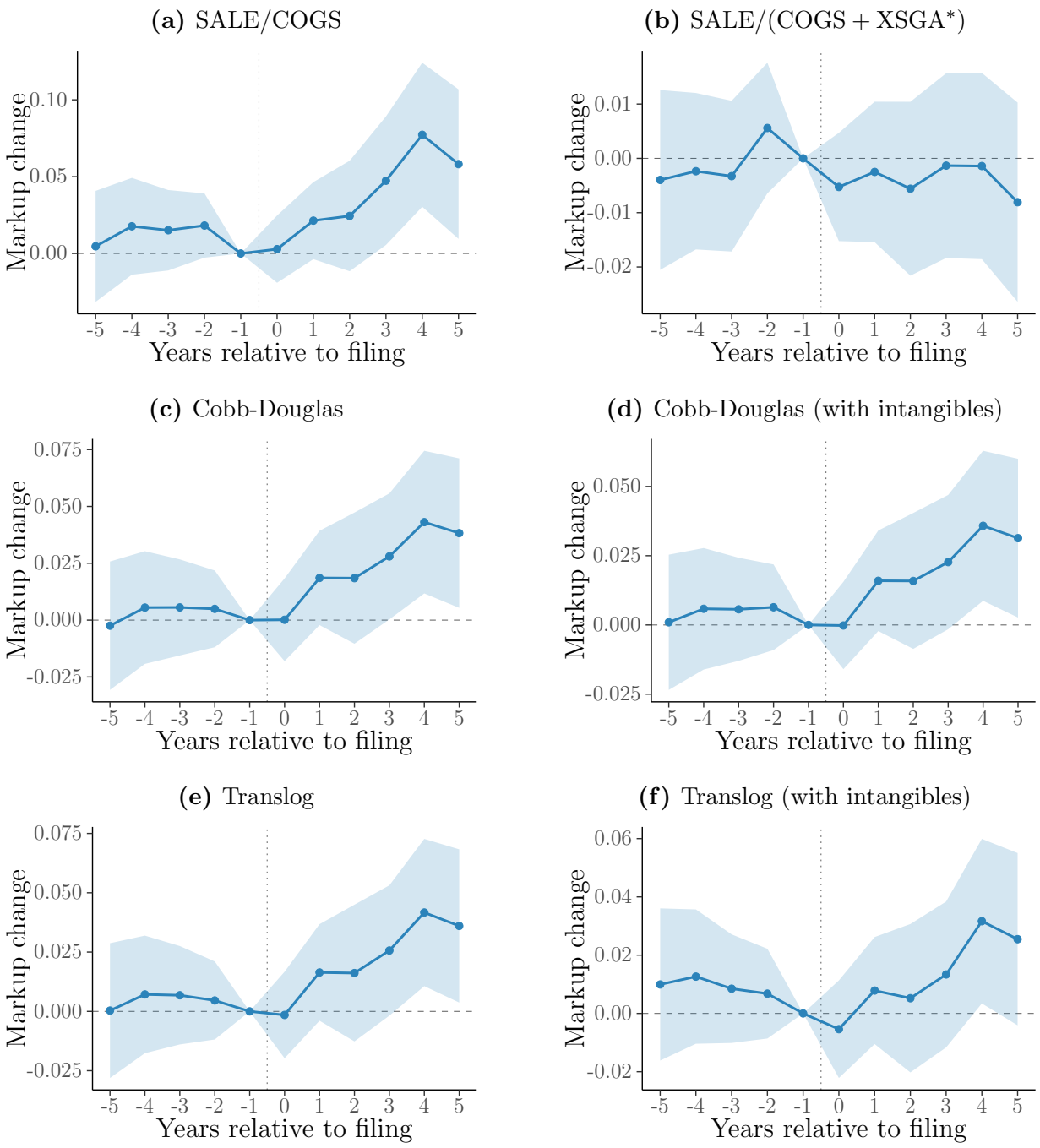


Figure IA.5: Event study around patent filing: highly cited patents

This figure repeats the event study of Figure 2 (Panels (a) and (b)) around patent filing, for all six markup measures. The specification is identical; see the note to Figure 2.

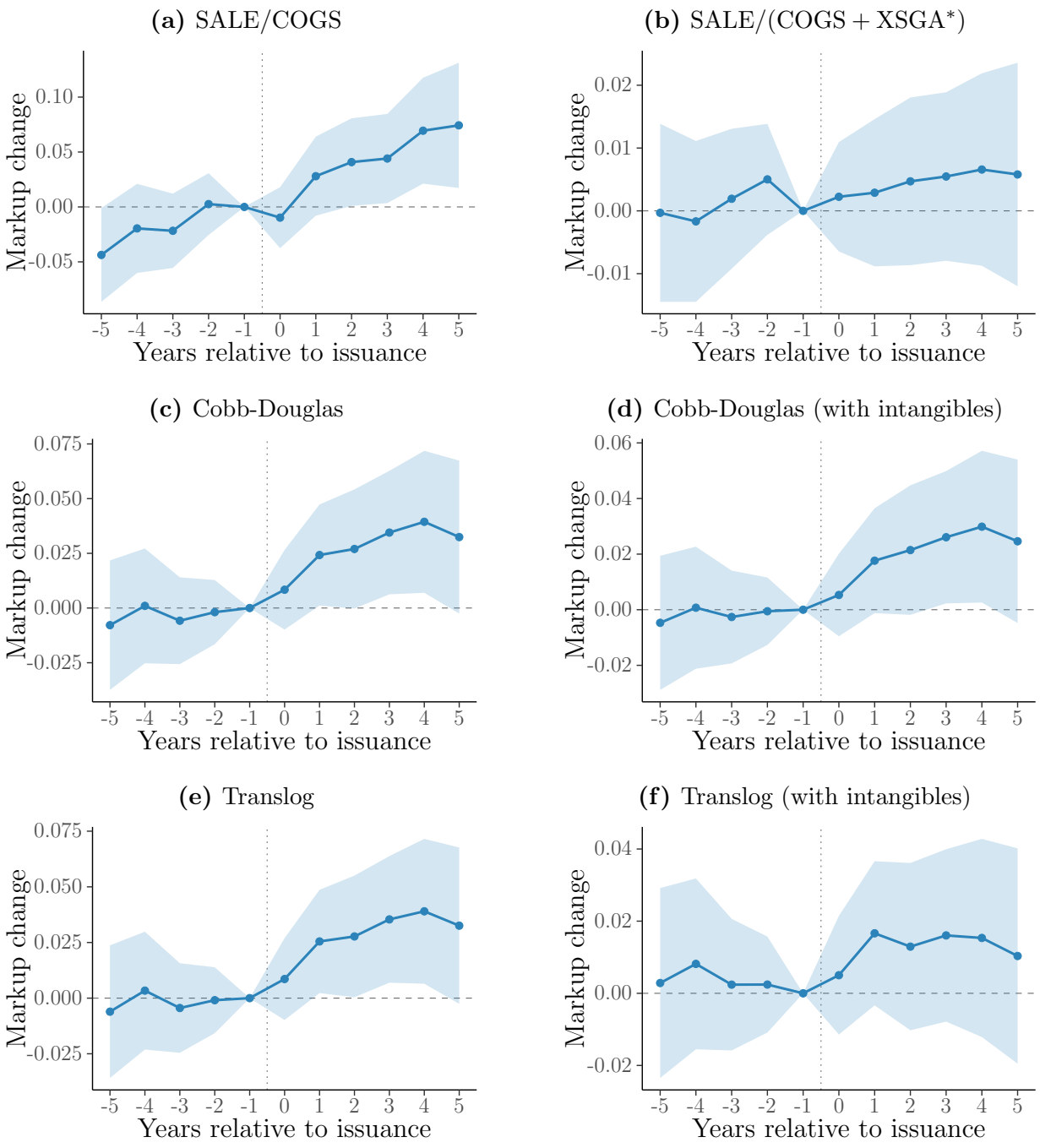


Figure IA.6: Event study around patent issuance: highly cited patents

This figure repeats the event study of Figure 2 (Panels (a) and (b)) around patent issuance, for all six markup measures. The specification is identical; see the note to Figure 2.

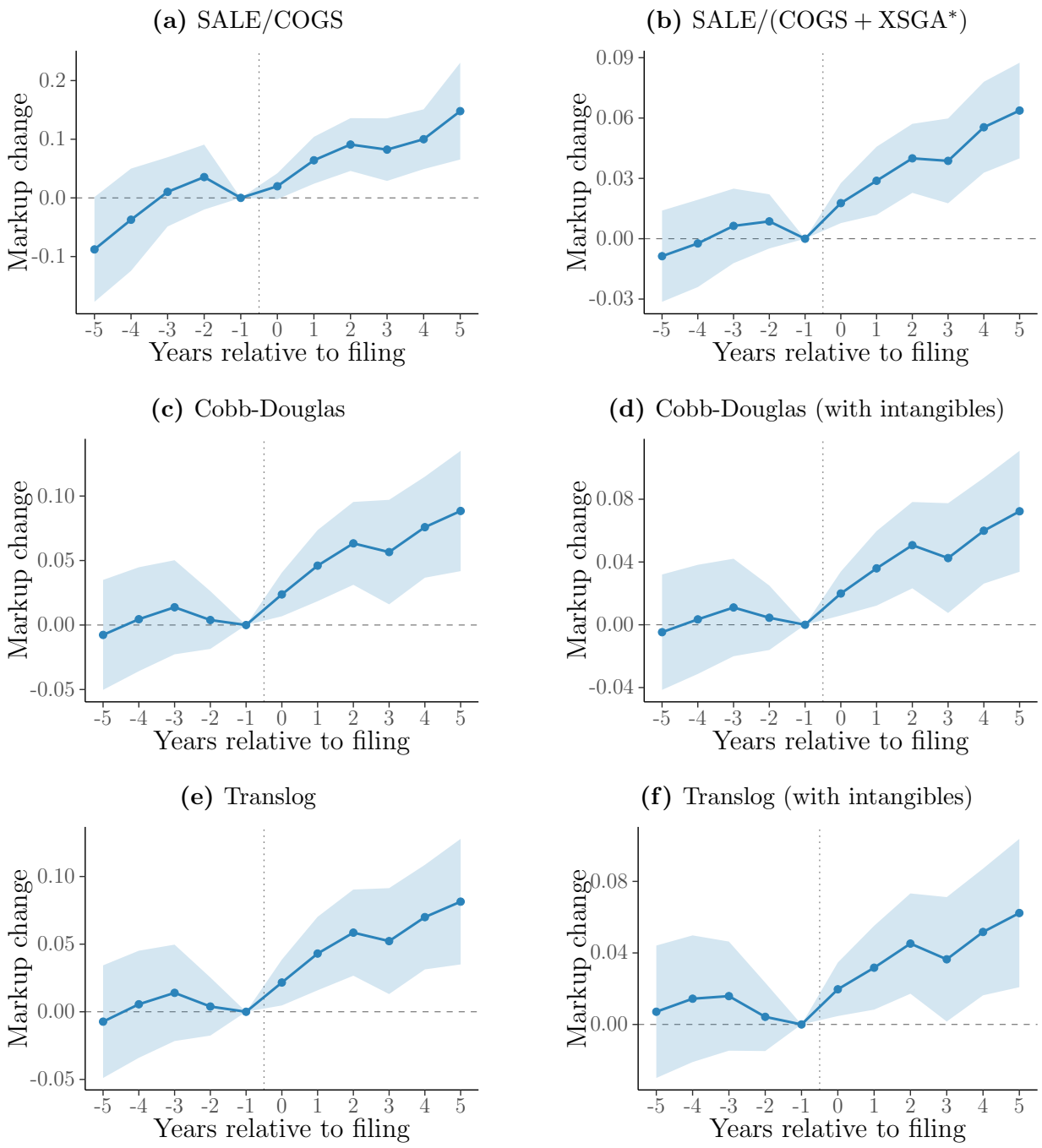


Figure IA.7: Event study around patent filing: high-value patents

This figure repeats the event study of Figure 2 (Panels (c) and (d)) around patent filing, for all six markup measures, where important patents are defined by the Kogan et al. (2017) stock-market value rather than forward citations. The specification is identical; see the note to Figure 2.

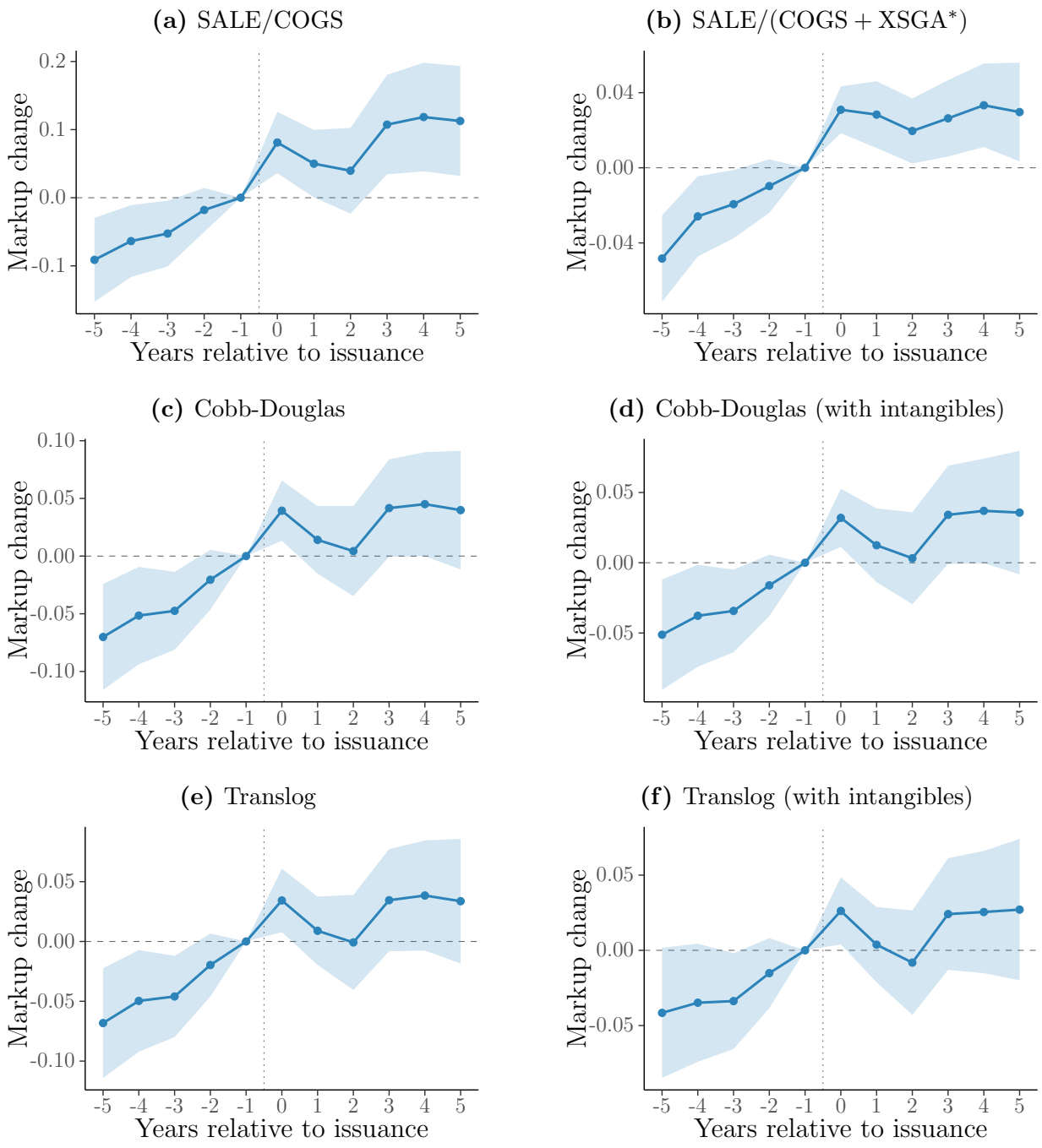


Figure IA.8: Event study around patent issuance: high-value patents

This figure repeats the event study of Figure 2 (Panels (c) and (d)) around patent issuance, for all six markup measures, where important patents are defined by the Kogan et al. (2017) stock-market value rather than forward citations. The specification is identical; see the note to Figure 2.

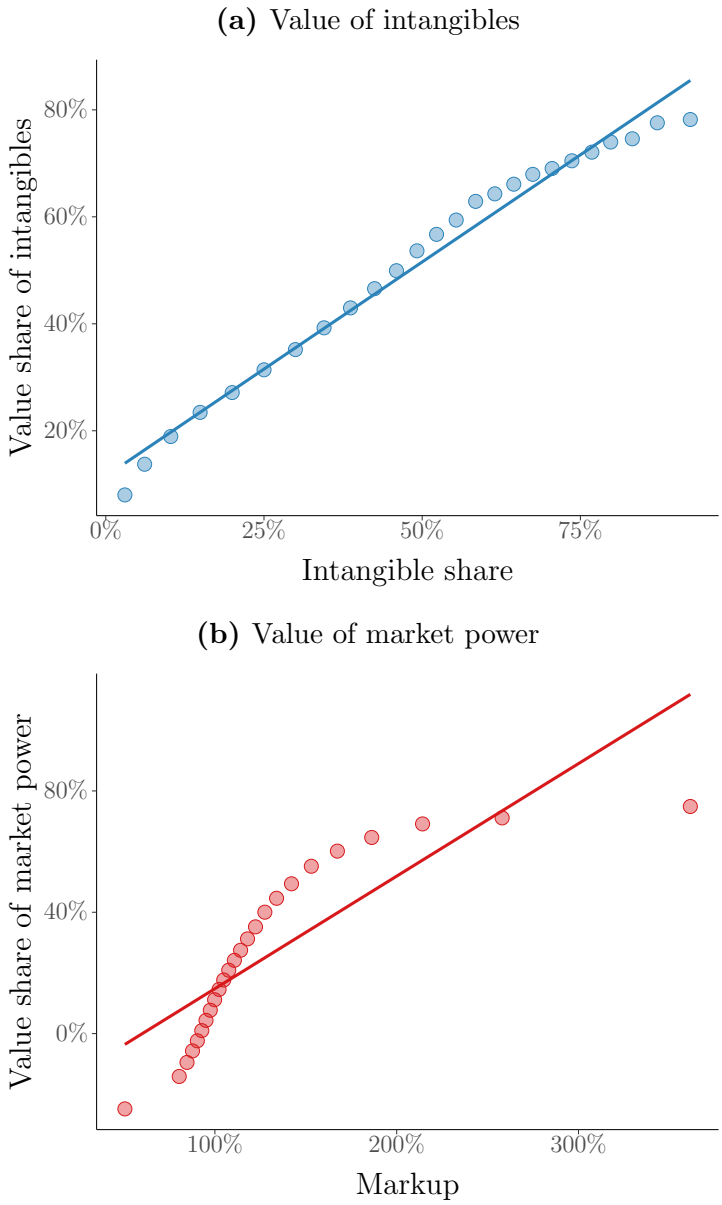


Figure IA.9: Value shares against intangible share and markup

This figure plots binned scatter plots of model-implied value shares against firm characteristics. Panel (a) plots the share of firm value attributable to intangibles ($V_{it}^{\text{prod-int}} + V_{it}^{\text{mkp-int}}$, scaled by V_{it}^{model}) against the firm's intangible share. Panel (b) plots the share of firm value attributable to market power ($V_{it}^{\text{mkp-int}} + V_{it}^{\text{mkp-non}}$, scaled by V_{it}^{model}) against the firm's markup. Observations are grouped into equal-sized bins based on the x-axis variable, and bin means are plotted. The solid lines show the linear fit.

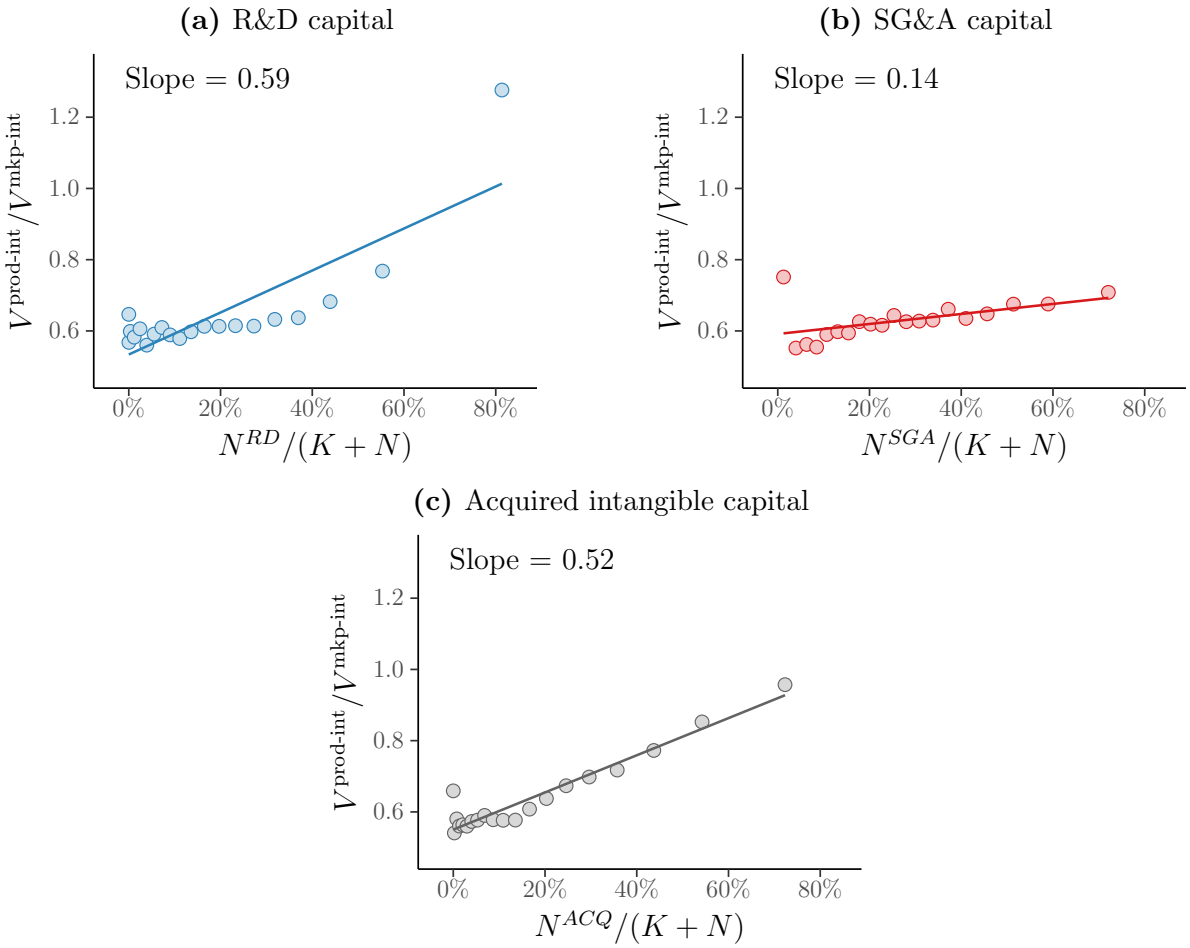


Figure IA.10: Intangible production relative to market power by type of intangible capital

This figure presents binned scatter plots of the ratio of the value of intangibles in production to the value of intangible market power against each type of intangible capital, scaled by total capital $K + N$. Intangible capital is split into capitalized R&D, capitalized SG&A, and acquired intangible capital, each measured at $t - 1$. Industry (2-digit SIC) $\times$ year fixed effects are removed from the ratio before binning, and the sample mean is added back. The ratio is winsorized at the 1st and 99th percentiles. The reported slope is the linear fit through the bin means.

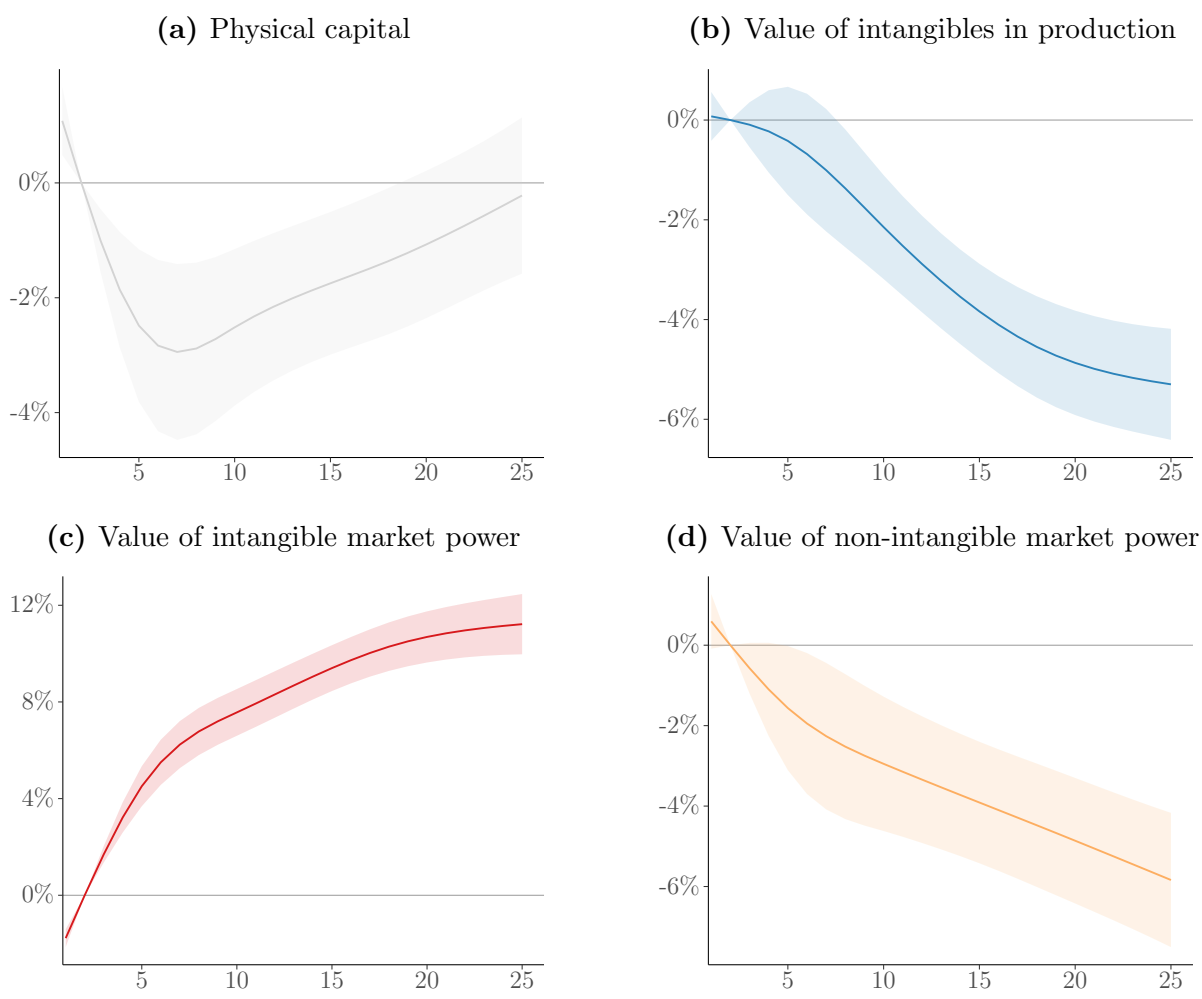


Figure IA.11: Life cycle changes

This figure plots the evolution of each value component's share over a firm's life cycle. For each component, I regress the ratio of the component to total model-implied value on firm age splines with year and cohort fixed effects, restricting the sample to firms with at least 15 years of data. The solid line shows the estimated coefficients and the shaded area represents the 95% confidence interval.